

COMPREHENSIVE NETWORK SECURITY DESIGN AND IMPLEMENTATION FOR AVALANCH COOPERATIONS

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Executive Summary

This report is a detailed plan and execution of a secure hierarchical network infrastructure to be used by the new branch of the Avalanch Cooperations in Negombo, Sri Lanka. The project was aimed at linking 358 hosts in eight different departments with the Main Branch through a secure Site-to-Site IPSec VPN, by which data confidentiality and integrity would be ensured. The network design is based on the high availability, scalability, and high security with the usage of redundancy in multilayer switches, rigorous VLAN segmentation and Access Control Lists (ACLs) to enforce isolation of sensitive Finance and HR activities. The integration of centralized management services was done, such as AAA, DHCP, and Syslog, so that the administration becomes easier and the same security policies apply across the organization. The last design addresses the need to achieve 99.9% uptime and offers a scale base to expand in the future.

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1. Introduction

Avalanch Cooperations is a technology services provider that deals with IT consulting, network design and deployment, security solutions, cloud services, and network monitoring, and has recently opened a new branch in Negombo, Sri Lanka. The new location will require the implementation of efficient and effective network infrastructure that will be able to sustain eight different departments with their respective operation needs [2].

1.1. Project Objectives

1. To develop the hierarchical network infrastructure to support both the present and future growth.
2. To introduce effective IP address schemes based on VLSM to maximize the use of address spaces.
3. To implement safe connectivity amid the chief branch and the new one using VPN technology.
4. To implement the segmentation of the network through VLANs to improve security and performance.
5. To centralize network services, such as DHCP, NTP, Syslog and AAA.
6. To implement security policy using ACLs, SSH, and port-security.
7. To provide high availability using redundancy.

1.2. Scope of Work

- Network topology design and documentation
- VLSM-based IP addressing scheme for efficient address allocation
- Cisco Packet Tracer implementation of the complete network infrastructure
- Configuration of eight departmental VLANs with appropriate security controls
- VPN tunnel establishment for secure inter-branch communication
- Implementation of routing protocols for connectivity
- Security mechanism deployment (ACLs, SSH, port security)
- Comprehensive testing and verification procedures
- Detailed documentation of all configurations and implementations

1.3. Organizational Context

Negombo branch should have 358 hosts serviced by eight departments, each with different security need. Finance and HR department handle very sensitive information and hence require insulation with other departments and yet need to be connected to other vital services [3]. The network design should thus be able to attain scalability, reliability and security without interfering with the optimum performance.

2. Network Requirements Analysis

2.1. Departmental Requirements

The new branch has eight departments, which have unique host requirements and security needs.

Department	Host Count	Primary Function	Security Level
WA1	130	Workgroup Area 1	Standard
WA2	72	Workgroup Area 2	Standard
IT	60	Technology Infrastructure	Elevated
C&C	35	Command & Control	Standard
Finance	18	Financial Operations	High (Isolated)
HR	18	Human Resources	High (Isolated)
SOC	15	Security Operations Center	Elevated
Admin	10	Administrative Functions	Standard

Table 1: Department Distribution of Avalanche Negombo Branch

2.2. Security Requirements

The following security requirements have been determined according to best practice policies in the organization and the industry [4]

- 1. Network Segmentation:** The different departments need to work with disconnected broadcast domains (VLANs) to restrict the attack surface and restrict a possible security attack.
- 2. Access Control:** The Finance and HR departments need to be highly isolated ensuring that
 - No inter-departmental communication.
 - No contact with other departments.

- Only the main-branch and internet access.
- 3. Secure Communication:** VPN technology should be used to encrypt all the inter-branch communications with a view of ensuring confidentiality and integrity of data as it passes [5].
- 4. Authentication:** To control administrative access, administrative access must be authenticated centrally with the help of AAA servers.
- 5. Physical Security:** Access switches have physical security that is used to prevent unauthorized connections to devices.
- 6. Security:** SSH; all SSH access is restricted to security reasons [6].

2.3. Connectivity Requirements

- 1. Inter Branch Connectivity:** There needs to be a secure and reliable connection between the main branch (192.168.1.0/24) and the new branch (10.10.0.0/16).
- 2. Internet Access:** The internet access will be controlled by allowing all the departments to use the main-branch router.
- 3. Service Access:** The departments of the new branch should be able to access centralized services (AAA, DHCP, NTP, Syslog) that are hosted in the primary branch.
- 4. Inter-VLAN Routing:** Communication between departments should be provided in a controlled way with exceptions being the restriction applied to Finance and HR.
- 5. Redundancy:** There should also be backup connections and redundancy paths so that the availability is 99.9% [7].

2.4. Performance Requirements

According to organizational criteria and industry standards:

- **Latency:** Inter-branch communication with a latency of less than 50ms.
- **Availability:** 99.9% uptime, which is up to 8.76 hours of downtime per year.
- **Throughput:** 500Mbps to desktop connections with 1Gbps backbone.
- **Scalability:** 50% growth (2 more hosts) without significant redesign.

3. Network Design and Architecture

3.1. Topology Selection and Justification

3.1.1. Hierarchical Network Model

The network follows the hierarchical three layers design model of cisco with its hierarchical structure providing a modularity, scalability, and simplified troubleshooting [8].

This architecture is made up of:

1. Core Layer:

- High-speed packet switching and routing;
- branch router and main branch router provide WAN connectivity;
- inter-distribution layer connectivity;
- no security policies or packet manipulation to satisfy the best performance.

2. Distribution Layer:

- Main Branch: single multilayer switch (Cisco 3560) to offer core switching and connection to servers;
- New Branch: two multilayer switches (Cisco 3560) - Neg-Core-MLS01 and Neg-Core-MLS02 - to support inter-VLAN routing and redundancy;
- HSRP/VRRP set up between the two core switches to provide first-hop redundancy;
- Policy-based connectivity control
- Access layer switches include the access layer connect to the core switch.

3. Access Layer:

- End user connectivity through access switches (Cisco 2960);
- 8 departmental switches within her new branch (one switch per

department);

- VLAN tagging and port security;
- optional Quality of Service (QoS) tagging;
- Power over Ethernet (POE) option of expansion.

3.1.2. Improved Redundancy Design in Negombo Branch:

The new branch adopts dual-core architecture with two multilayer switches in active standby configuration.

Primary Core Switch (NegcoreMLS01):

- Routes all inter-VLAN routing traffic in the active mode
- HSRP priority 110 (active router)
- Connected to all eight access switches through trunk connections
- Connected with the second core switch through Gi0/22 -24 on redundancy basis.

Secondary Core Switch (Neg -Core -MLS02):

- Standby mode can be instantly failed over
- HSRP priority 90 (meaning the standby mode)
- Configuration and routing tables are synchronized
- Takes over the previous state within 3 seconds when the primary goes offline.

Interconnection of Main Switches

- Redundant connections between the two core switches (represented as dashed red lines)
- High-speed interconnection is done using EtherChannel/StackWise technology

- Spanning Tree Protocol (Rapid PVST+) is used to avoid loops but leave redundant paths.

Benefits of Dual-Core Design:

- 1. High Availability:** In case Neg-Core-MLS01 is unavailable, Neg-Core-MLS02 automatically goes online and minimally disruptive (less than 3 seconds of downtime) [7]
- 2. Load Balancing:** Specific VLAN can be configured to use a different switch as the primary gateway, and this is used to load balance traffic
- 3. Maintenance Flexibility:** one switch can be brought offline to allow maintenance, and the other switch continues, which is critical to business survival
- 4. Fault Tolerance:** a single point of failure at the distribution layer is removed
- 5. Scalability:** Additional switches can be added as network grows

This increased redundancy within the distribution layer to guarantee adherence to a 99.9% uptime specification (maximum 8.76 hours of downtime per year), so that the network is now business critical [82].

3.1.3. Design Rationale

The hierarchical model was chosen due to a number of strong reasons:

- 1. Scalability:** It is possible to add new departments or expand on the existing ones without affecting the basic infrastructure [9].
- 2. Performance:** Traffic localization at the access layer to the network minimizes unnecessary core traffic, which is a better way of enhancing network performance.
- 3. Redundancy:** There are several paths between layers which offer high availability and load balancing.

4. **Security:** There is good delimitation of points which makes it possible to enforce security policy at the distribution layer.
5. **Manageability:** Logical segmentation and distinct traffic paths result in simplified troubleshooting [10].
6. **Cost -Efficiency:** The location of equipment optimizes spending, such that expensive multilayer switches are deployed where needed.

3.1.4. Topology Type

Physical topology is a star topology where the access layer- devices are linked to central switches- whereas in the distribution and core layers, a partial-mesh topology is used to provide redundancy. This is a combination of cost and performance in this hybrid method coupled with reliability [11]. The benefits of the selected topology are the;

- Single points of failure are eliminated by redundancy.
- Easier trouble Shooting and fault isolation.
- Effective bandwidth usage.
- Assistance with the implementation of QoS.
- Future wireless integration accommodation.

3.2. VLSM Addressing Scheme Design

3.2.1. Address Space Allocation

The following address space has been allocated to the organization:

1. Main Branch: 192.168.1.0/24 (254 addresses in use).
2. New Branch: 10.10.0.0/16 (65,534 usable addresses).
3. WAN Link: 200.200.200.0/30 (2 usable addresses).

The 10.10.0.0/16 address space is spacious enough and efficient address utilization and VLSM are achieved [12].

3.2.2. VLSM Calculation Methodology

Variable Length Subnet Masking (VLSM) allows effective assignment of IP addresses by creating subnets of different sizes according to real needs, and in this way reducing the amount of wasted addresses [13].

Calculation Process:

1. Arrange sort departments in order of the number of hosts (most to the fewest).
2. Divide the number of required addresses by formula $2^n \geq (\text{hosts} + 2)$ where the two extra addresses compensate for network address and broadcast address.
3. Give subnet masks according to the requirements calculated.
4. Assign adjacent blocks of addresses so that route summarization can be done.

Department Sorting (Largest to Smallest):

1. WA1: 130 hosts $\rightarrow 2^8 = 256$ addresses $\rightarrow /24$ (254 usable)
2. WA2: 72 hosts $\rightarrow 2^7 = 128$ addresses $\rightarrow /25$ (126 usable)
3. IT: 60 hosts $\rightarrow 2^6 = 64$ addresses $\rightarrow /26$ (62 usable)
4. C&C: 35 hosts $\rightarrow 2^6 = 64$ addresses $\rightarrow /26$ (62 usable)
5. Finance: 18 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
6. HR: 18 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
7. SOC: 15 hosts $\rightarrow 2^5 = 32$ addresses $\rightarrow /27$ (30 usable)
8. Admin: 10 hosts $\rightarrow 2^4 = 16$ addresses $\rightarrow /28$ (14 usable)

3.2.3. Complete VLSM Table

Department	Hosts	Network Address	Subnet Mask	CI DR	Usable Range	Default Gateway	Broadcast	VLAN
WA1	130	10.10.0.0	255.255.255.0	/24	10.10.0.1 - 10.10.0.254	10.10.0.1	10.10.0.255	10
WA2	72	10.10.1.0	255.255.255.128	/25	10.10.1.1 - 10.10.1.126	10.10.1.1	10.10.1.127	20
IT	60	10.10.1.128	255.255.255.192	/26	10.10.1.129 - 10.10.1.190	10.10.1.129	10.10.1.191	30
C&C	35	10.10.1.192	255.255.255.192	/26	10.10.1.193 - 10.10.1.254	10.10.1.193	10.10.1.255	40
Finance	18	10.10.2.0	255.255.255.224	/27	10.10.2.1 - 10.10.2.30	10.10.2.1	10.10.2.31	50
HR	18	10.10.2.32	255.255.255.224	/27	10.10.2.33 - 10.10.2.62	10.10.2.33	10.10.2.63	60
SOC	15	10.10.2.64	255.255.255.224	/27	10.10.2.65 - 10.10.2.94	10.10.2.65	10.10.2.95	70
Admin	10	10.10.2.96	255.255.255.240	/28	10.10.2.97 - 10.10.2.110	10.10.2.97	10.10.2.111	80

Table 2: Complete VLSM Table for Avalanch Corporations

Additional Subnets:

Purpose	Network Address	Subnet Mask	CIDR	Usable Range	Usage
Management	10.10.3.0	255.255.255.240	/28	10.10.3.1 - 10.10.3.14	Switch management interfaces
WAN Link	200.200.200.0	255.255.255.252	/30	200.200.200.1 - 200.200.200.2	Router-to-router connection
Main Branch	192.168.1.0	255.255.255.0	/24	192.168.1.1 - 192.168.1.254	Main branch LAN

Table 3: Additional Subnets for the Avalanch Corporations

3.2.4. Address Space Efficiency Analysis

- Total Addresses Allocated: 638 addresses
- Total Addresses Used: 358 hosts
- Efficiency: 56.1%
- Growth Capacity: 280 additional hosts (78.2% expansion possible)

This efficiency ratio is acceptable as it provides significant growth capacity while maintaining hierarchical address structure for route summarization [14].

3.2.5. Route Summarization Opportunities

The nearby allocation of addresses allows effective summarization of routes:

- Summary Route for New Branch: 10.10.0.0/16
- Advertised to Main Branch: One route that will be used to represent all the eight departments.
- Advantage: Smaller routing table, faster convergence and easier routing [15].

3.3. Equipment Selection

3.3.1. Router Selection: Cisco 2911

Both branches have been selected as having Cisco 2911 Integrated Services Router on the following ground [16]:

- **VPN Capabilities:** Hardware-accelerated IPsec with encryption (AES-256) with a throughput of 100Mbps.
- **Port Density:** WAN and LAN Multiple Gigabit Ethernet interfaces.
- **Security Features:** VPN concentrator, IPS features, and firewall built-in.
- **Scalability:** Modular construction that can be expanded to large interface cards in the future.
- **Reliability:** Enterprise-grade hardware with MTBF of over 200,000 hours.
- **Cost-Effectiveness:** Best tradeoff between functionality and financial limitations.

Technical Specifications:

- Throughput: 300Mbps (no encryption), 100Mbps (IPsec encrypted)
- Interfaces: 3 Gigabit Ethernet (built-in) and modules extendable.
- Memory: 512 MB DRAM, 256MB Flash (default)
- Routing Protocols RIPv2, EIGRP, OSPF, BGP.
- Security: IPsec, IPS VPN, Firewall, IPS.

3.3.2. Multilayer Switch Selection: Cisco 3650-24PS

Core/distribution switch The Cisco Catalyst 3650 24PS was chosen as the core/distribution switch in the two branches [17]:

- **Layer3 Routing:** Hardware routing that supports inter-VLAN routing at wire rate.
- **Performance:** 32GB switching, sufficient to support departmental traffic.
- **VLAN Support:** 1,024 VLANs (which is much more than eight VLANs, as is needed)
- **Routing Protocols:** Complete support of OSPF and EIGRP as inter-branch routing.
- **Security Features:** Port security, 802.1X authentication, ACL support.
- **PoE Support:** Power over ethernet in the future to be used with wireless AP.
- **Stackability:** StackWise technology in case of future redundancy and expansion.

Technical Specifications:

- Switching Capacity: 32 Gbps
- Forwarding Rate: 23.8 Mpps
- Port Configuration: 24- 10/100+ 2x uplinks SFP.
- Routing Performance: 8.8Mpps (Layer 3)
- VLAN Support: 1,024 VLANs, 4,096 VLAN IDs
- ACL Capacity: 1,024 entries

The common router-on-a-stick networks use a router interface to route all inter-VLAN traffic through one router, thus forming a performance bottleneck [18]. Multilayer switch approach has the following benefits:

- **Better Performance:** Hardware vs. Software Routing.
- **Reduced Latency:** Traffic exchanged within a local area without the participation of the router.
- **Scalability:** It is less difficult to add VLANs without compromising performance.
- **Reduced Configuration:** Switch Virtual Interfaces (SVIs) less complex to do.
- **Economical:** The switching is done by one device that is also used as a router.

3.3.3. Access Switch Selection: Cisco 2960-24TT

Cisco Catalyst 296024TT has been chosen to be installed on access-layers [19]:

- **Cost-Effective:** A suitable feature configuration at the access layer without allowing extraneous cost.
- **Port Density:** 24 Fast Ethernet ports to meet the needs of departments.
- **VLAN Support:** Complete 802.1Q VLAN support of segmentation.
- **Security Features:** Port security, DHCP snooping, DAI (Dynamic ARP Inspection).
- **Reliability:** Established platform under 10 failure reasons.
- **Energy Efficiency:** Low power consumption which is appropriate to stay on constantly.

Technical Specifications:

- Switching Capacity: 16 Gbps
- Port Type: 24×10/100 + 2x 10/100/1000T uplinks
- VLAN Support: 255 active VLANs
- MAC Address Table: 8,192 entries

3.3.4. Server Specifications

1. AAA Server : Unified network device access control.

Software: Cisco ISE or RADIUS compatible.

Protocols: RADIUS, TACACS+

2. DHCP Server: Automation of IP address assignment to every VLAN.

Capacity: 1,000+ DHCP leases.

Capabilities Scope management, lease tracking, conflict detection.

3. NTP Server: Time synchronization on a network-wide basis.

Accuracy: Stratum 2 or better (within 50ms of UTC)

Significance: Crucial to the log correlation and security event analysis [20].

4. Syslog Server: Observation and evaluation of logs.

Capacity: 10000 messages/ sec. and above.

Storage: 500GB is required to store a 90D retention.

Functionality: Log filtering, alerting, reporting functionality.

3.3.5. Complete Equipment List

Main Branch:

Device	Model	Quantity	Purpose	IP Address
Router	Cisco 2911	1	Edge routing, VPN termination, internet gateway	192.168.1.1
Main MLS	Cisco 3650-24PS	1	Core switching, inter-VLAN routing	192.168.1.254
NTP Server	Generic Server	1	Network time synchronization	192.168.1.5
Syslog Server	Generic Server	1	Centralized log management	192.168.1.8
AAA Server	Generic Server	1	Authentication, Authorization, Accounting	192.168.1.7
Backup Server	Generic Server	1	TFTP backup storage for configurations	192.168.1.20
End Devices	Generic PC	2	Administrative workstations (static IP)	192.168.1.0/24

Table 4: Equipment List for Avalanch Main Branch

New Branch (Negombo)

Device	Model	Quantity	Purpose	Location/VLAN
Router	Cisco 2911	1	Branch routing, VPN termination	10.10.3.2
Main MLS (Primary)	Cisco 3650- 24PS	1	Inter-VLAN routing, HSRP Active, Root Bridge	Neg-MLS01
Main MLS (Secondary)	Cisco 3650- 24PS	1	Inter-VLAN routing, HSRP Standby	Neg-MLS02
Access Switch - WA1	Cisco 2960- 24TT	1	Department access layer	VLAN 10
Access Switch - WA2	Cisco 2960- 24TT	1	Department access layer	VLAN 20
Access Switch - IT	Cisco 2960- 24TT	1	Department access layer	VLAN 30
Access Switch - C&C	Cisco 2960- 24TT	1	Department access layer	VLAN 40
Access Switch - Finance	Cisco 2960- 24TT	1	Department access layer (ISOLATED)	VLAN 50
Access Switch - HR	Cisco 2960- 24TT	1	Department access layer (ISOLATED)	VLAN 60
Access Switch - SOC	Cisco 2960- 24TT	1	Department access layer	VLAN 70
Access Switch - Admin	Cisco 2960- 24TT	1	Department access layer	VLAN 80
DHCP Server	Generic Server	1	Local DHCP services for Negombo branch	VLAN 30 (IT)
End Devices	Generic PC	358	Departmental workstations	Distributed across 8 VLANs

Table 5: Equipment List for Avalanch Negombo Branch

3.4. Network Diagram

The comprehensive network diagram illustrates a hierarchical three-layer architecture connecting the Main Branch and Negombo Branch via an encrypted IPSec VPN tunnel.

Key elements include:

1. Main Branch (Left Section):

- Edge router (Main-Router) that is linked to the internet/WAN cloud.
- Main Core switching (Main-Sw1) offers switching services.
- Three central servers (AAA,NTP, Syslog)
- Additional Backup server with TFTP feature on
- Testing and administrative end-user workstations.

2. WAN Connection (Top Section):

- Internet cloud which represents the public network.
- IPSec VPN tunnel (encrypted tunnel) between the two routers.
- Serial WAN connections.
- GRE Tunnel (172.16.1.0/30) for additional VPN redundancy
- Addressing: 200.200.200.0/30

3. New Branch- Negombo (Right Section):

- VPN termination Edge router (Negombo-Router)
- Dual core Architecture in Multi-Layer Switches
 - Primary Core Switch - Neg-MLS01:
 - Active HSRP router for VLANs 10, 20, 30, 40
 - Root Bridge for Spanning Tree Protocol
 - Primary gateway for departmental traffic
 - Connected to all eight access switches via trunk links

- Secondary Core Switch - Neg-MLS02:
 - Standby HSRP router for VLANs 50, 60, 70, 80
 - Backup spanning tree bridge
 - Automatic failover capability (sub-3 second convergence)
 - Redundant connections to all access switches
- Eight access switches in two row:
 - Top Row:- WA1-SW VLAN 10 WA2-SW VLAN 20 IT-SW VLAN 30 C&C-SW VLAN 40
 - Bottom Row Finance-SW (VLAN 50), HR-SW (VLAN 60), SOC-SW (VLAN 70), Admin-SW (VLAN 80)
- Departmentally dispersed end devices.

Connection Types:

- Trunk Links: Connects between access switches and multilayer switches
- Access Links: Switches to end devices (single VLAN)
- Redundant Links : Multi layer switches interaction with HSRP + LACP +STP
- WAN Links: Green dashed line implied VPN tunnel encryption.

3.4.1. Network Segmentation and Security Zones:

Zone 1 - Main Branch

- Corporate services and internet gateway
- Centralized management servers
- Trusted network segment

Zone 2 - WAN/Internet

- Public network infrastructure
- Encrypted VPN tunnel overlay
- Untrusted network segment

Zone 3 - Negombo Branch

- Eight departmental VLANs with logical segmentation
- Finance and HR departments isolated from other departments
- Local DHCP services in IT department

3.4.2. IP Addressing Summary

Network Segment	Network Address	Purpose
Main Branch LAN	192.168.1.0/24	Main branch devices and servers
WAN Link	200.200.200.0/30	Router-to-router connection
Backup GRE Tunnel	172.16.1.0/30	Secondary WAN connection
Negombo Branch	10.10.0.0/16	All departmental networks
Management VLAN	10.10.3.0/28	Switch management interfaces

*Table 6: IP addressing Summary Table***Design Notes:**

- Every trunk port that is encapsulated 802.1Q.
- Access ports dedicated to certain VLANs (None of VLANs 1)
- High availability uplinks that are redundant.
- VLAN switch administration (VLAN 99).
- Finance and HR switches were pointed out to denote ACL protection.

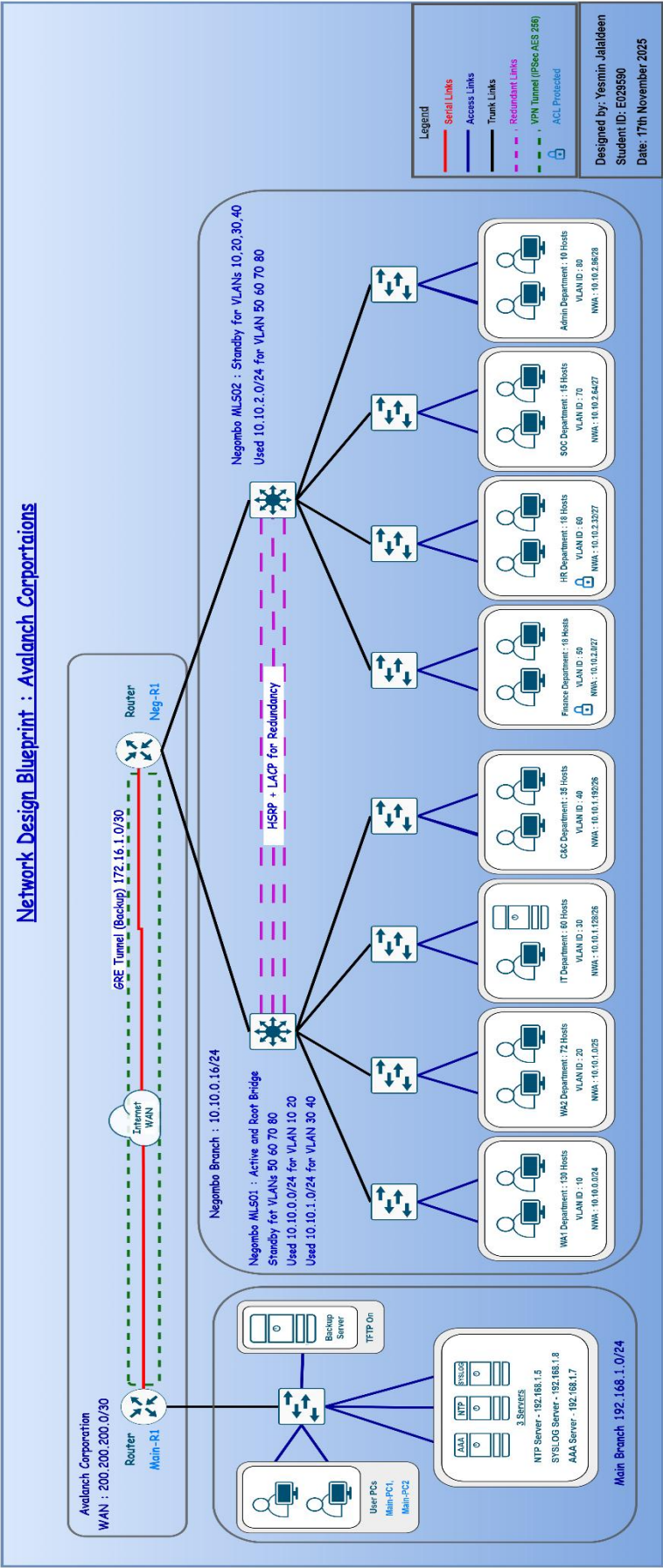


Figure 1: Network Design Blueprint for Avalanch Corporation

4. Network Design Implementation

The network topology has been designed and visualized with the draw.io tool in order to render correct and industry-standard documentation of a hierarchical design, which has given a clear logical picture. Cisco Packet Tracer Version 9 was used in practical implementation, configuration and testing of the network infrastructure. Every protocol, such as EIGRP routing, HSRP redundancy, and IPsec VPN tunnels, could be tested within this simulation environment before deploying in the real world, and, therefore, all the configurations were verified to comply with the high security and performance standards of Avalanch Cooperations.

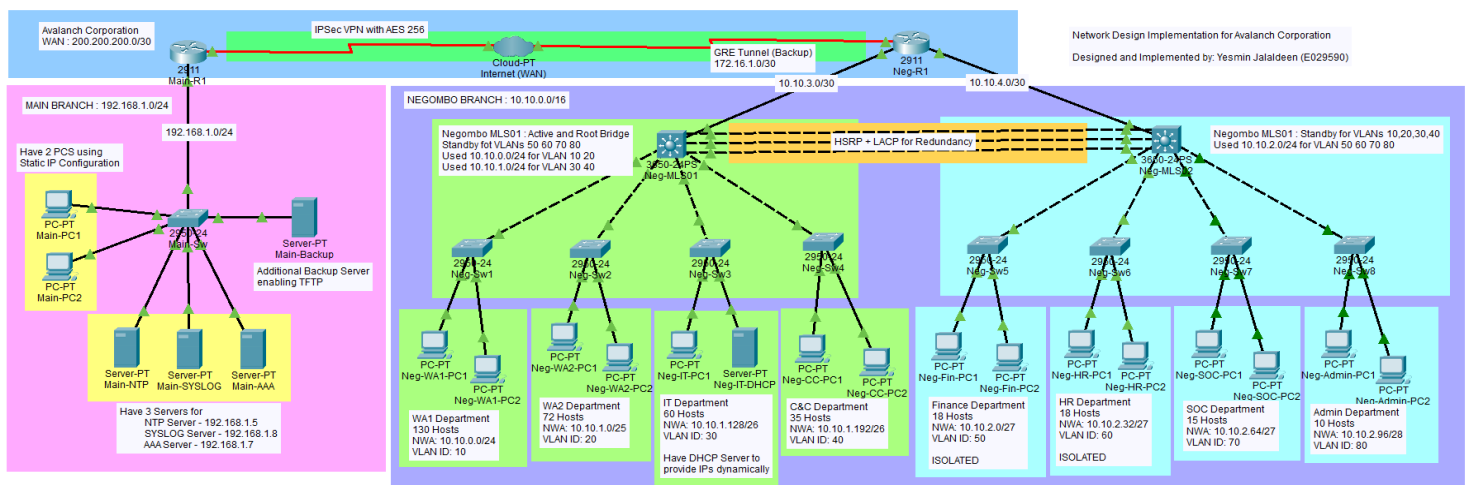


Figure 2: Labelled Network Design in PKT

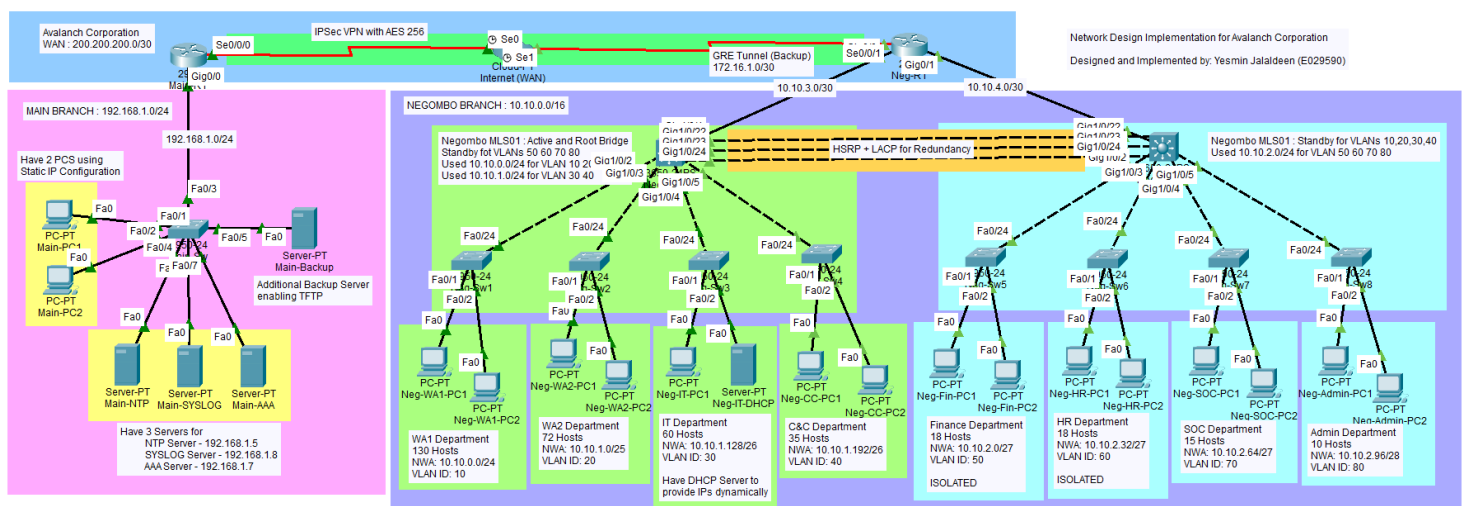


Figure 3: Labelled Network Diagram in PKT with Visible Interfaces

4.1. VLAN Design Strategy

VLANs allow logical physical networks segmentation, which has many advantages that are vital to organizational security and performance [21]:

- **Security Isolation:** There is separation of broadcast domains to reduce the threat of unauthorized cross-departmental access.
- **Traffic Management:** The decrease in broadcast traffic is associated with the enhanced performance of the whole network.
- **Eased Administration:** Logical grouping is not tied to physical location and as such, administration is made easier.
- **Policy Enforcement:** It is possible to apply various security policies on a departmental basis.
- **Scalability:** New users can be added without physical reconfiguration being required.

4.1.1. VLAN Assignment Strategy

Department VLANs (Data Plane)

VLAN ID	VLAN Name	Network	Department	Security Level
10	WA1-VLAN	10.10.0.0/24	WA1	Standard
20	WA2-VLAN	10.10.1.0/25	WA2	Standard
30	IT-VLAN	10.10.1.128/26	IT	Elevated
40	CC-VLAN	10.10.1.192/26	C&C	Standard
50	FINANCE-VLAN	10.10.2.0/27	Finance	High (Isolated)
60	HR-VLAN	10.10.2.32/27	HR	High (Isolated)
70	SOC-VLAN	10.10.2.64/27	SOC	Elevated
80	ADMIN-VLAN	10.10.2.96/28	Admin	Standard

Table 7: Department VLAN Breakdown of Avalanch Corporation

Infrastructure VLANs (Management/Control Plane):

VLAN ID	VLAN Name	Network	Purpose
99	MANAGEMENT	10.10.3.0/28	Switch management interfaces
100	NATIVE	-	802.1Q native VLAN (tagged)

Table 8: Infrastructure VLANs for Avalanche Corporation

Design Rationale:

- **VLAN 1 Avoided:** VLAN 1 was not used in order to reduce the possibility of VLAN hopping attacks [22].
- **Separate Management VLAN:** The traffic of the management was separated to user traffic and hence increased security.
- **Native VLAN Tagged:** Trunk traffic was set to be tagged to curb VLAN hopping.
- **Use of logical numbering:** VLAN identifiers were assigned as multiples of ten to help in making it easier to understand and easy to expand in future.
- **High-Security VLANs:** The Finance (50) and the HR (60) were numbered in order to enhance the ACL performance.

Additionally, Switch Management interfaces are configured with IP addresses for remote administration via SSH, following security best practices.

The screenshot displays a Cisco IOS CLI session. At the top, a window title bar reads 'Multilayer Switch0'. Below it, a tabbed interface shows 'Physical', 'Config', and 'CLI' (which is selected and highlighted in blue). The 'CLI' tab contains the text 'Attributes'. The main area of the window is titled 'IOS Command Line Interface' and shows a series of commands and their outputs. The commands are entered in a monospaced font, and the outputs are displayed in a standard font. The session shows the configuration of a VLAN named 'Avalanch' on a switch named 'Neg-MLS01'. The commands include setting the VTP domain name, enabling VTP, setting the VTP mode to 'SERVER', and configuring the VTP password. The session ends with a 'copy' command to save the configuration to the startup configuration file.

```

Neg-MLS01(config)#
Neg-MLS01(config)#vtp s Avalanch
Changing VTP domain name from NULL to Avalanch
Neg-MLS01(config)#vtp domain VTP_DOMAIN_NAME_CHG: VTP domain name changed to
Avalanch.

Neg-MLS01(config)#vtp p Avalanch
Setting device VLAN database password to Avalanch
Neg-MLS01(config)#vtp m s
Device mode already VTP SERVER.
Neg-MLS01(config)#vtp v 2
Neg-MLS01(config)#vtp m t
Setting device to VTP TRANSPARENT mode.
Neg-MLS01(config)#ex
Neg-MLS01#
%SYS-5-CONFIG-I: Configured from console by console

Neg-MLS01#
Neg-MLS01#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Neg-MLS01(config)#vlan 10
Neg-MLS01(config-vlan)#name WA1_Dep:
Neg-MLS01(config-vlan)#ex
Neg-MLS01(config)#vlan 20
Neg-MLS01(config-vlan)#name WA2_Dep:
Neg-MLS01(config-vlan)#ex
Neg-MLS01(config)#vlan 30
Neg-MLS01(config-vlan)#name IT_Dep:
Neg-MLS01(config-vlan)#ex
Neg-MLS01(config)#vlan 40
Neg-MLS01(config-vlan)#name Ctc_Dep:
Neg-MLS01(config-vlan)#ex
Neg-MLS01(config)#
Neg-MLS01(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Neg-MLS01(config)#
Neg-MLS01(config)#ex
Neg-MLS01#

```

At the bottom of the window, there is a status bar showing 'Top' and a taskbar with various icons, including a clock showing '2 cm of rain' and 'Saturday'.

The screenshot displays a Windows desktop environment. In the foreground, a Cisco IOS Command Line Interface (CLI) window titled "Neg-MLS02" is open. The window has a title bar with standard Windows controls and a menu bar with "Physical", "Config", "CLI" (highlighted), and "Attributes". The main area shows the CLI prompt "Neg-MLS02(config)#" and the following commands and outputs:

```

Neg-MLS02(config)#ntp d Avalanch
Changing VTP domain name from NULL to Avalanche
Neg-MLS02(config)#SVL_VLAN-6-VTP_DOMAIN_NAME_CFG: VTP domain name changed to
Avalanch.

Neg-MLS02(config)#ntp s Avalanch
Setting device VLAN database password to Avalanche
Neg-MLS02(config)#vtp m s
Device mode already VTP SERVER.
Neg-MLS02(config)#vtp v 2
Neg-MLS02(config)#
Neg-MLS02(config)#
Neg-MLS02(config)#vlan 50
Neg-MLS02(config-vlan)#name Finance
Neg-MLS02(config-vlan)#ex
Neg-MLS02(config-vlan)#vlan 60
Neg-MLS02(config-vlan)#name HR
Neg-MLS02(config-vlan)#ex
Neg-MLS02(config-vlan)#vlan 70
Neg-MLS02(config-vlan)#name SOC
Neg-MLS02(config-vlan)#ex
Neg-MLS02(config-vlan)#vlan 80
Neg-MLS02(config-vlan)#name Admin
Neg-MLS02(config-vlan)#ex
Neg-MLS02(config)#
Neg-MLS02(config)#
Neg-MLS02(config)#int r gig1/0/2-5
Neg-MLS02(config-if-range)#sw m tr
Neg-MLS02(config-if-range)#ex
Neg-MLS02(config)#
Neg-MLS02(config)#int gig 1/0/1
Neg-MLS02(config-if)#no sw
Neg-MLS02(config-if)#
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state
to down
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state
to up

```

At the bottom of the CLI window, there is a status bar showing the IP address "10.10.10.1" and the MAC address "00:00:00:00:00:00". Below the CLI window, a Windows taskbar is visible with the Start button, a search bar, and several application icons including File Explorer, Microsoft Edge, and a game icon.

Physical Config CLI Attributes

IOS Command Line Interface

```

40 C&c                active
50 Finance            active
60 HR                active
70 SOC               active
80 Admin             active
1002 fddi-default    active
1003 token-ring-default active
1004 fddinet-default active
1005 trnet-default   active
Neg-MLS01#
Neg-MLS01#
Neg-MLS01#sh vl ?
  brief VTP all VLAN status in brief
  id    VTP VLAN status by VLAN id
  name  VTP VLAN status by VLAN name
  <cr>
Neg-MLS01#sh vl

VLAN Name                Status Ports
---
1  default                active Po1, Gg1/0/6, Gg1/0/7, Gg1/0/8
                        Gg1/0/9, Gg1/0/10, Gg1/0/11, Gg1/0/12
                        Gg1/0/13, Gg1/0/14, Gg1/0/15, Gg1/0/16
                        Gg1/0/17, Gg1/0/18, Gg1/0/19, Gg1/0/20
                        Gg1/0/21, Gg1/1/1, Gg1/1/2, Gg1/1/3
                        Gg1/1/4

10 WA1                  active
20 WA2                  active
30 IT                   active
40 C&c                 active
50 Finance              active
60 HR                   active
70 SOC                 active
80 Admin               active
1002 fddi-default       active
1003 token-ring-default active
1004 fddinet-default    active
1005 trnet-default      active

--More-- |

```

Copy Paste

Config CLI Attributes

VLAN Configuration

VLAN Number

VLAN Name

Add Remove

VLAN No	VLAN Name
1	default
10	WA1
20	WA2
30	IT
40	C&c
50	Finance
60	HR
70	SOC
80	Admin

IOS Commands

```

Authorized Network Personnel Only
Monitoring and Logging Are Enabled
=====

Less Verification

:
:2>enable
:
:
:2>configure terminal
:configuration commands, one per line. End with CNTL/Z.
:2(config)#
:2(config)#

```

27°
4:42 PM

Windows Taskbar Icons
System Tray Icons

Page 36 of 107

4.3. VTP Configuration

VLAN Trunking Protocol (VTP) is a proprietary protocol in Cisco which is used to forward VLAN configuration between switches and thus saves administrative overhead [23]. Nevertheless, VTP may pass on the misconfigurations, and this may cause network outages to spread out.

4.3.1. VTP Mode Selection

To do this implementation, the following reasons were chosen with **VTP Client/Server Mode**:

- **Safety:** The switch is not involved in VTP advertisements and therefore, there is no possibility of deleting VLANs accidentally.
- **Control:** VLAN database changes are operational either within a single switch, and therefore, it has more control.
- **Flexibility:** Custom VLAN can also be used with specific switches without interfering with others.
- **Security:** This means that the processing of VTP advertisements is not done, which decreases the attack surface [24].
- **Reliability:** The threat of conflict of VTP domain name or revision number is removed.

4.3.2. VTP Configuration Parameters

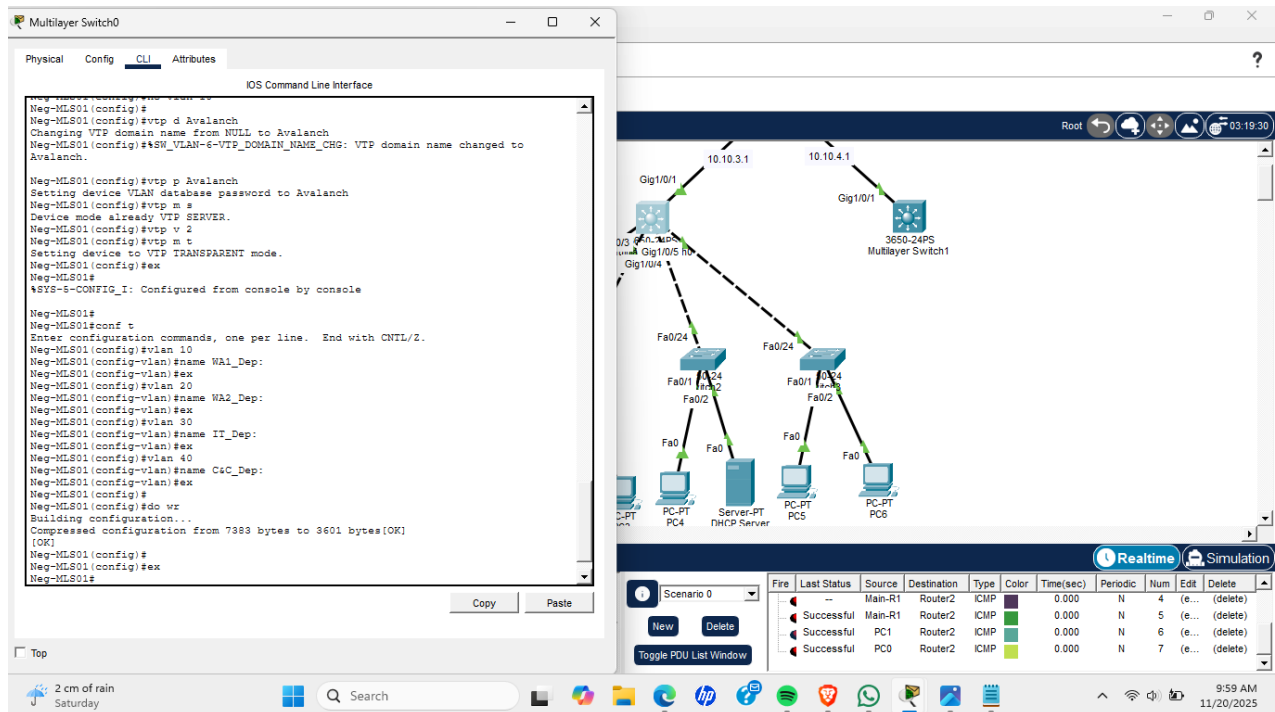


Figure 7: VTP Server and VLAN Database in Neg-MLS01

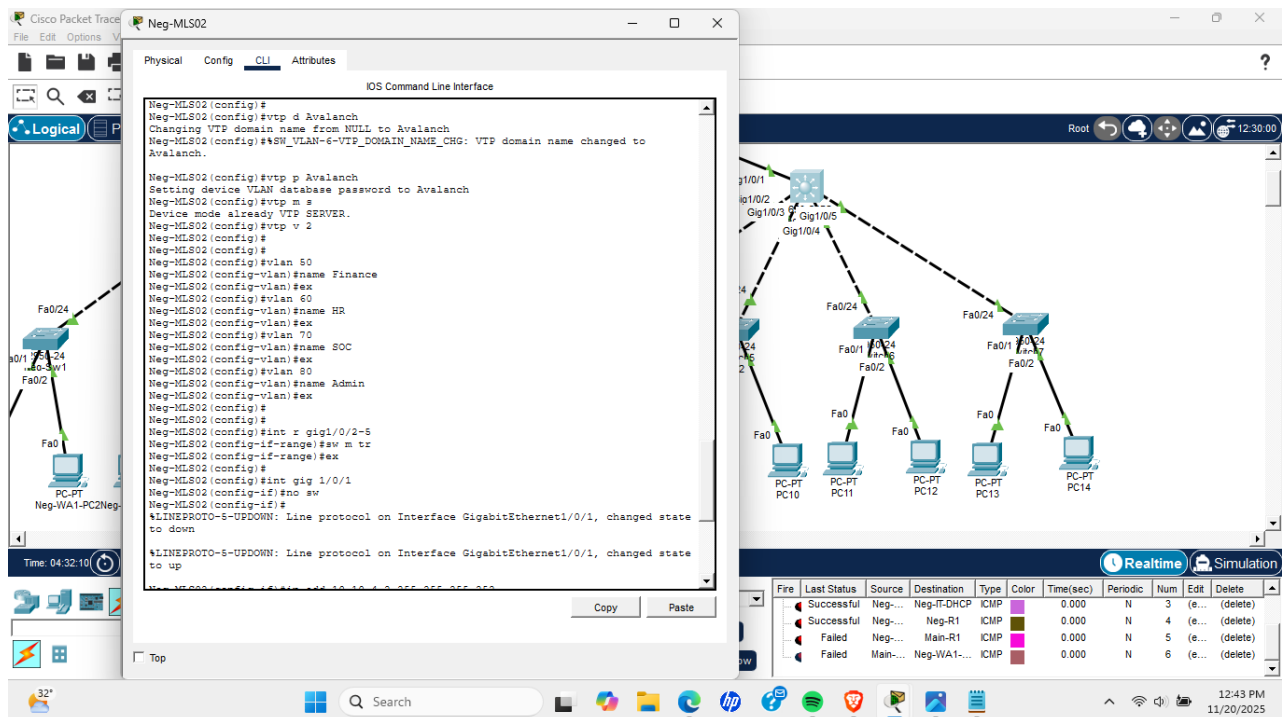


Figure 8: VTP Server and VLAN Databases in Neg-MLS02

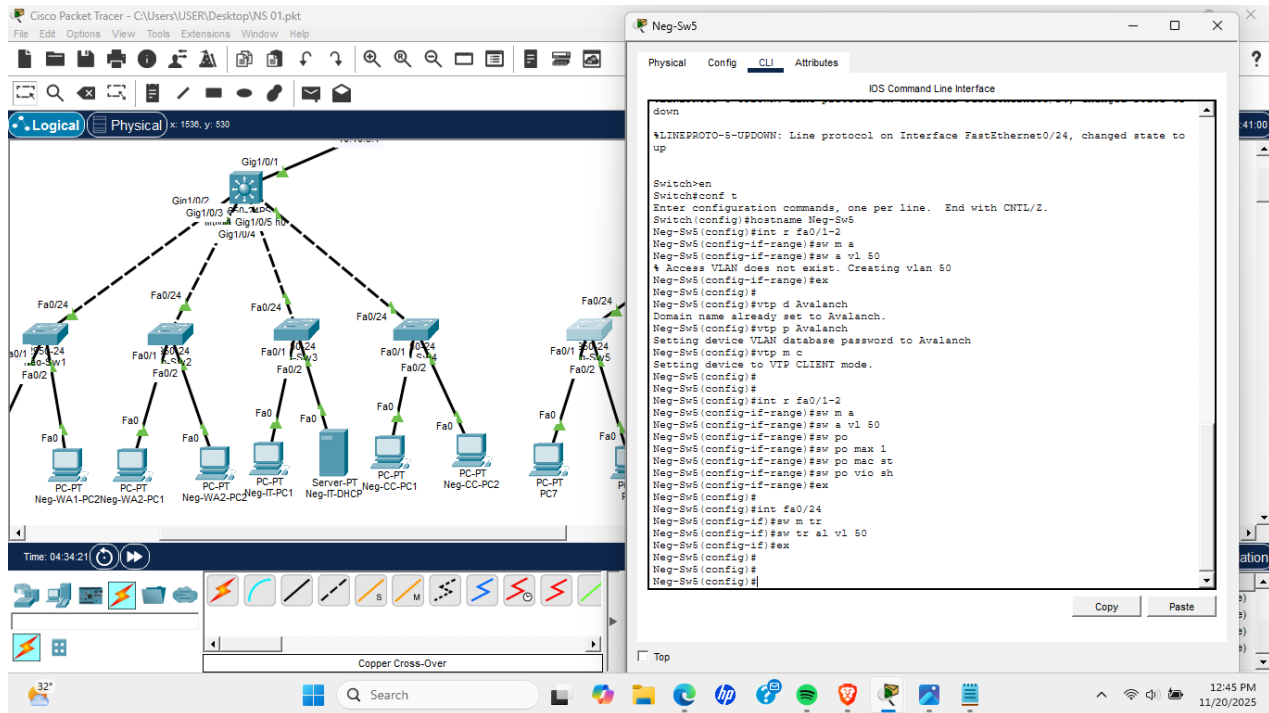


Figure 10: VTP Client in Negombo Branch

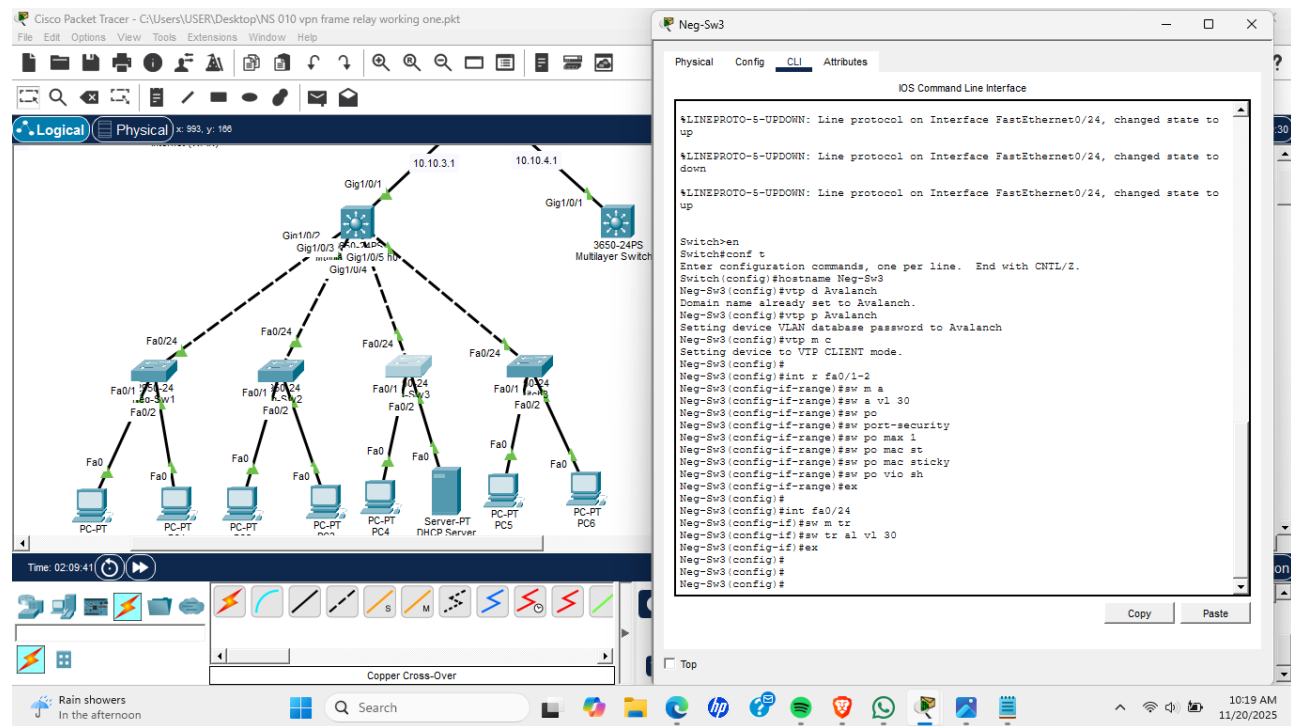


Figure 9: VTP Client in Negombo Branch

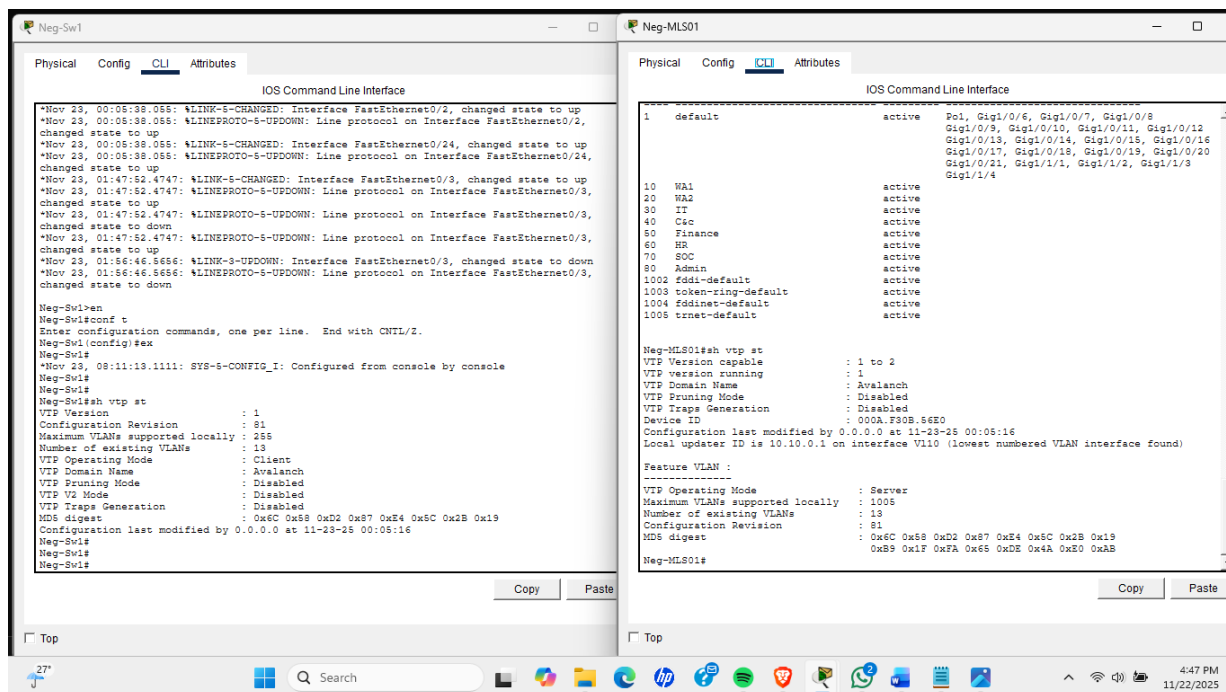


Figure 11: How VTP displays

4.4. Trunk and Access Port Configuration

4.4.1. Trunk Port Configuration

The trunk ports carry traffic across switches that are part of multiple VLANs over 802.1Q encapsulation [26]. In below, the trunk ports are allowed and assigned to relevant VLAN

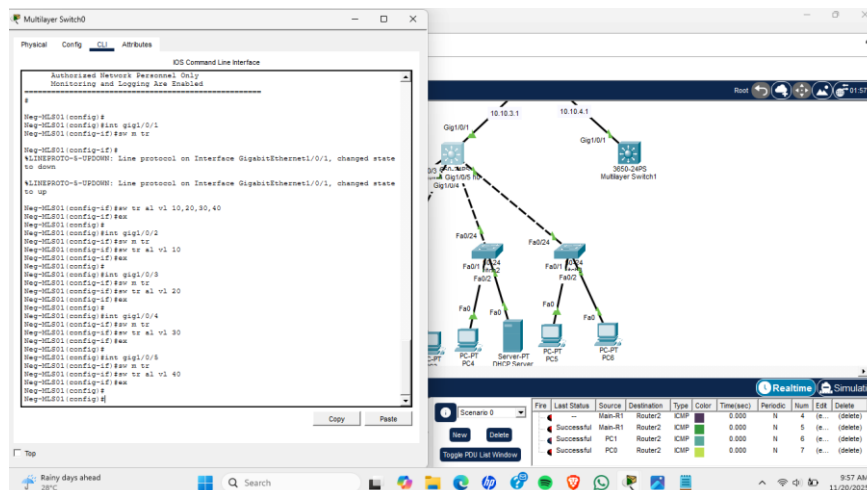


Figure 12: Trunk Ports in Neg-MLS01

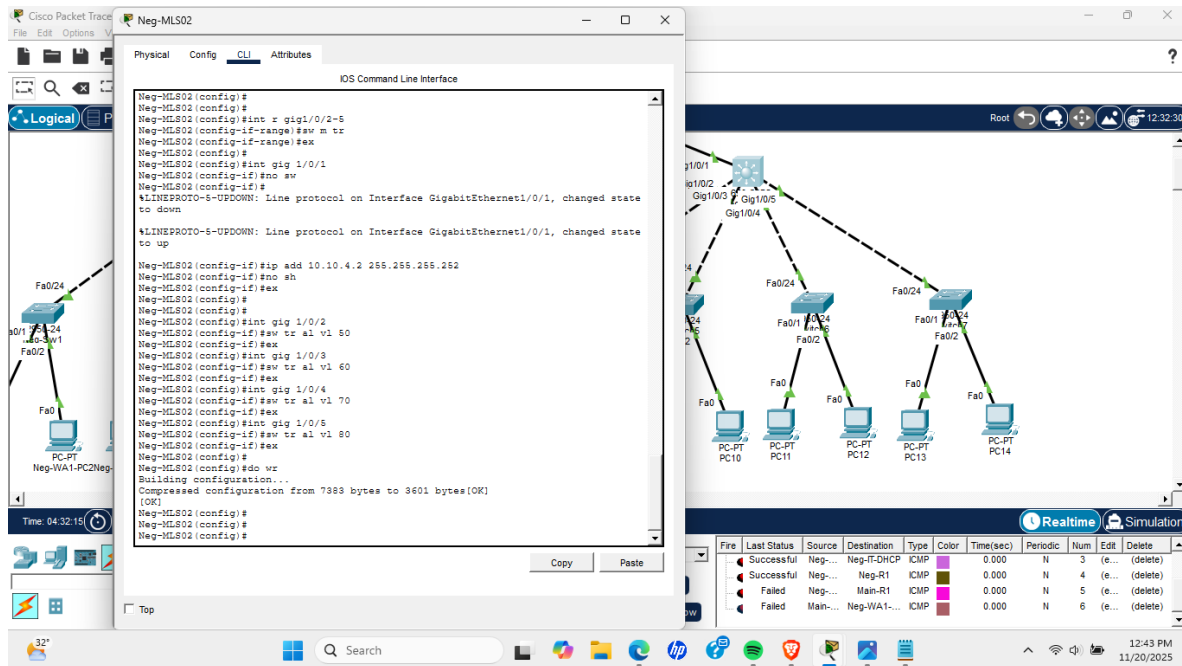


Figure 13: Trunk Port in Neg-MLS02

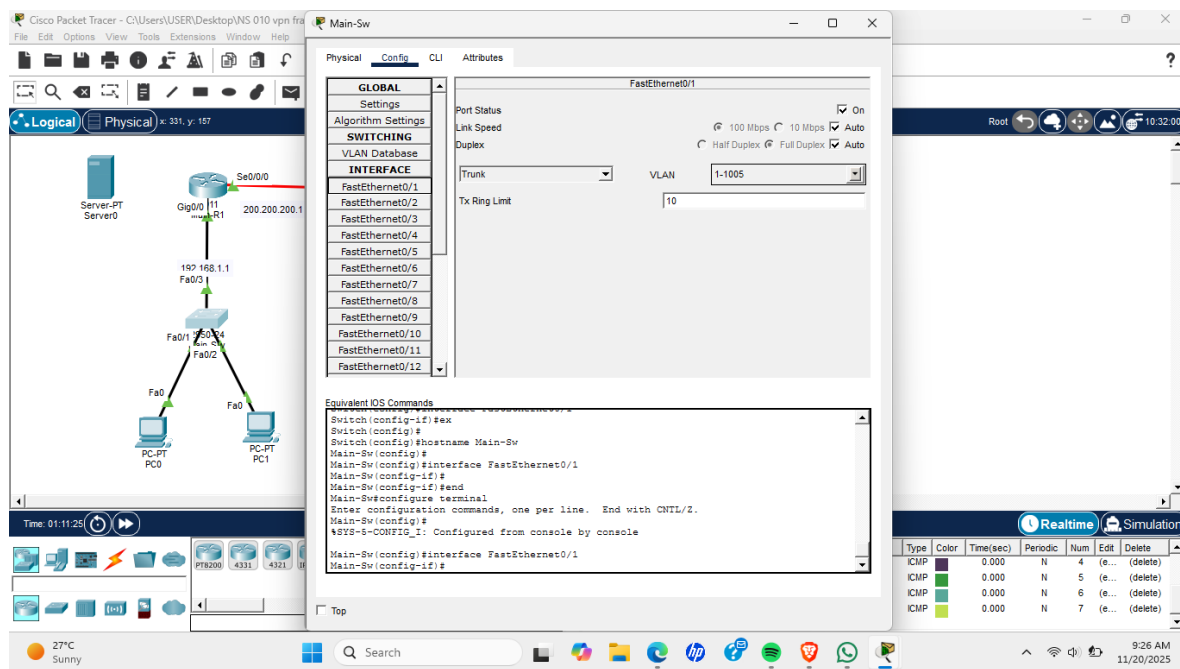


Figure 14: Trunk Ports in Main Branch

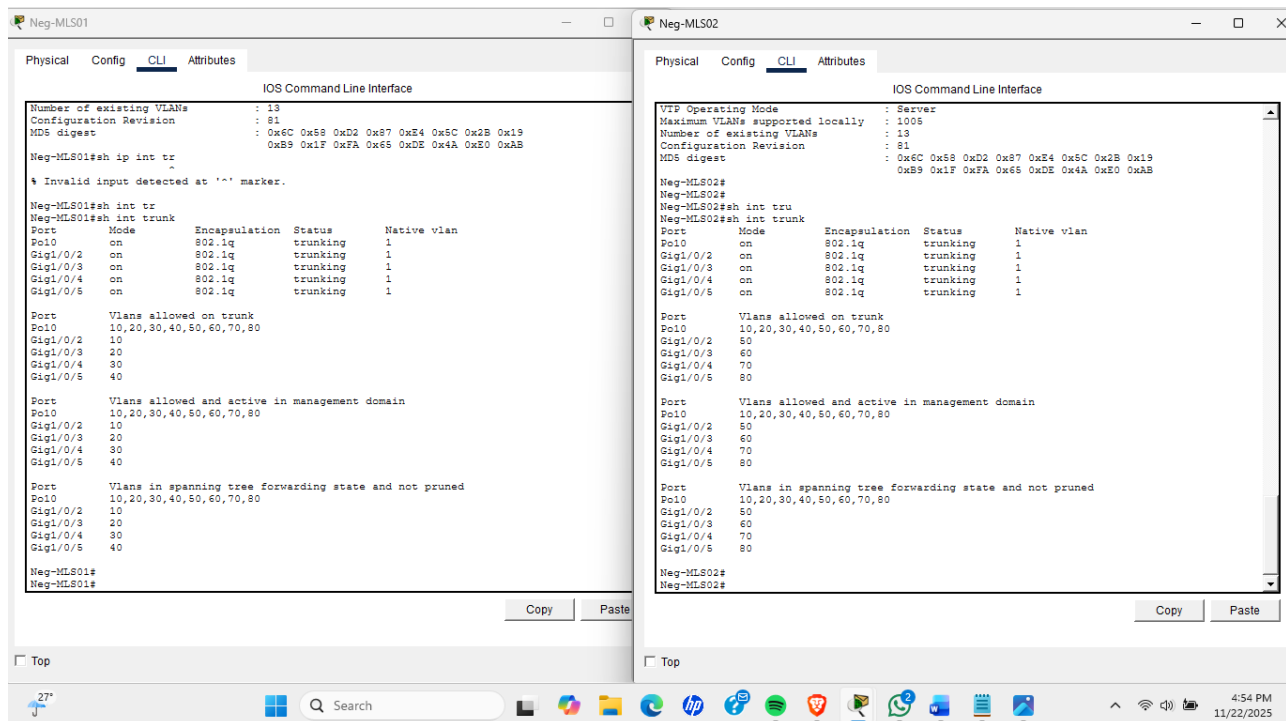


Figure 15: How Trunk Ports assigned

Security Hardening

- **Native VLAN Changed:** VLAN default (VLAN 1) is not the native VLAN to reduce VLAN-hopping attacks.
- **Native VLAN Tagged:** The `vlan dot1q tag native` command makes sure that even native VLAN is tagged.
- **Limit VLANs Limited:** VLANs Only the VLANs necessary to each service are authorized on trunk links, hence observing the principle of least privilege.
- **DTP Disabled:** Dynamic Trunking Protocol is switched off using `switchport non-negotiate` to avoid negotiation attack [27].
- **Unused Ports Disabled:** Trunk Ports that are not used are administratively disabled.

4.4.2. Access Port Configuration

Access ports are used to connect the end devices, and they are dedicated to a single VLAN [28].

4.4.3. Port Security Configuration

This port can be configured to either allow or deny entry to the host based on a predetermined set of rules

Port security limits the number of MAC addresses accepted on a port to stop unauthorized connections to the device [29].

In configuration I have done both at the same time for easiness

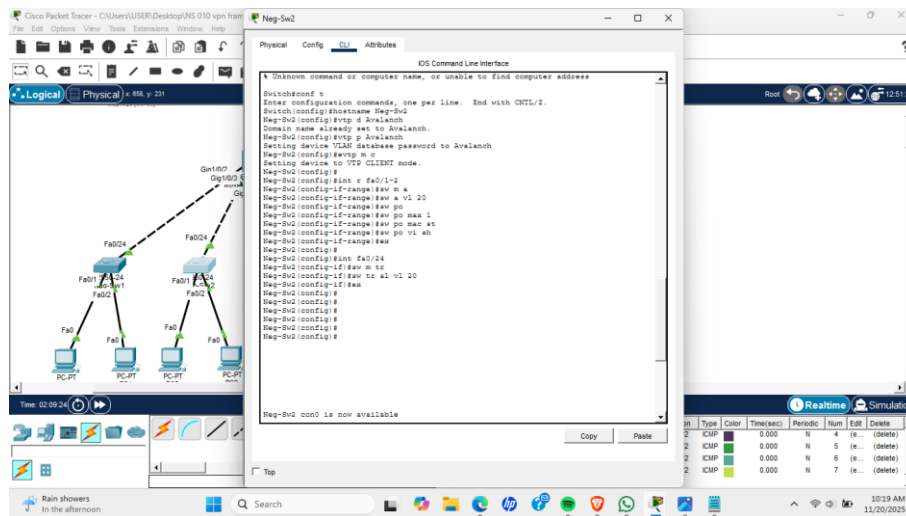


Figure 16: Access ports and Port Security in Neg-Sw2

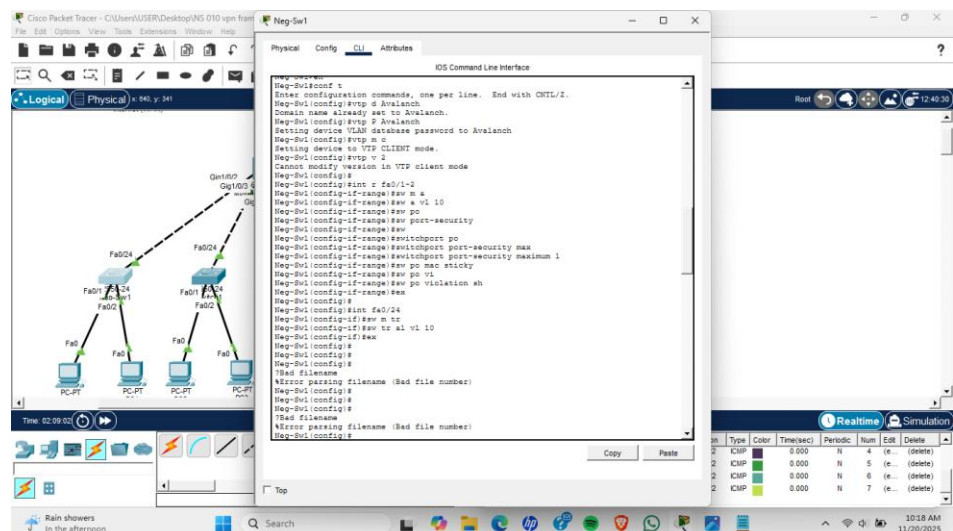


Figure 17: Access ports and Port Security In Neg-Sw1



- **Prevent MAC Flooding Attacks:** The limiting of the number of MAC addresses assigned to a port suppresses flooding.
- **Unauthorized Device Prevention:** Devices can be only connected to pre-registered devices.
- **Compliance and Audit Trail:** Port Security meets security policy standards by use of syslog; messages are caused by violations.

5. Network Devices Configuration

5.1. Router Configuration and Interconnections

5.1.1. Routing Protocol Selection

There is a need of a dynamic routing protocol to exchange route information and give automatic failover features to connect the main branch and the new branch.

EIGRP was chosen as the routing protocol because of the following reasons:

- **Fast Convergence:** Sub-second convergence in networks of small size, which is essential to sustain the VPN tunnels stability [77].
- **Low Overhead:** It does not need periodic updates as periodic updates done by RIP, but just when topology is changed.
- **Automatic Route Summarization:** VLSM and manual summarization assist in efficient routing.
- **Cisco Optimized:** Excellent performance on Cisco machines with little CPU usage.
- **Simplicity:** The OSPF configuration is easier than in two-site topology.
- **Unequal Cost Load Balancing:** Is capable of using many paths with varied metrics (extension in the future).

5.1.2. Router Configurations – Main

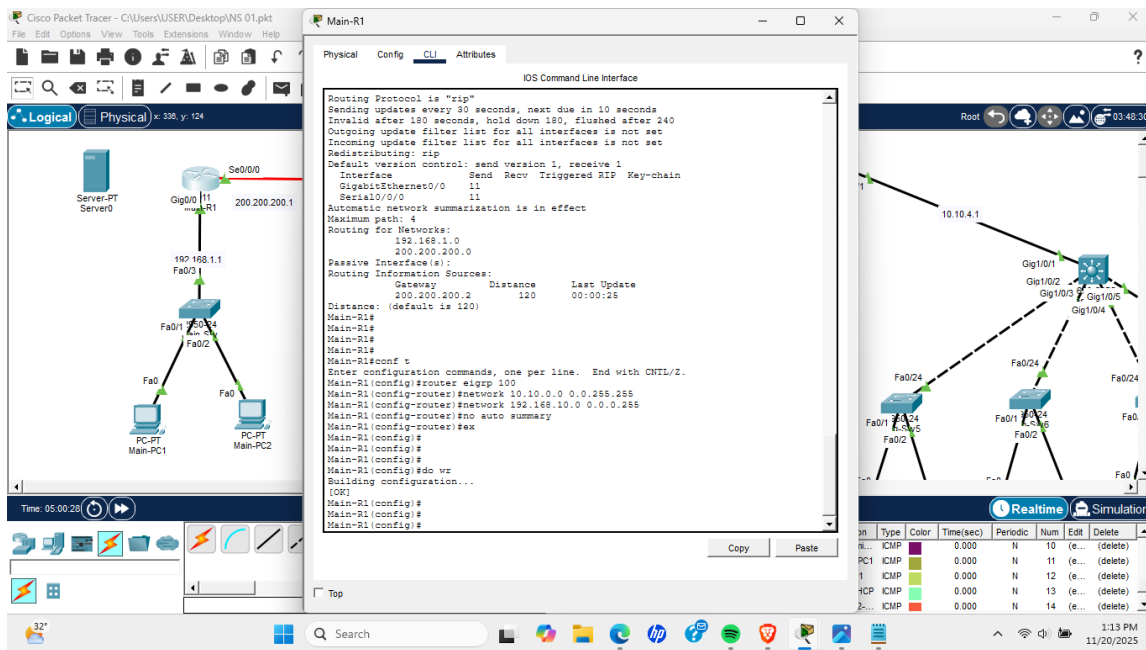


Figure 20: EIGRP in Main-RL

Configuration Explanation:

- AS Number 100: The number of an Autonomous System should be the same across all EIGRP routers.
- Network Statements: Advertise LAN (192.168.1.0/24) and WAN link (200.200.200.0/30)
- No Auto-Summary: Turns off classful summarization, which is needed by VLSM [78].
- Passive Interface Default: Disables default EIGRP sending hellos on all interfaces (security).
- No Passive on Gi0/1: this is used to allow the EIGRP neighbor formation on the WAN interface only.
- Router ID: Unique identifier of EIGRP process (follows a similar format like OSPF)



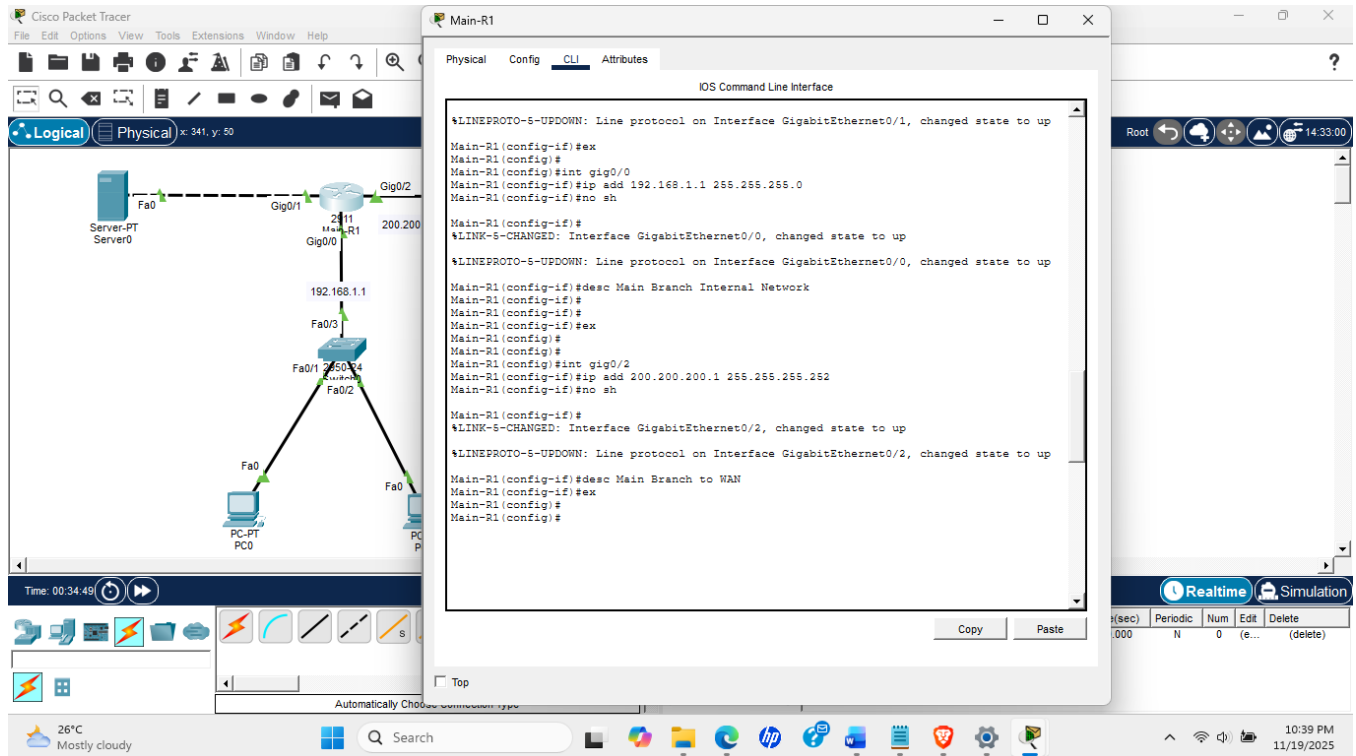


Figure 23: IP Route in Main-R1

5.1.3. Router Configurations -Negombo Branch

Same as the main branch the configurations are done

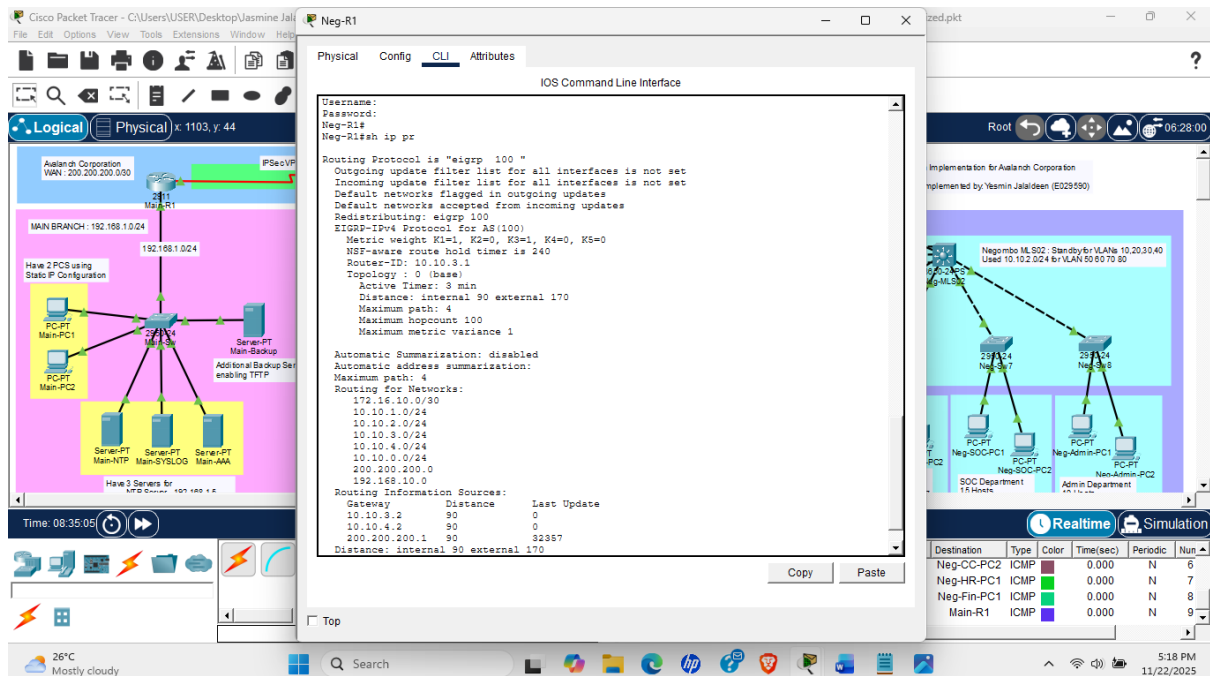


Figure 24: EIGRP Result in Neg-R1

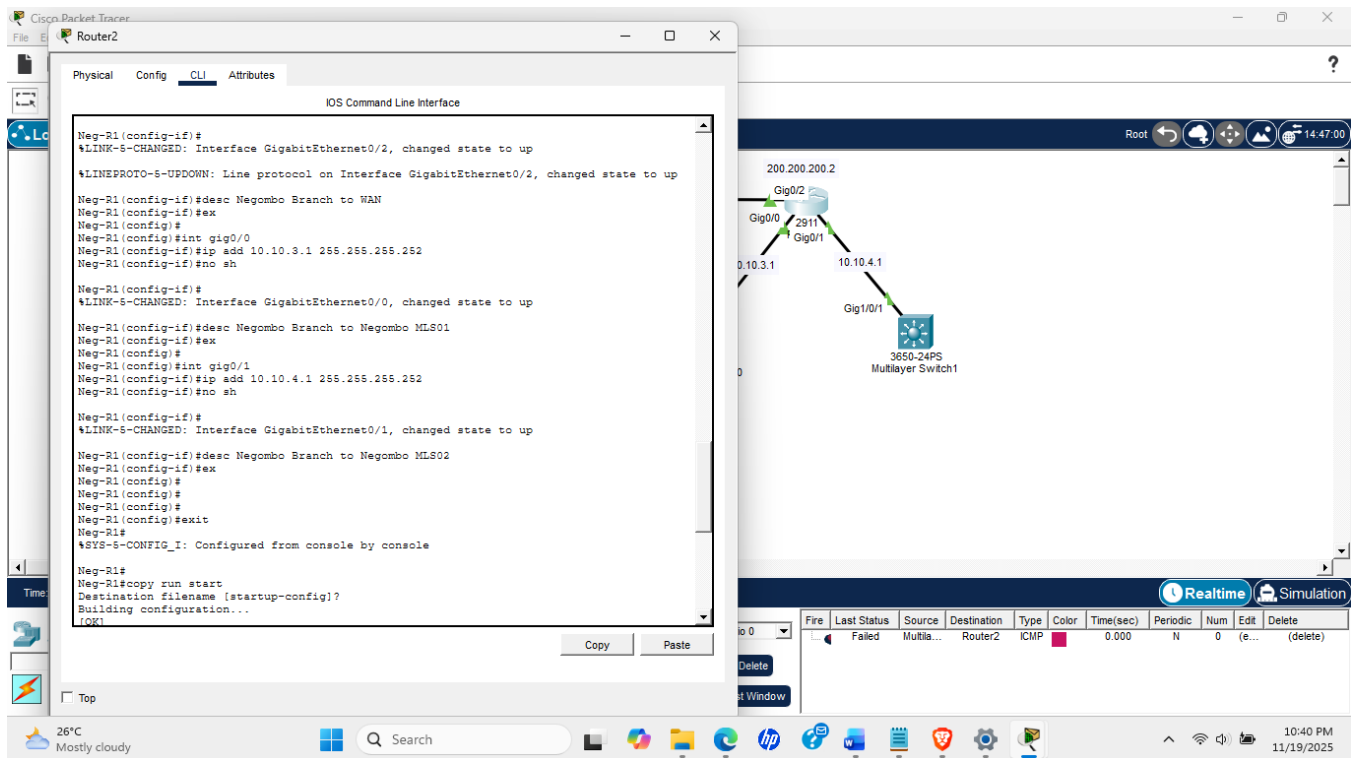
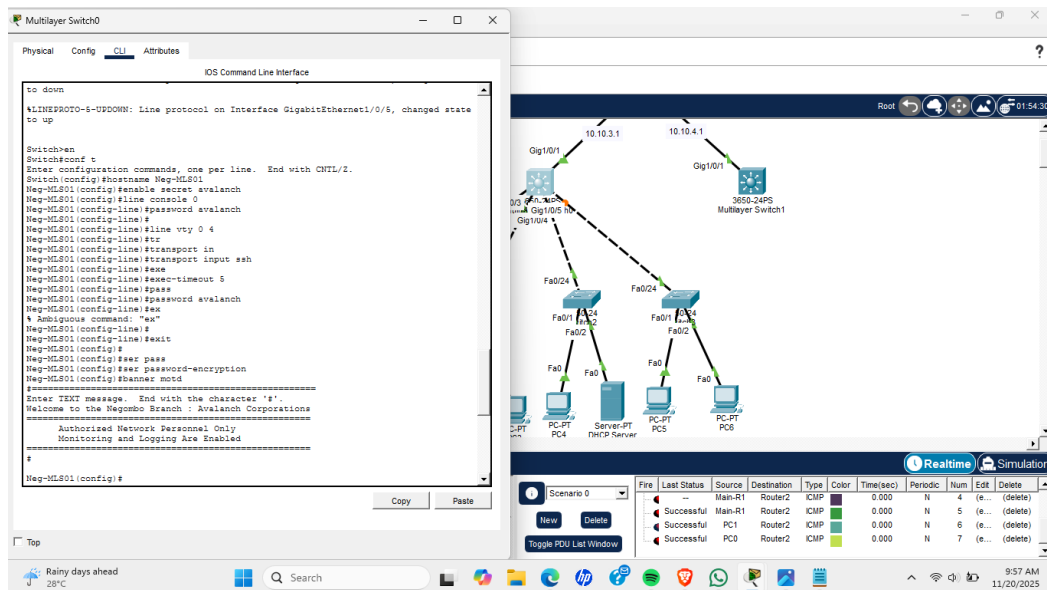


Figure 25: IP Route in Neg-R1

5.2. Multi-Layer Switch Configurations

5.2.1. Negombo Branch (Neg-MLS01 and Neg-MLS02)





STP is a network protocol (IEEE 802.1D) available in Layer 2, which can be used to eliminate loops in a network topology by blocking redundant paths when there are two or more paths between two points. It guarantees a path loop free up to the root bridge and the redundant links remain in an inactive state until an automatic activation occurs in the event of a primary path failure.



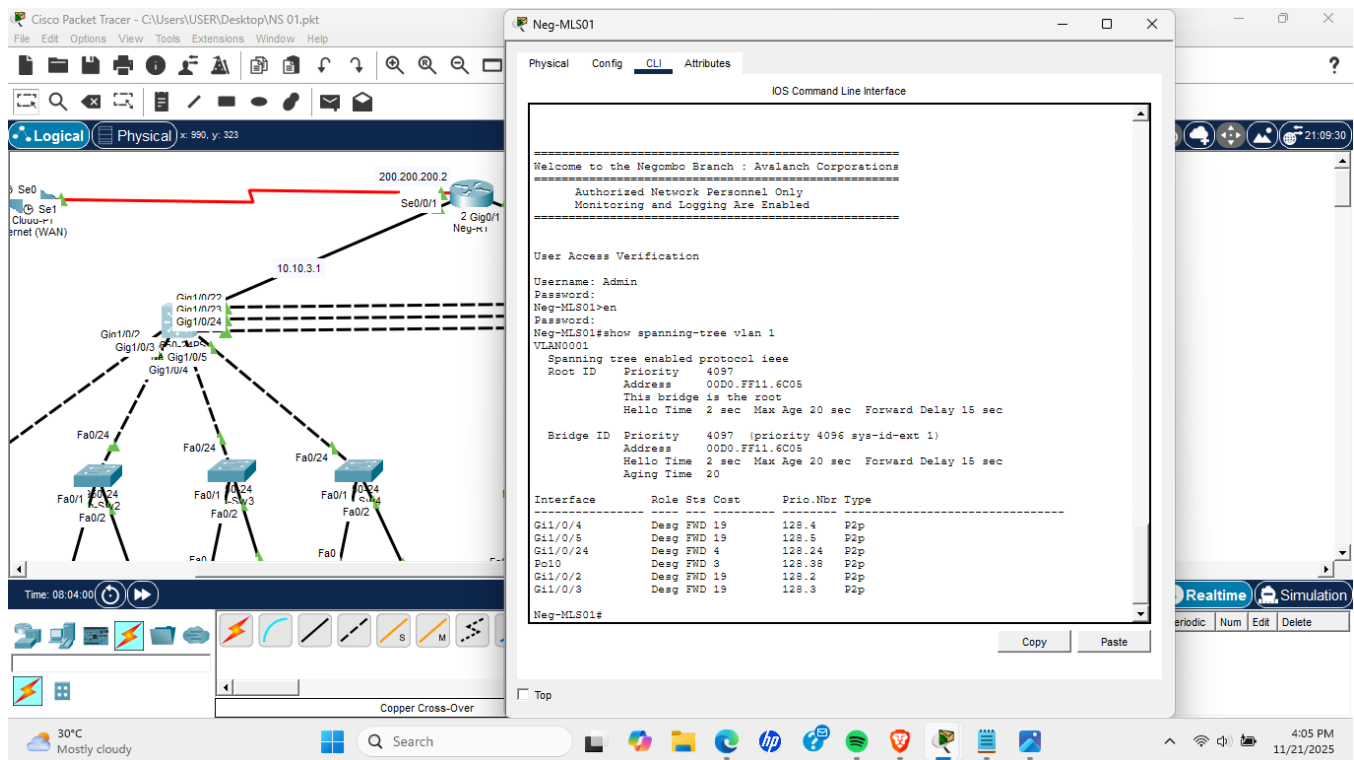


Figure 29: STP and STP Results in Neg-MLS01

5.4. Inter-VLAN Routing (SVI)

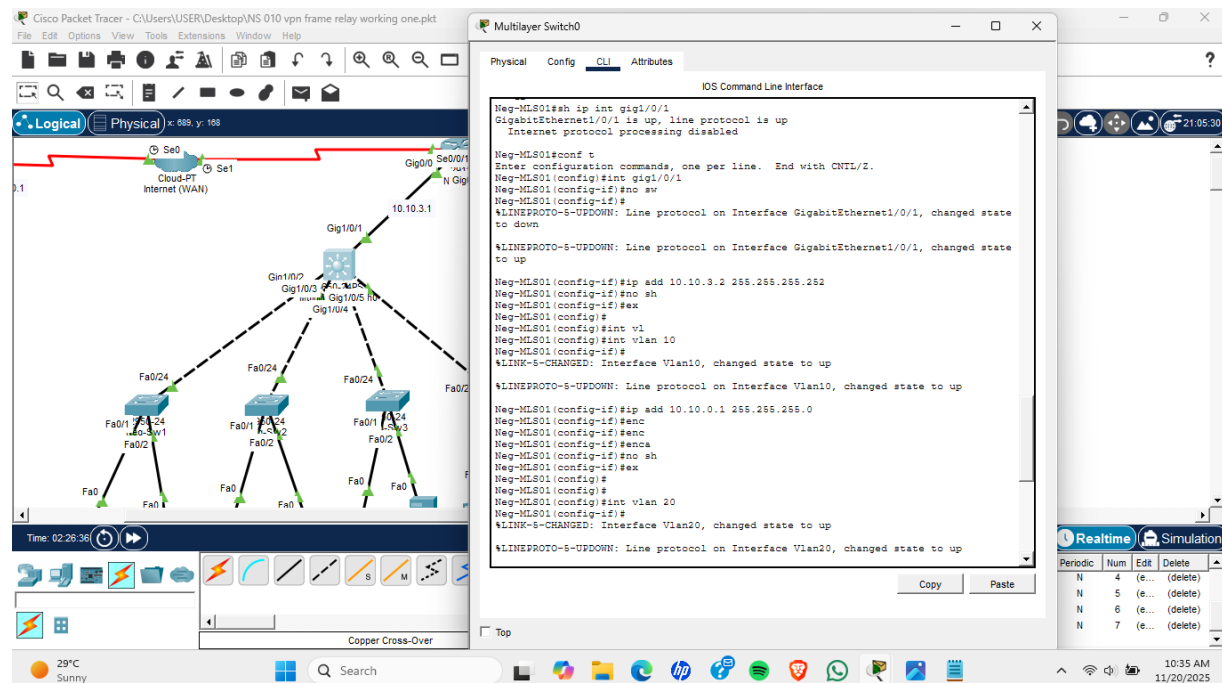


Figure 30: Inter-VLAN Neg-MLS01

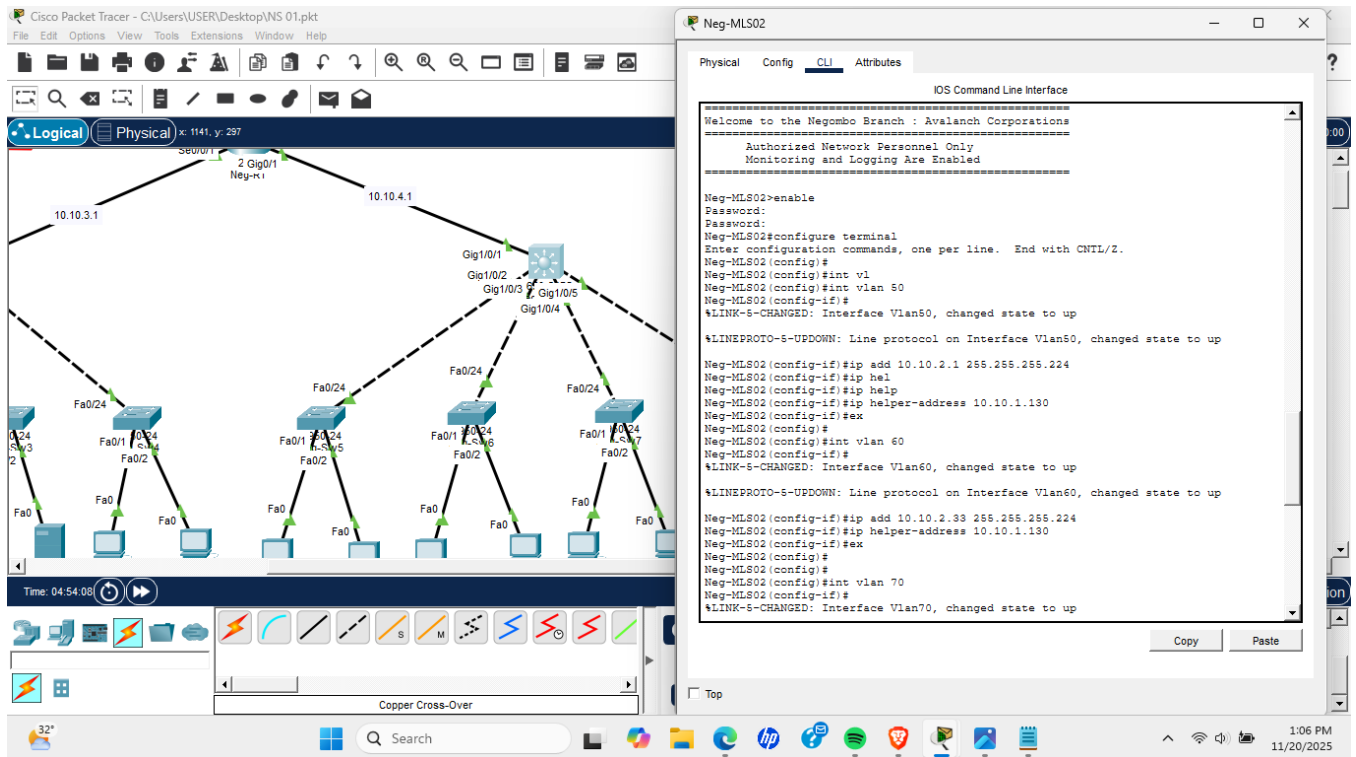


Figure 31: Inter Vlan Neg-MLS02

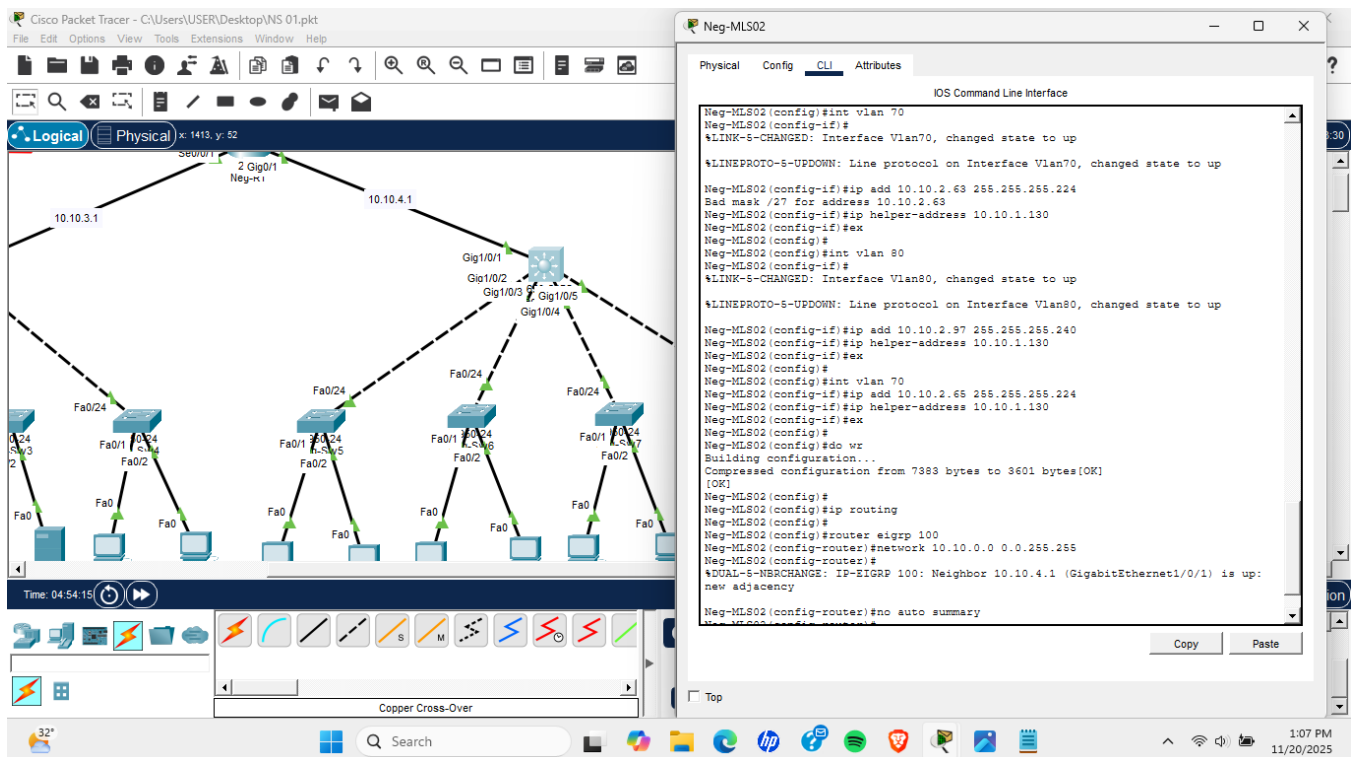


Figure 32: Inter VLAN Neg-MLS02

5.5. HSRP for Redundancy

HSRP is a redundancy protocol that is designed to group several routers into one virtual router and share an IP address. It guarantees network connectivity by assigning one router to be active and the rest to be standby so that in case primary router fails, the other router will automatically assume control without interrupting connections of the users.

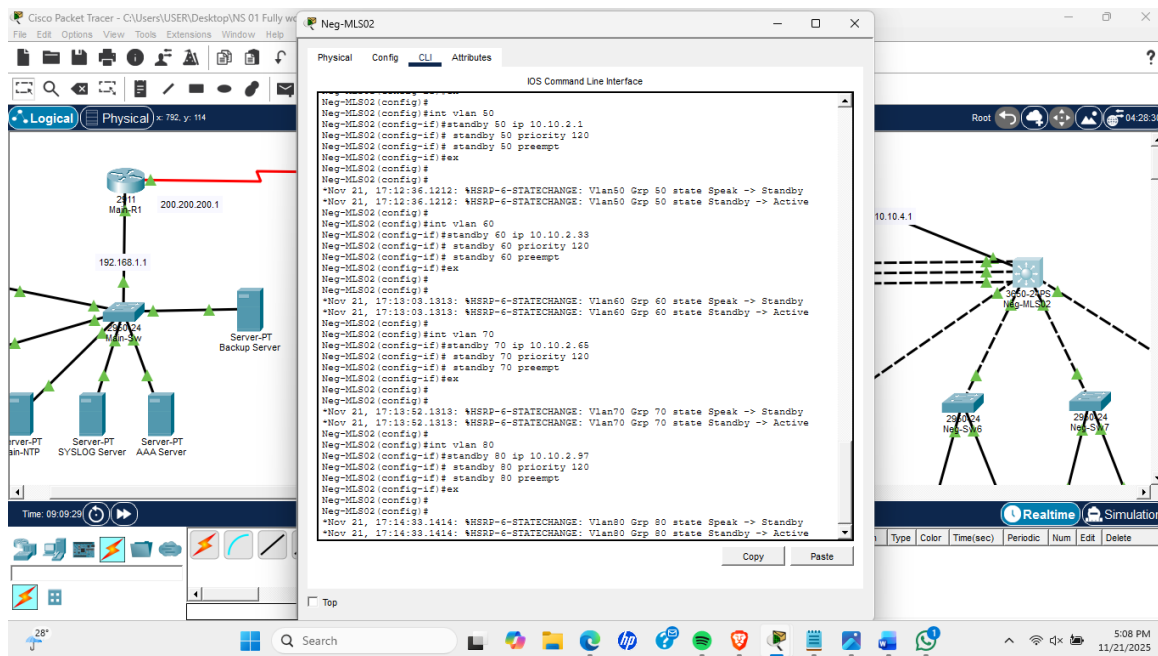


Figure 34: HSRP in Neg-MLS01

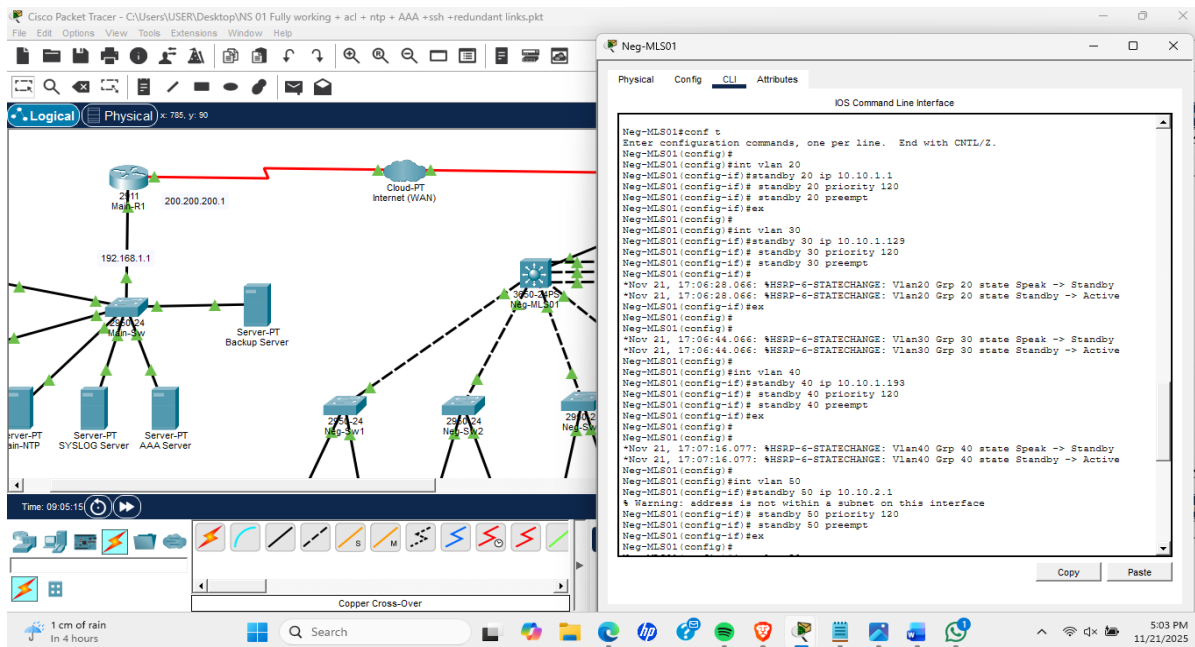


Figure 33: HSRP in Neg-MLS02

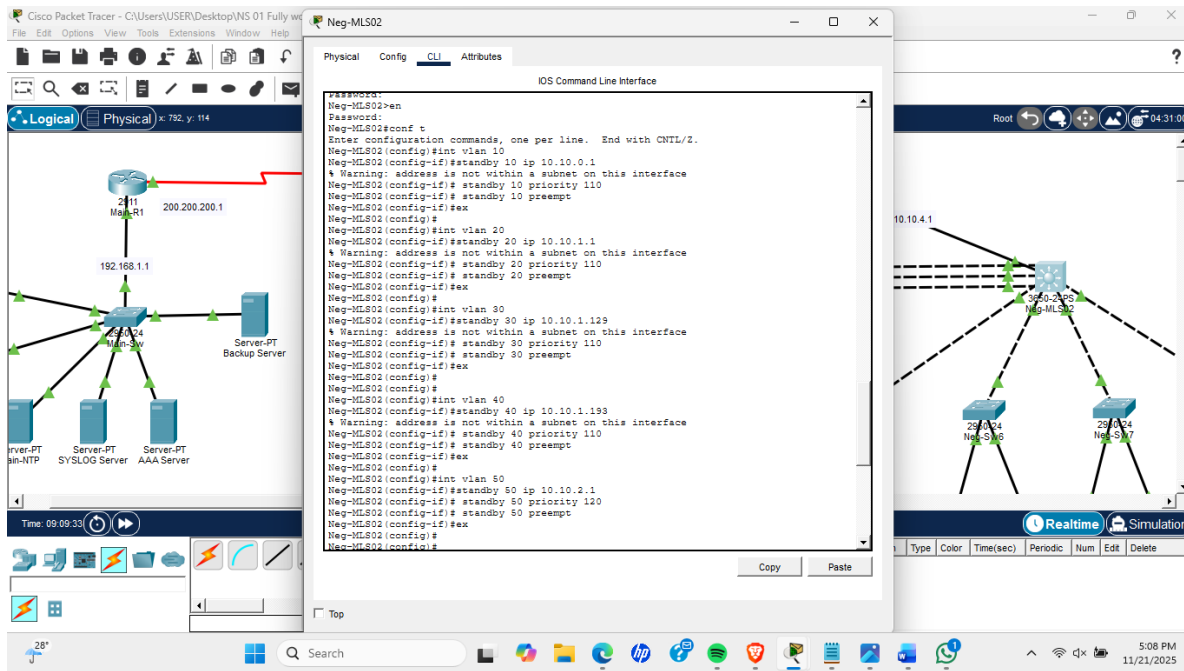


Figure 35: HSRP in Neg-MLS02

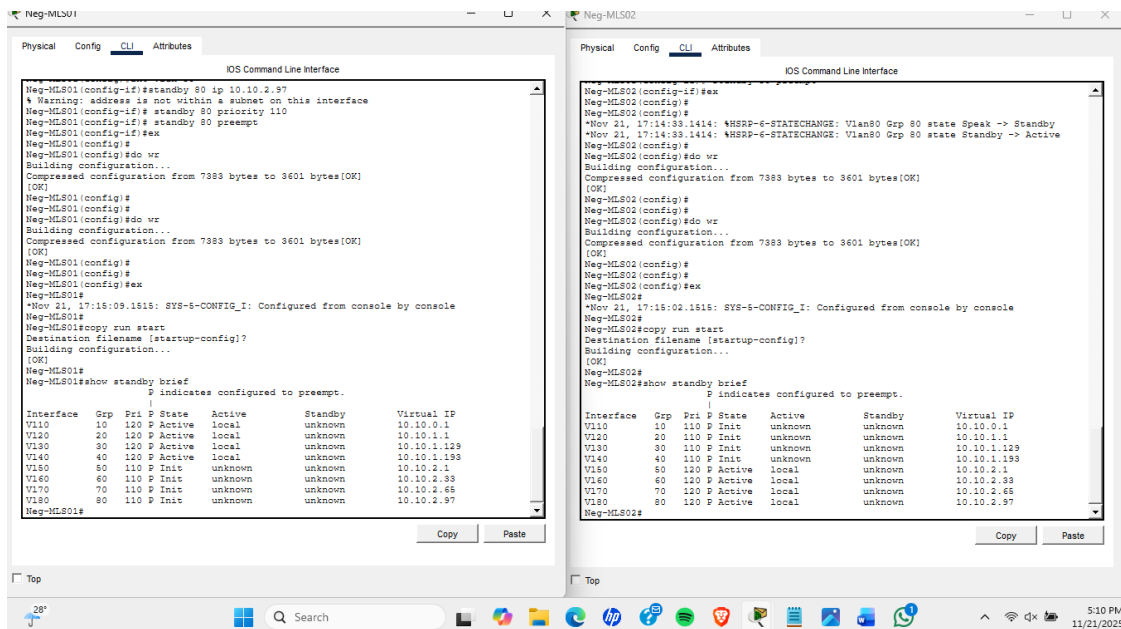


Figure 36: HSRP Results

5.6. Link Aggregation Control Protocol (LACP)

LACP is a protocol (IEEE 802.3ad) that is an open standard and is designed to provide the functionality of combining several physical switch ports into one logical channel, referred to as an Etherchannel. This routing aggregation enhances the total bandwidth and redundancy, whereby in case one physical link fails, the traffic will be routed in a different link that remains active.

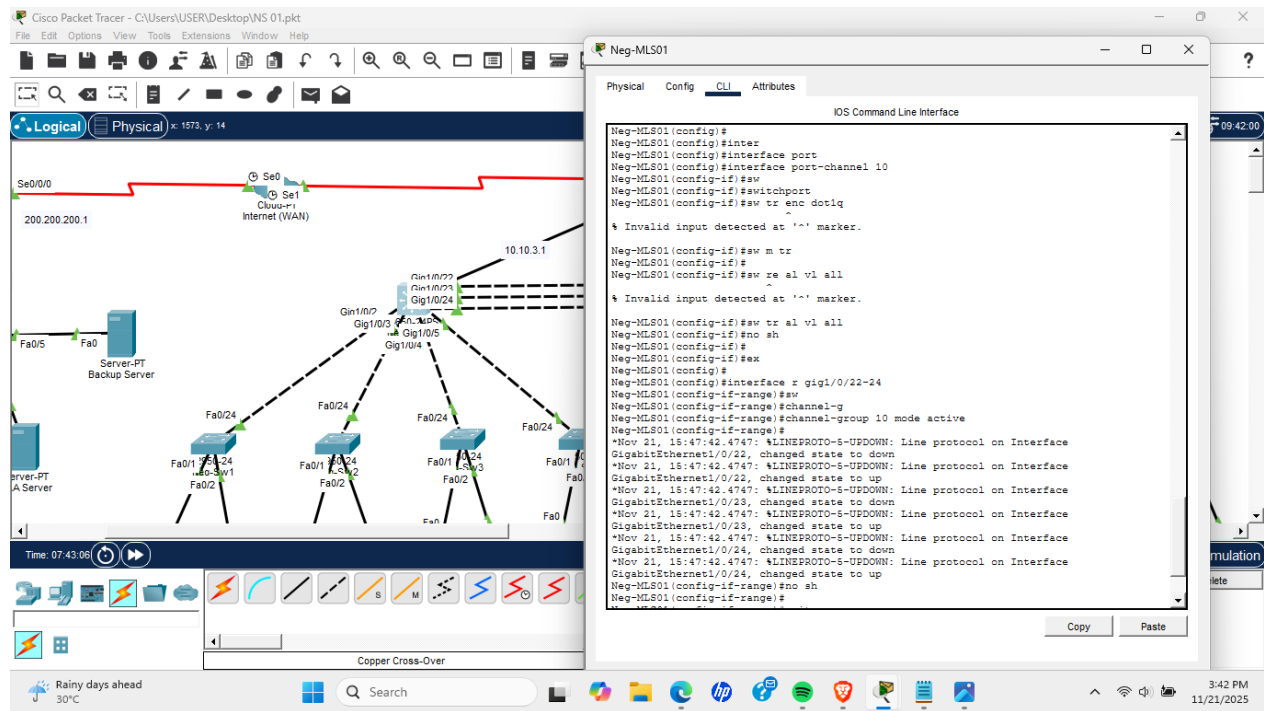


Figure 37: LACP in Neg-MLS01

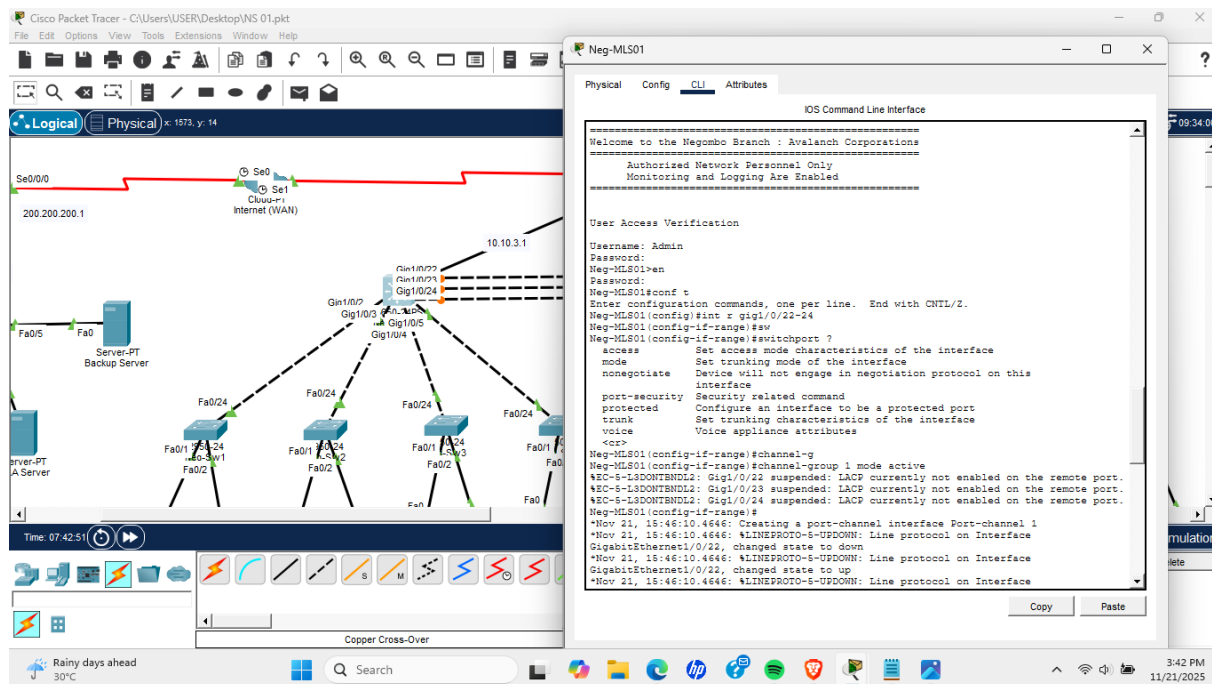


Figure 38: LACP in Neg-MLS01 pt 2

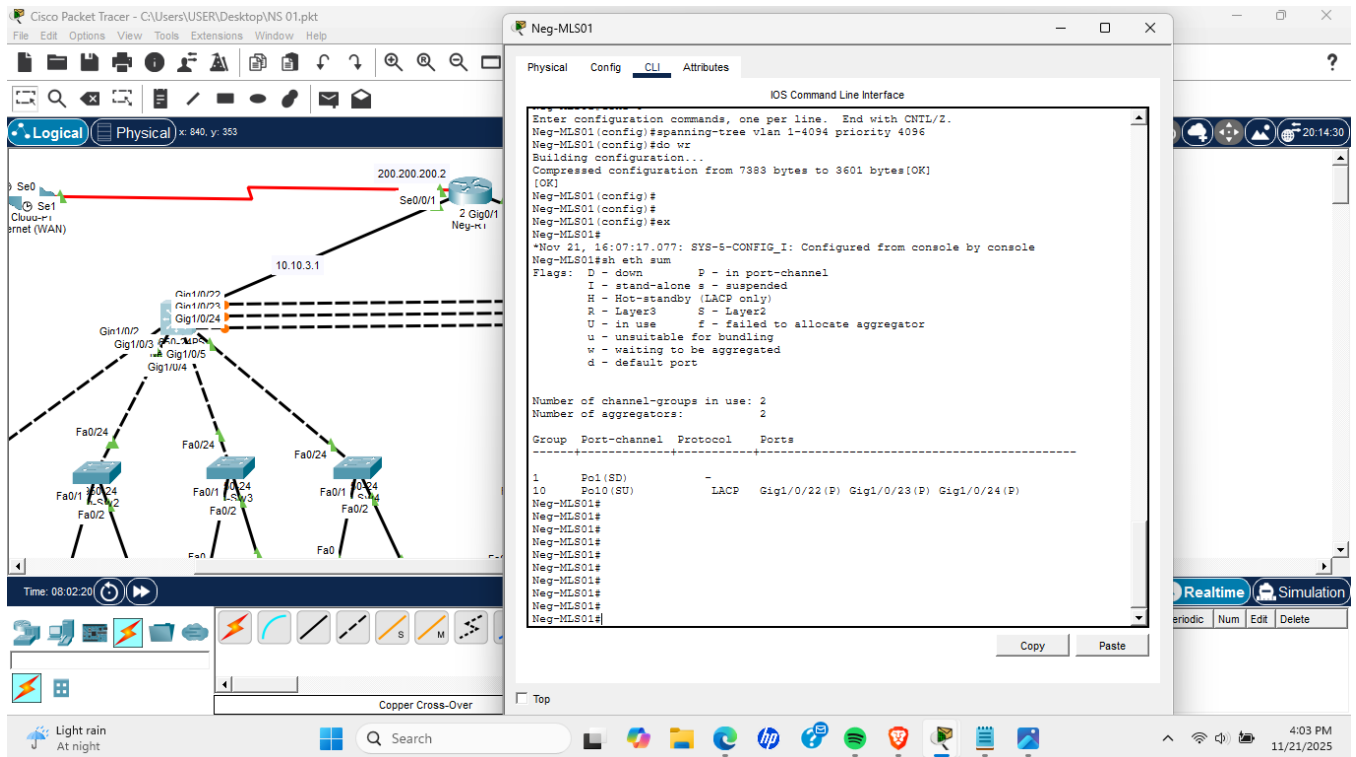


Figure 39: LACP results

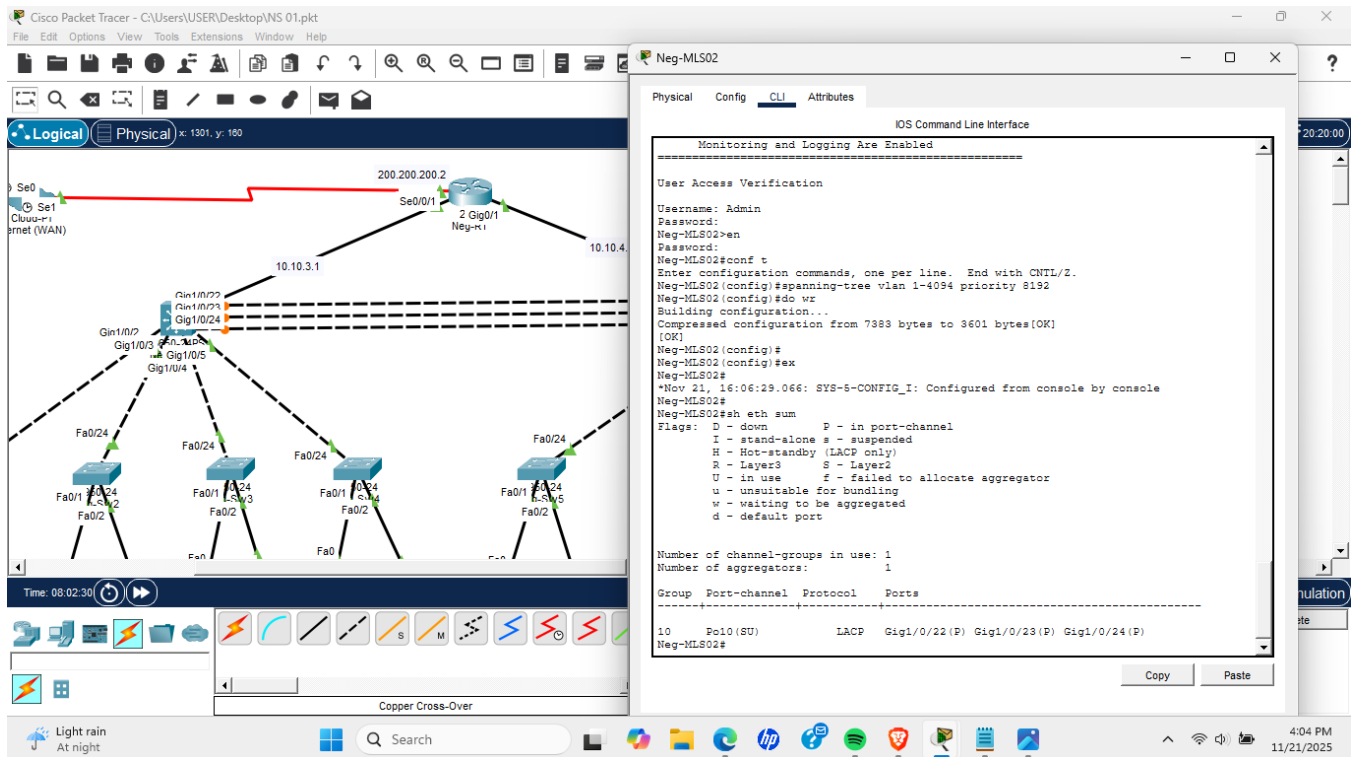


Figure 40: LACP Results in Neg-MLS02

6. Network Services Implementation

6.1. DHCP Configuration

6.1.1. DHCP Design Strategy

Dynamic Host Administration Protocol (DHCP) is used to automatically assign IP addresses, which saves the administrative work and configuration mistakes [30]. The main branch has a centralized DHCP server where issues are addressed to all the departments on the two branches.

Advantages of Centralized DHCP

- **Easy to Manage:** There is a single point of configuration that is used across all network.
- **Consistency:** There are uniform policies of address allocation which apply throughout the organization.
- **Low overhead:** Eliminates the use of DHCP servers at each site.
- **Centralized Logging:** Transactions of the DHCP are stored at a single point.
- **Cost-Efficient:** The use of a single server is effective in the management of multiple scopes [31].

6.1.2. DHCP Pool Configuration

The screenshot shows the Neg-IT-DHCP configuration window. The 'Services' tab is active, and the 'DHCP' service is configured for the 'FastEthernet0' interface. The 'Service' is set to 'On'. The 'Pool Name' is 'serverPool'. The 'Default Gateway' is '0.0.0.0'. The 'DNS Server' is '0.0.0.0'. The 'Start IP Address' is '0' and the 'Subnet Mask' is '0'. The 'Maximum Number of Users' is '0'. The 'TFTP Server' is '0.0.0.0' and the 'WLC Address' is '0.0.0.0'. Below the configuration fields is a table of DHCP pools.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	0.0.0.0	0.0.0.0	0	0.0.0.0	0.0.0.0
Admin Dep:	10.10.2.97	0.0.0.0	10.10.2.100	255.255.255.240	12	0.0.0.0	0.0.0.0
SOC Dep:	10.10.2.63	0.0.0.0	10.10.2.65	255.255.255.224	31	0.0.0.0	0.0.0.0
HR Dep:	10.10.2.33	0.0.0.0	10.10.2.35	255.255.255.224	29	0.0.0.0	0.0.0.0
Finance Dep:	10.10.2.1	0.0.0.0	10.10.2.5	255.255.255.224	27	0.0.0.0	0.0.0.0
C&C Dep:	10.10.1.193	0.0.0.0	10.10.1.195	255.255.255.192	61	0.0.0.0	0.0.0.0
IT Dep:	10.10.1.129	0.0.0.0	10.10.1.130	255.255.255.192	61	0.0.0.0	0.0.0.0

Figure 41: DHCP Server Pools

6.1.3. DHCP Relay Configuration

Due to the distance between the DHCP server (main branch) and the clients (new branch), a DHCP relay (IP helper-address) is required in the VLAN interfaces of the core switch [32].

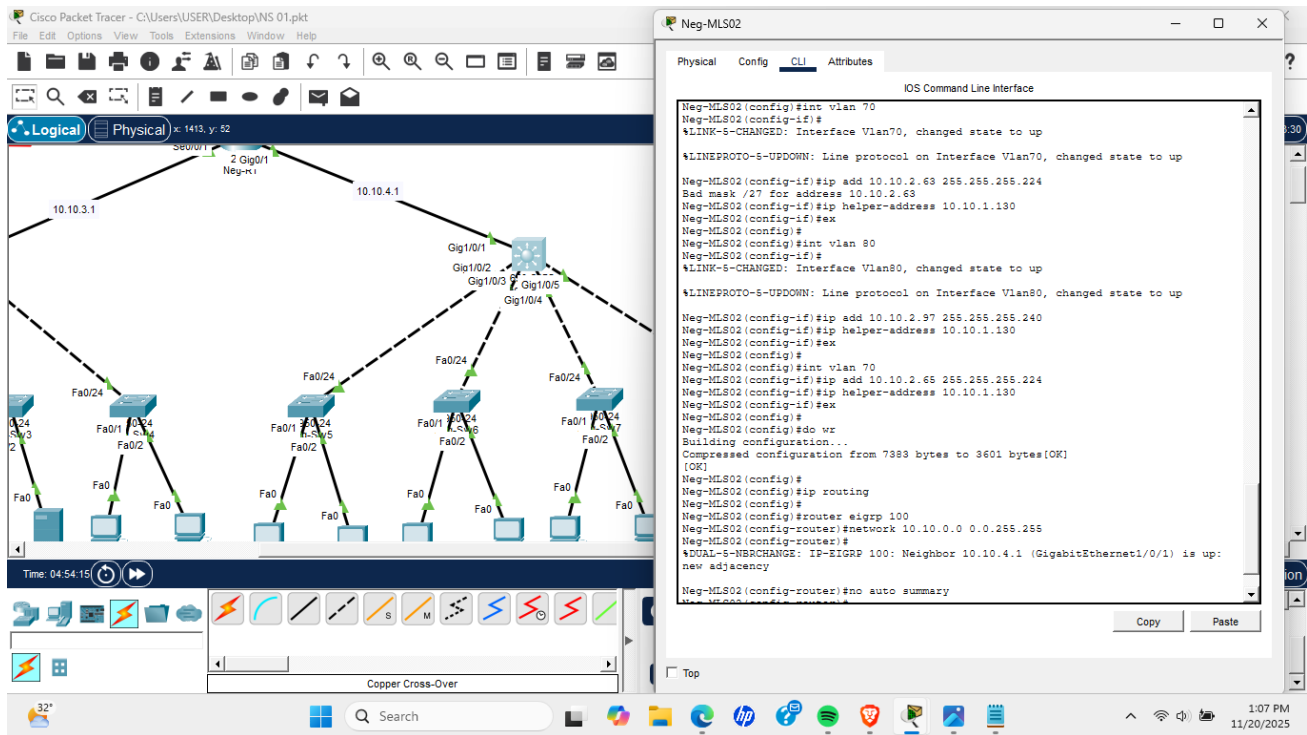


Figure 42: DHCP Relay on the Neg-MLS02

How does DHCP Relay work?

1. Client sends a DHCP DISCOVER in its local network.
2. The router/ multilayer switch accepts the broadcast and routes it as a unicast to the DHCP server at 192.168.1.10 through IP helper address.
3. The server of the DHCP responds with an address that belongs to the corresponding pool and the switch sends the answer of the server to the client [33].

6.2. NTP Configuration

Network Time Protocol (NTP) keeps devices on network in synchronization of time which is necessary in:

- **Security Event Correlation:** Proper correlation of security events is possible with proper timestamps.
- **Log Analysis:** This service brings sequential sorting of events across devices.
- **Authentication:** Time based authentication protocols (Kerberos, certificate

validation).

- **Compliance:** Met regulatory requirements to have accurate audit trails.
- **Troubleshooting:** Provides proper time sequencing of network troubleshooting.

Time Discrepancy Effect : Studies have determined that a ten seconds drift is enough to make security event correlation futile [37]. Therefore, security operations need less than a second accuracy.

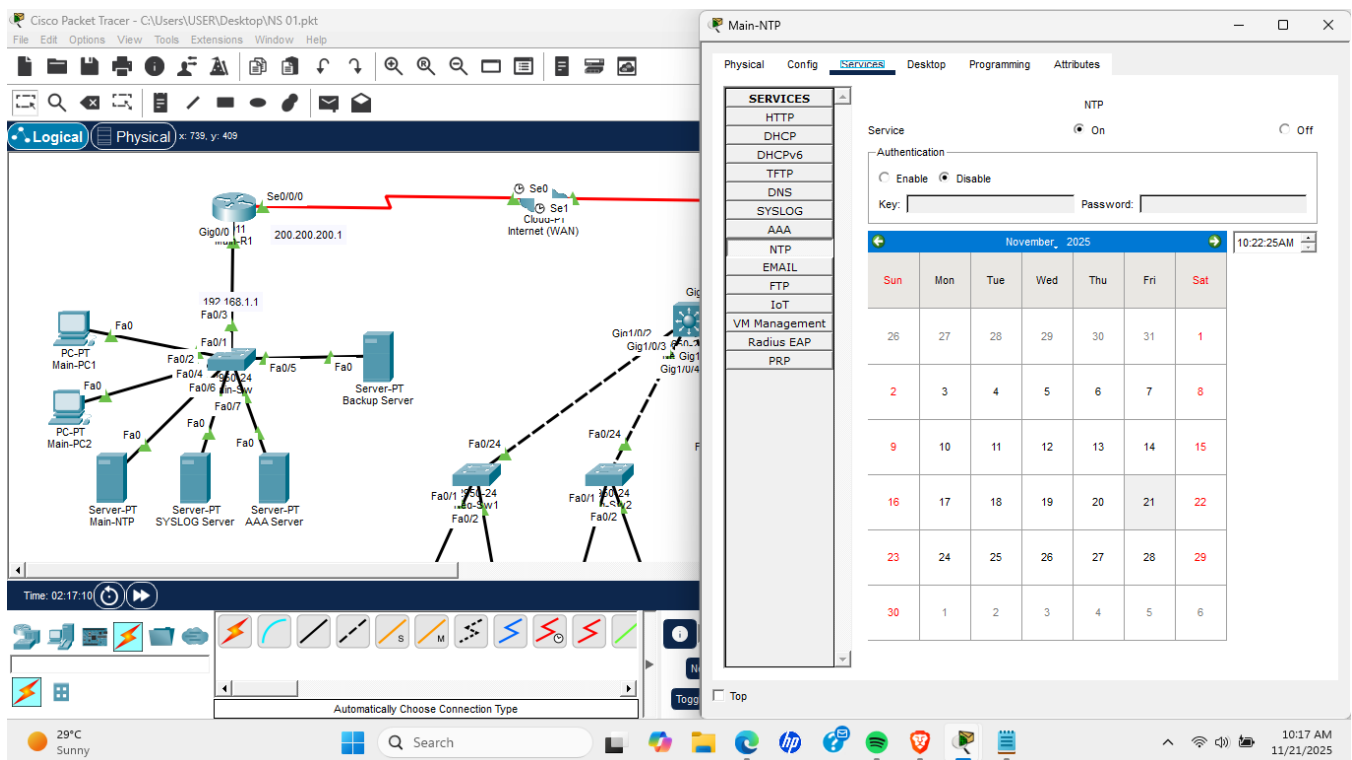


Figure 43: NTP Server

Every network equipment (switch and router) is synchronized to the configured NTP sources

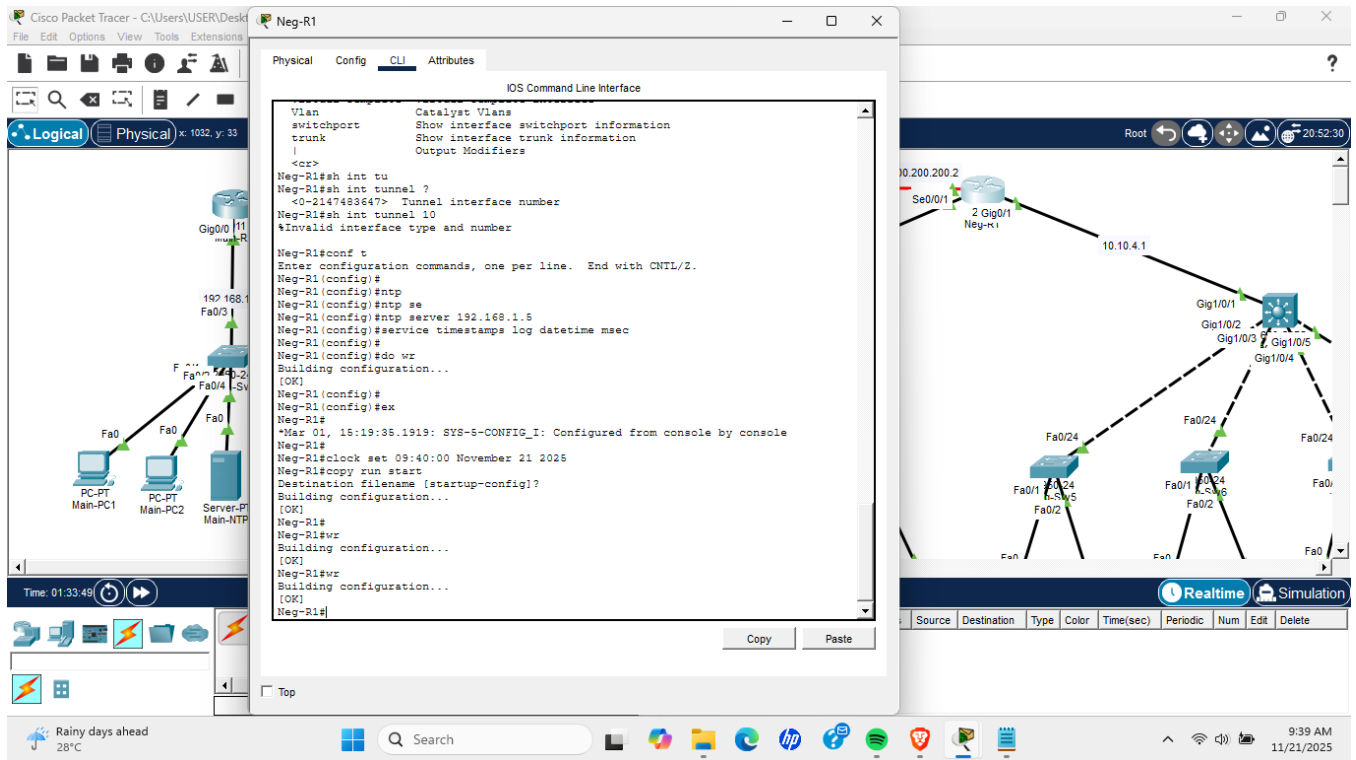


Figure 44: NTP Configuration Neg-R1

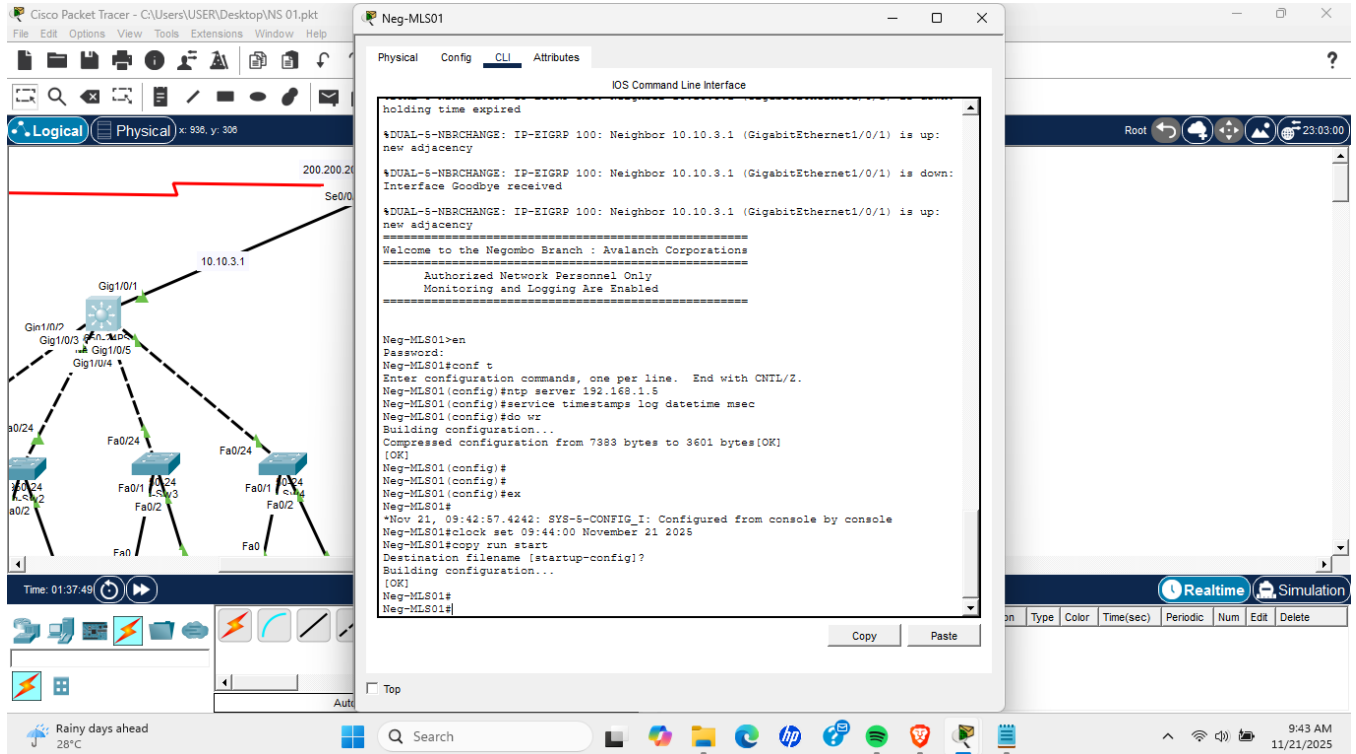


Figure 45: NTP Configuration Neg-MLS01

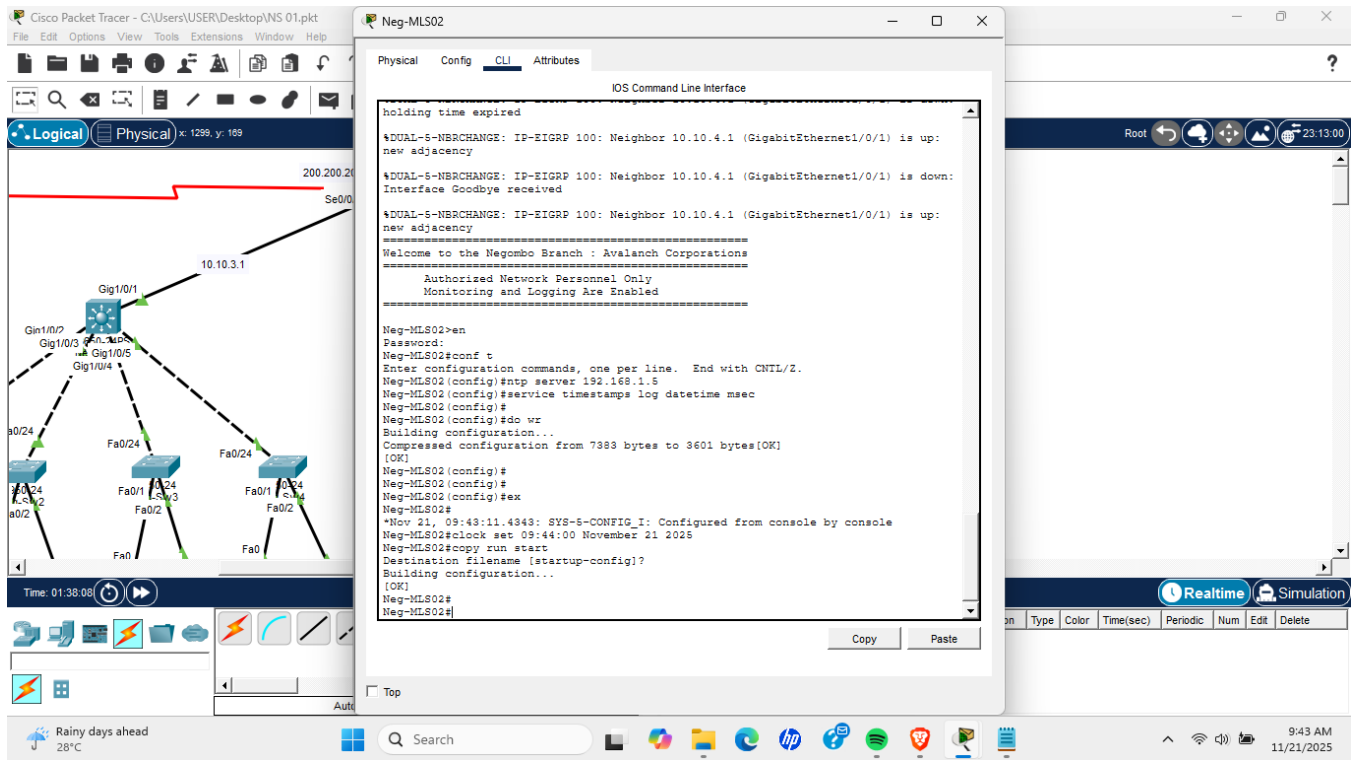


Figure 46: NTP Configuration Nrg-MLS02

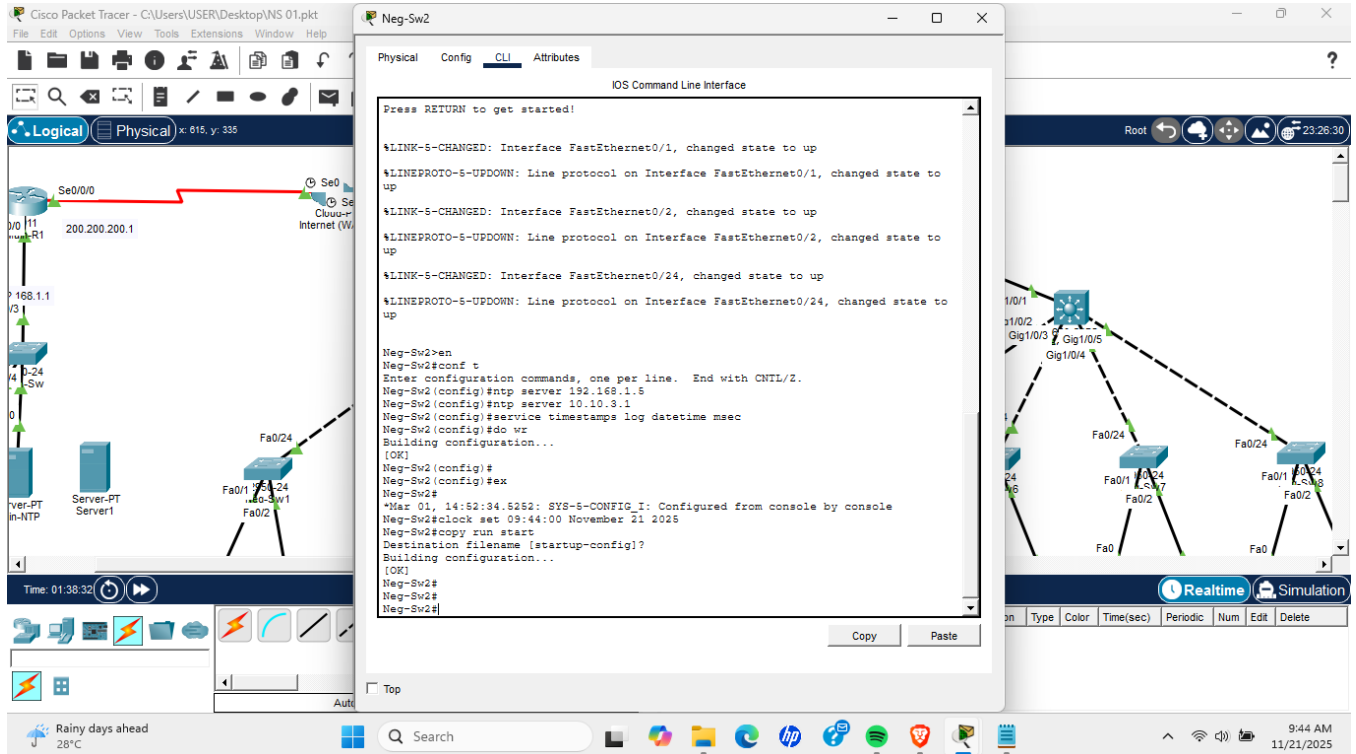


Figure 47: NTP Configuration Neg-Sw2

6.2.1. NTP Verification and Monitoring

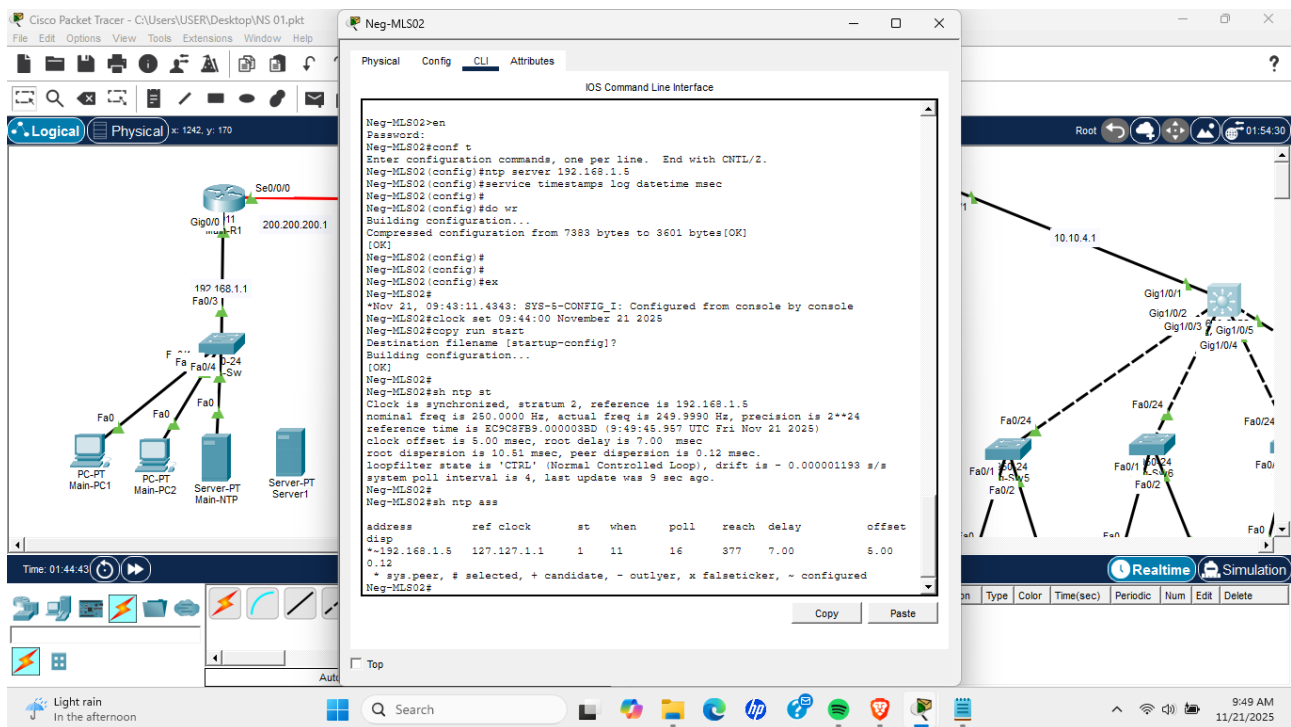


Figure 48: NTP Statistics Neg-MLS02

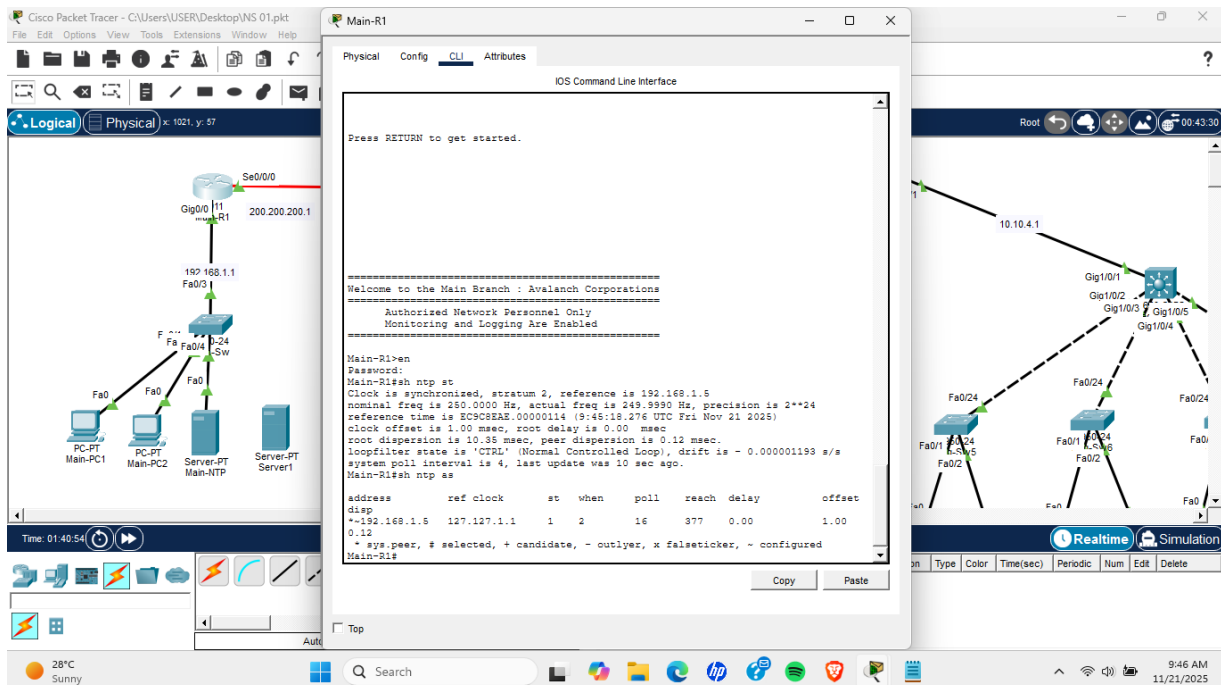


Figure 49: NTP Result Main-R1

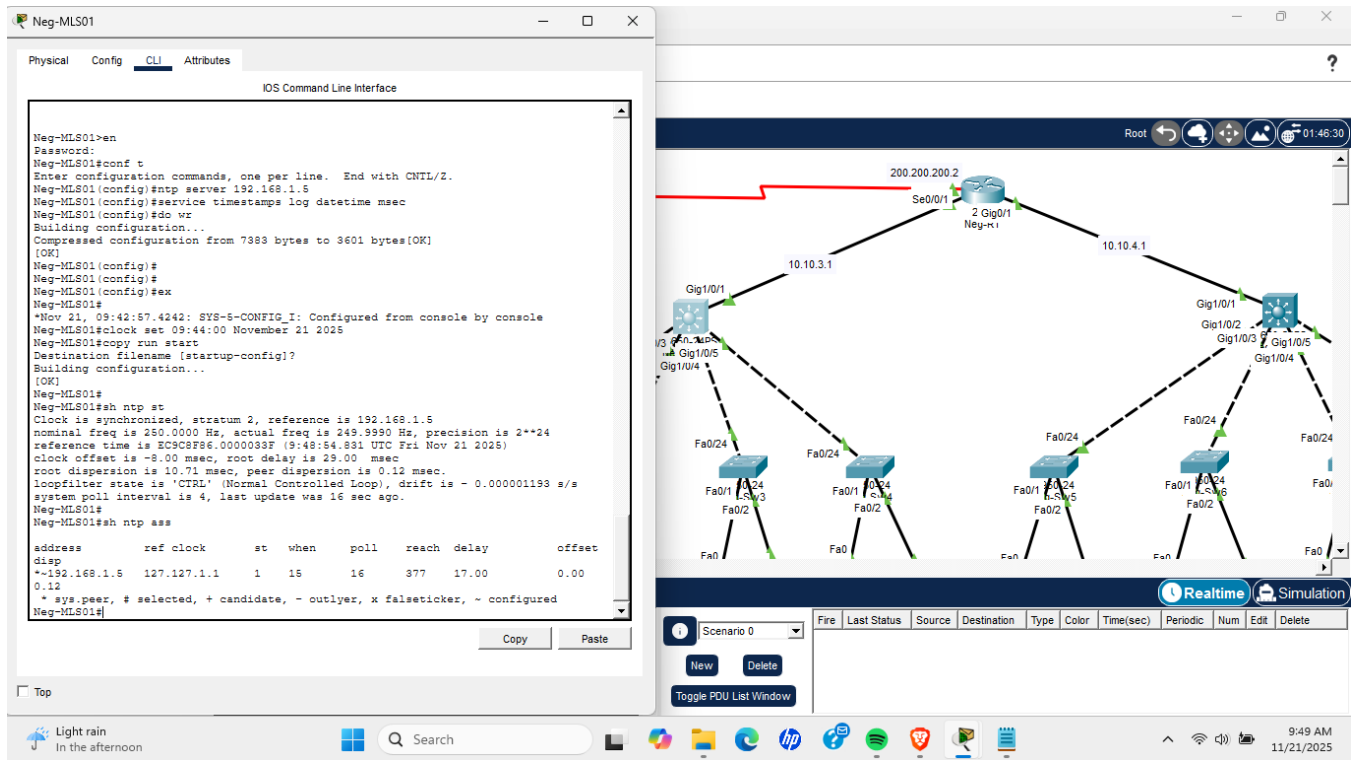


Figure 50: NTP result Neg-MLS01

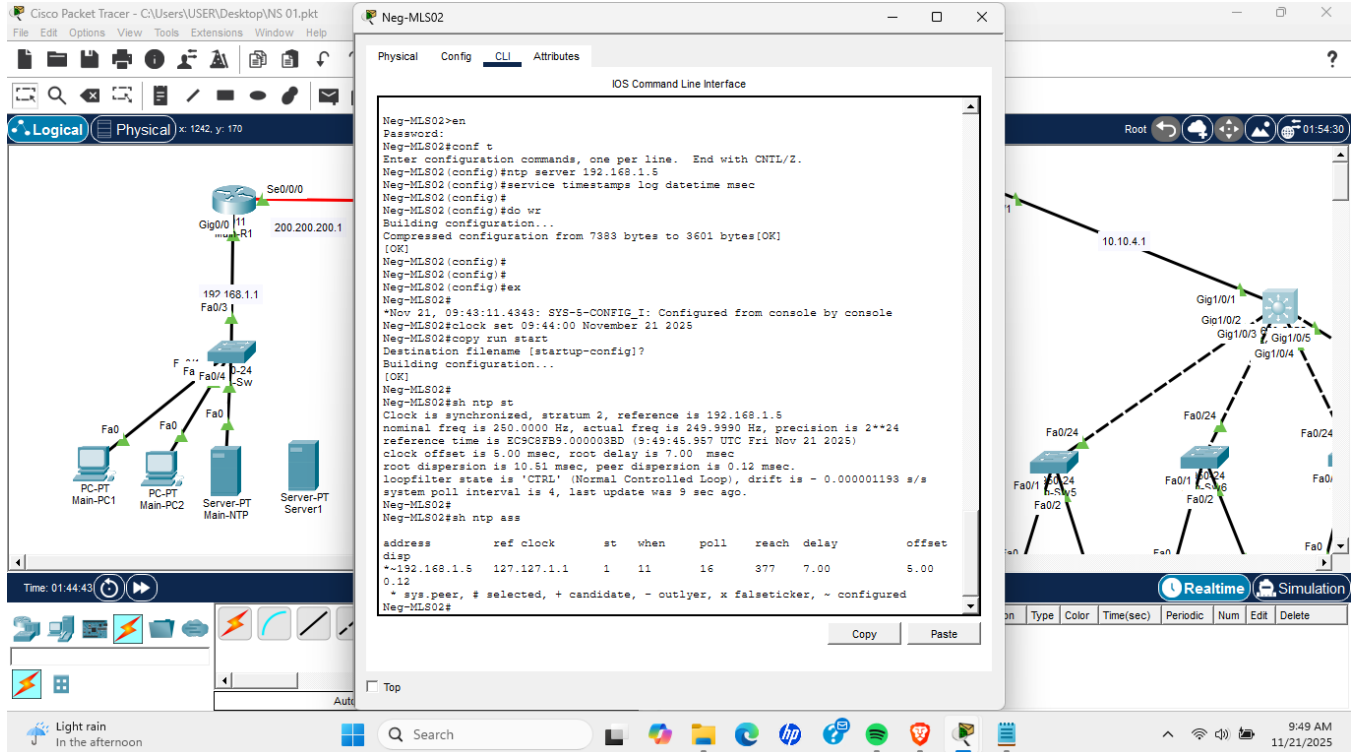


Figure 51: NTP Results in Neg-MLS02

6.3.Syslog Configuration

Syslog can use all messages that are sent by network devices and logs them under a single repository, therefore, supporting:

- **Security Monitoring:** Allows real time security event detection.
- **Troubleshooting:** It is a historical record of what has happened to the network.
- **Compliance:** Provides an audit trail which is mandated by regulatory frameworks (ISO 27001, PCI DSS).
- **Capacity Planning:** Enables us to analyze the trends of network usage.
- **Forensic Analysis:** Aids capabilities of post-incident investigation.

Regulatory Requirements

- The minimum length of log retention of PCI DSS is 90 days.
- HIPAA requires a period of six years, whereas SOX takes seven years.
- The GDPR retention times take different values based on the kind of data [40].

6.3.1. Syslog Server Implementation

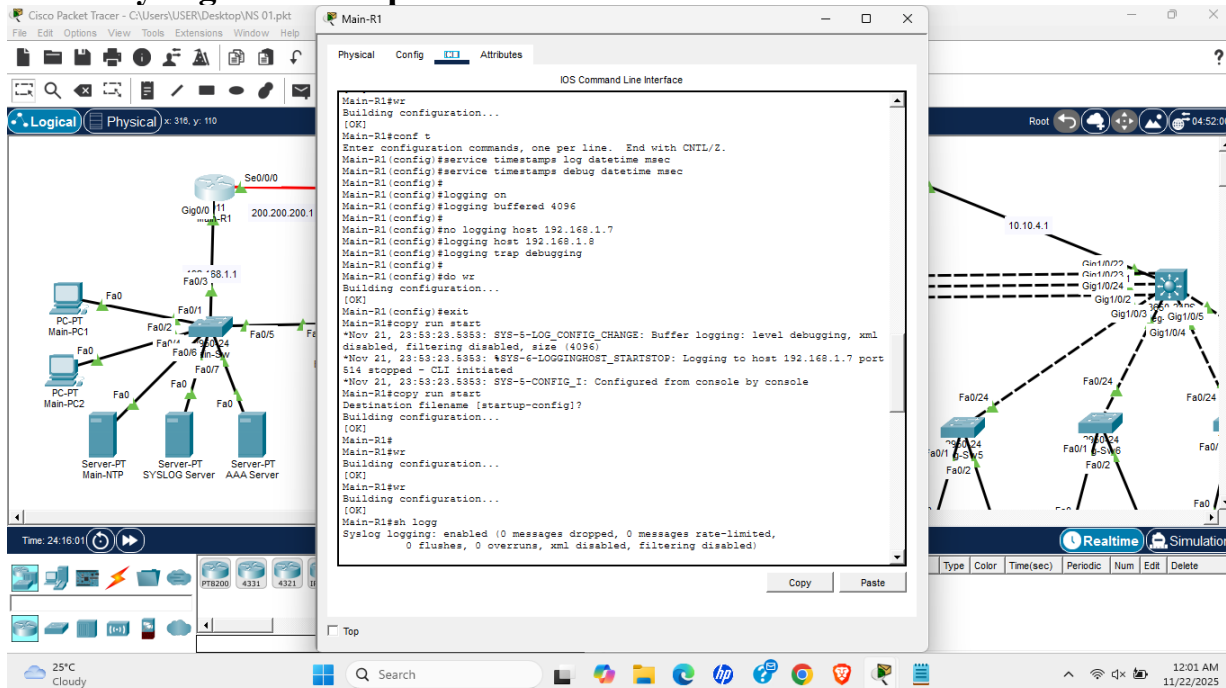


Figure 52: Syslog Config Main-RL

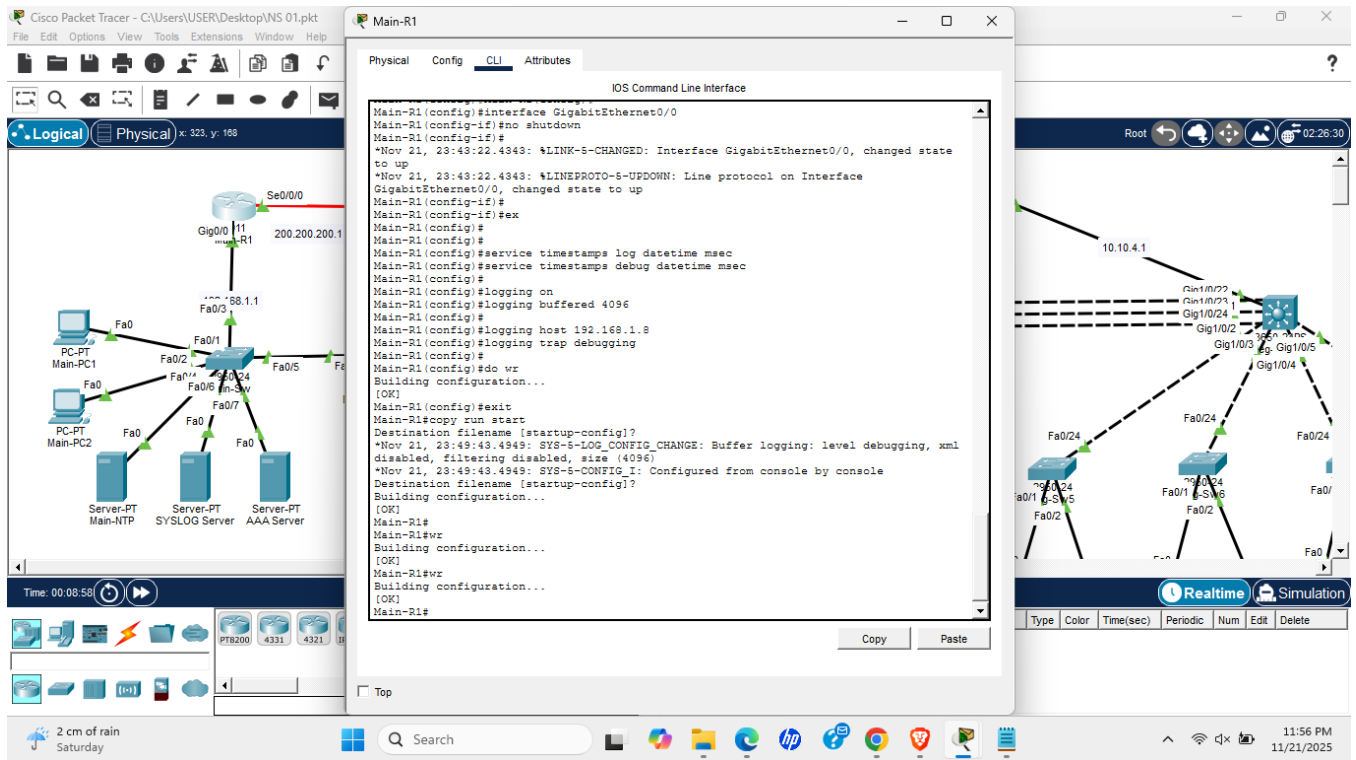


Figure 53: SYSLOG Config Main-R1 cont

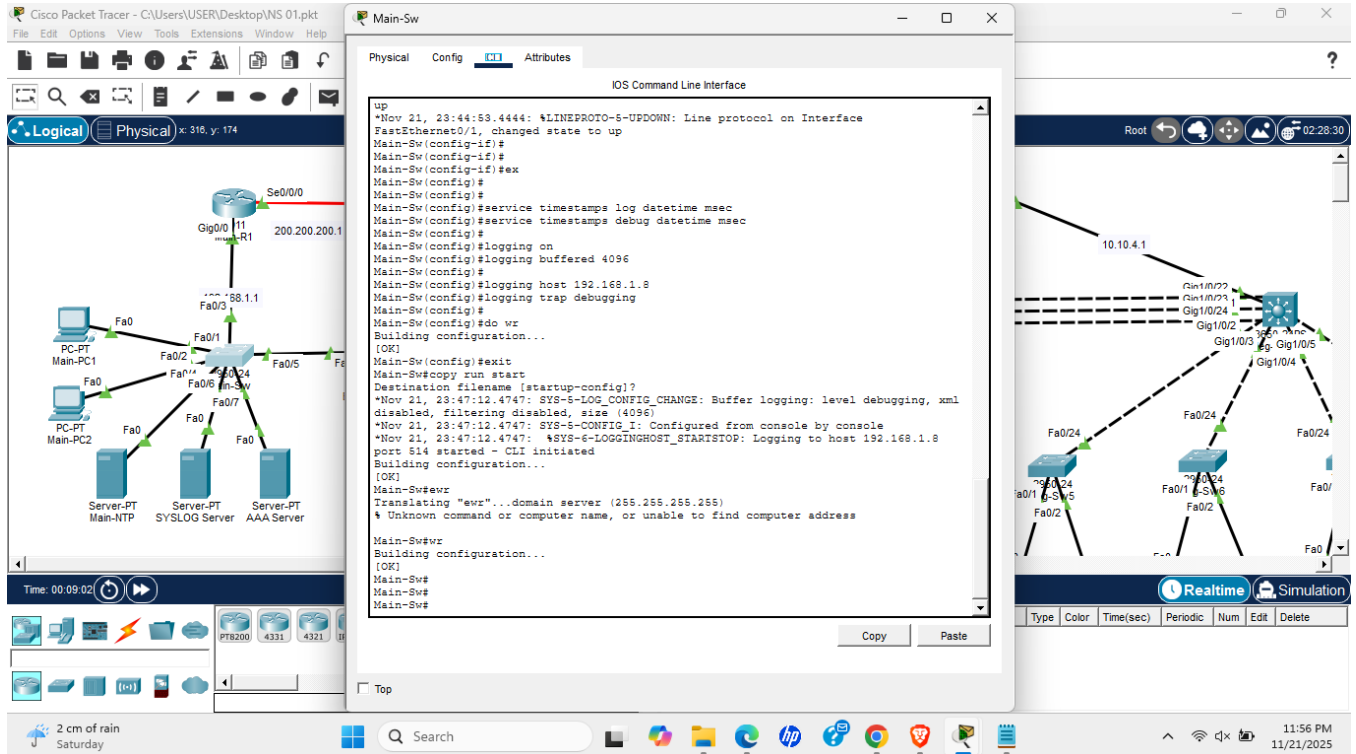


Figure 54: SYSLOG Config in Main-Sw

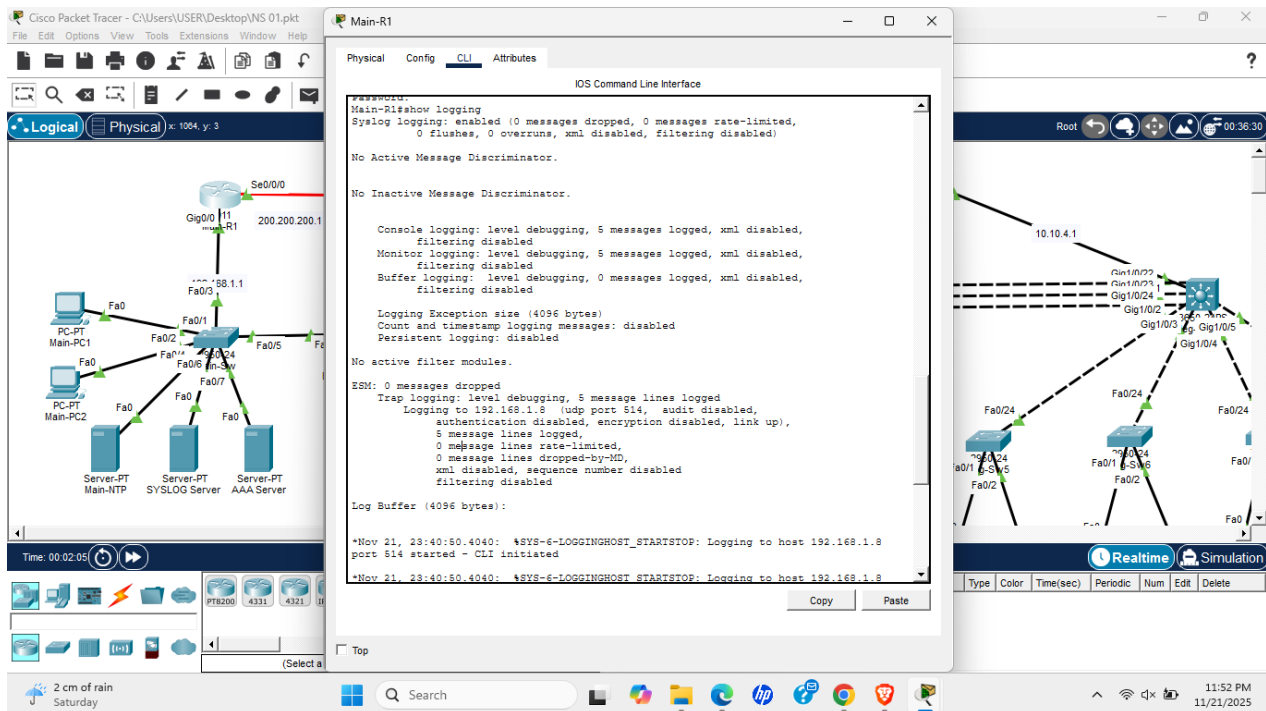


Figure 55: SYSLOG Result

Server Specifications:

- Storage: 500 GB (with a 90 day retention period)
- Message Rate: The accommodation of message rate of 10,000 messages per second.
- Rotation of logs: Every midnight UTC daily.
- Backup: (Weekly) to external storage.
- Retention Policy: Retention of 90 days, an extra one year of archived storage layer.

6.3.2. Log and Security Events Categories

1. Authentication Events

- Success and attempt to log in (successful and failed).
- Privilege level changes
- Things that break AAA authentication.

2. Configuration Changes:

- Configuration mode entry/exit mode.
- Specific command execution
- Configuration save

3. Interface Events:

- Link up/down transitions
- Error rate thresholds surpassed.
- Duplex and speed mismatches

4. Security Events:

- Port security violations
- ACL hits (denied traffic)
- DHCP snooping violations
- STP Topology Alterations

5. System Events:

- Device reboots
- Hardware failures
- Power supply issues

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is visible with a central router (R1) connected to several PCs (PC-PT Main-PC1, PC-PT Main-PC2) and two servers (Server-PT Main-NTP, Server-PT SYSLOG Server). The router is configured with GigabitEthernet 0/0/11 and 0/0/3 interfaces. The Syslog Server is connected to the router via its Fa0/0 interface.

The main window shows the 'Syslog Server' configuration and the 'Services' tab. The 'Syslog' service is enabled. Below the configuration, a table displays the Syslog log entries:

Time	HostName	Message
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-S-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, ...
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-S-CONFIG_I: Configured from console by console
11.21.2025 11:52:07.325 PM	200.200.200.2	%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.1.8 port 514 starte...
11.21.2025 11:52:51.168 PM	200.200.200.2	%SYS-6-LOGGINGHOST_STARTSTOP: ...
11.21.2025 11:53:23.896 PM	192.168.1.1	%SYS-S-LOG_CONFIG_CHANGE: Buffer logging: level debugging, xml disabled, ...
11.21.2025 11:53:23.896 PM	192.168.1.1	%SYS-6-LOGGINGHOST_STARTSTOP: ...
11.21.2025 11:54:19.540 PM	192.168.1.1	%SYS-S-CONFIG_I: Configured from console by console
11.22.2025 11:54:22.261 PM	192.168.1.1	%SYS-S-CONFIG_I: Configured from console by console
11.22.2025 11:54:22.134 PM	200.200.200.2	%LINEPROTO-S-UPDOWN: Line protocol on...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: IP-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: IP-EIGRP 100: ...
11.22.2025 11:54:22.134 PM	200.200.200.2	%LINEPROTO-S-UPDOWN: Line protocol on...
11.22.2025 11:54:22.134 PM	200.200.200.2	%DUAL-S-NBRCHANGE: IP-EIGRP 100: ...

The bottom of the screen shows the Windows taskbar with the system clock at 12:06 AM on 11/22/2025.

Figure 56: SYSLOG Server log

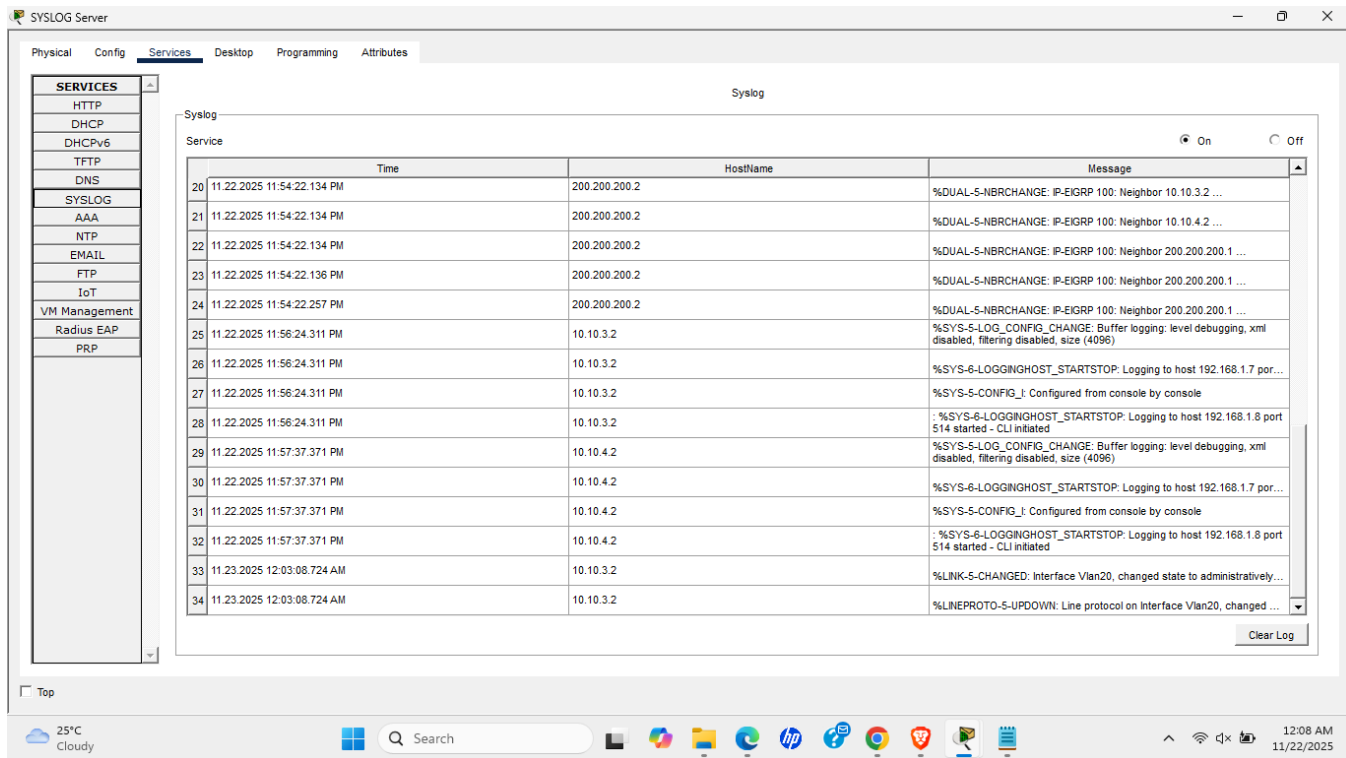


Figure 57:SYSLOG Server log cont.

6.4. AAA Configuration

Authentication, Authorization, and Accounting (AAA) is used to offer central security management to the access of network devices [42]. The AAA scheme applies the least privilege principle and accountability.

Three Components in AAA

1. Authentication: Verifying the person (Who are you?)
2. Authorization: Decides what you can do (What can you do?)
3. Accounting: Documenting user activity (What did you do?)

6.4.1. AAA Server Selection

Type of Server: RADIUS (Remote Authentication Dial-In User Service)

Justification:

- Industry Standard: ubiquitous protocol (RFC 2865)

- Encryption: Password encryption is used in transit.
- Flexibility: Multiple authentication options.
- Scalability: Processes thousands of authentication requests.
- Cost-Efficient: Open-source implementations: can be found (FreeRADIUS) [43].

TACACS+ Server not considered because of Cisco-proprietary but provides more fine-tuning command permissions.

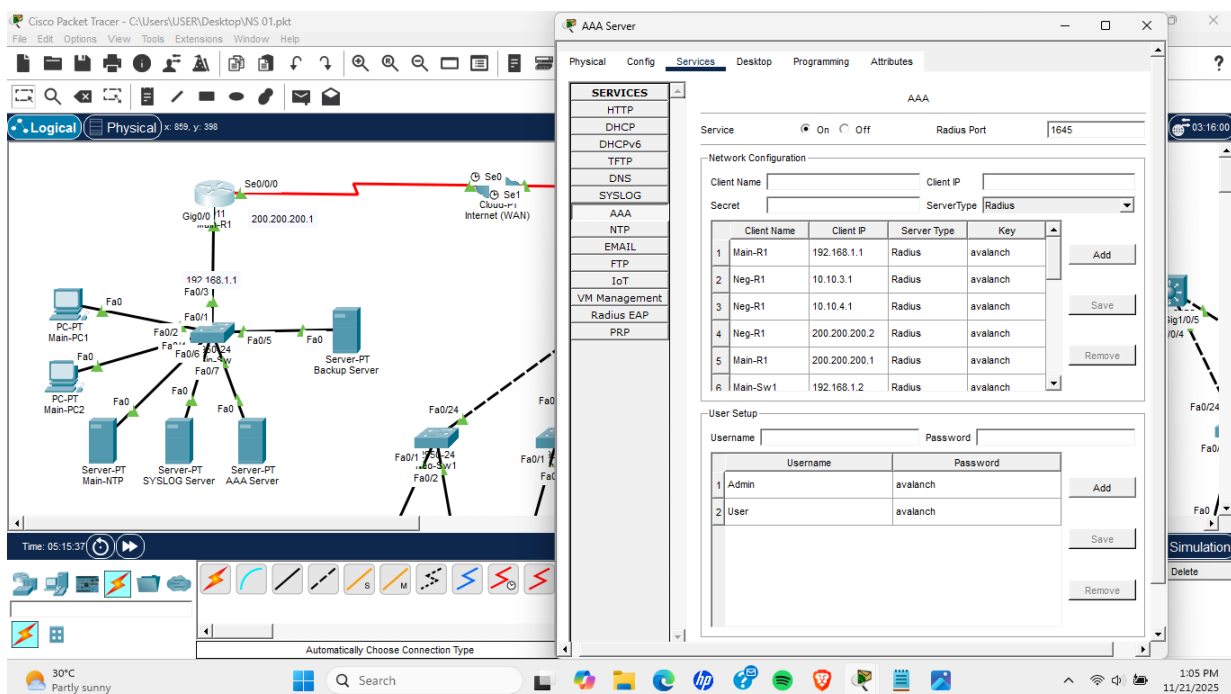


Figure 58: AAA Server

Authentication Flow

1. User connects through SSH to network device.
2. Device requests RADIUS server in order to get authenticated.
3. Credential validation is against database by RADIUS server.
4. And in case of RADIUS unavailable a local credentials fallback.
5. A successful authentication is one that is successful depending on the level of privilege [44].



6.4.2. Authorization Configuration



Command: Authorization

- Level 15: Ability to do reload, write memory, configuration.
- Level 7: Is able to perform show instructions, general troubleshooting.
- Level 1: Is only able to do show version, show ip interface brief.

6.4.3. Accounting Configuration

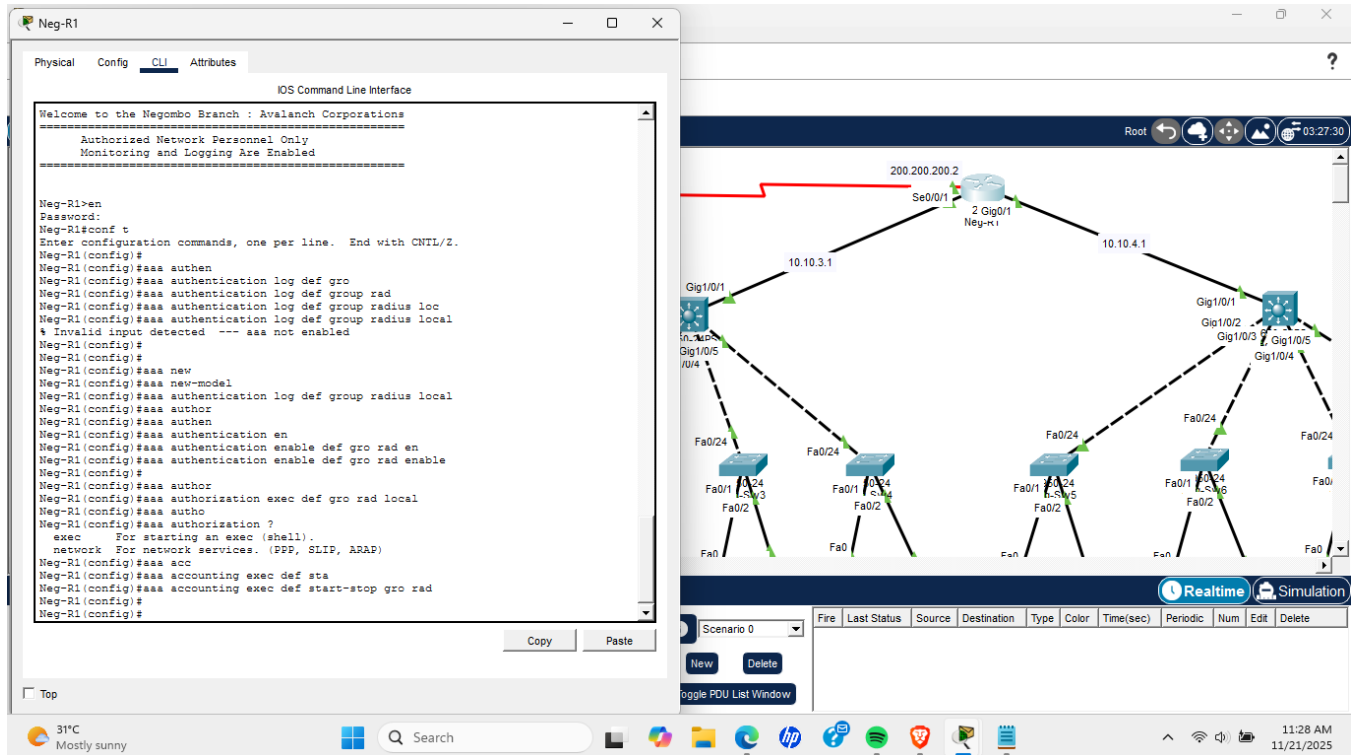


Figure 61: AAA Accounting

Accounting Records Include:

Session Information: Username, Source IP address, Login timestamp, Logout timestamp, Session duration

Command Execution: All the commands will be run at level 15., Timestamp of execution, Command output (optional)

Resource Usage: Bytes sent/received, Packets transmitted, Session statistics [45]

Accounting Storage:

- Primary: DB RADIUS server database.
- Secondary: Syslog messages to central logging server.
- Retention: A minimum of 1 year (compliance requirement)

6.4.4. AAA Security Benefits

1. Centralized Control: Single credential management point
2. Accountability: Full audit record of administrative procedures.
3. Compliance: Enforcement of ISO 27001, access control requirements of the PCI DSS.
4. Less risk: Does away with common passwords and local accounts.
5. Quick Revocation of Credentialing: Rapid credentialing revocation at the time of employee departure [46].

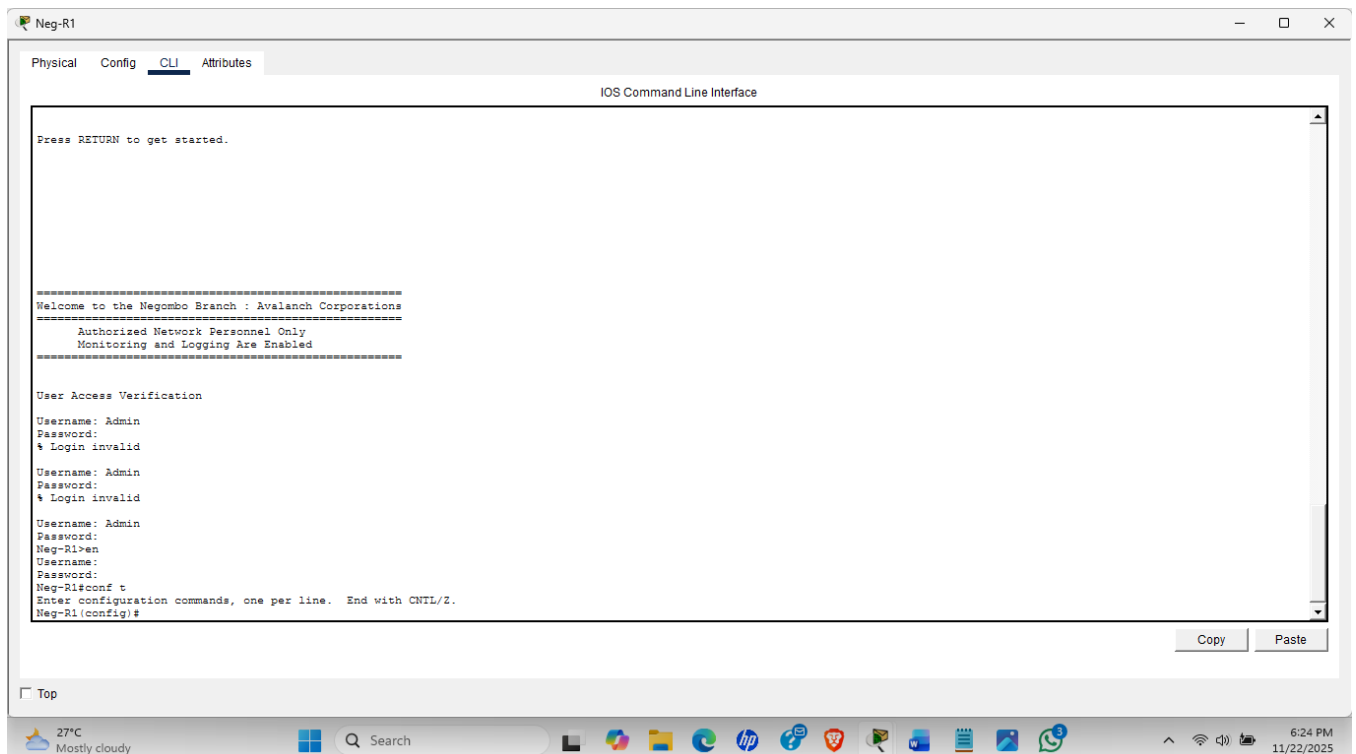


Figure 62: AAA results

7. Network Security Implementation

7.1. Access Control Lists (ACLs)

7.1.1. ACL Design Strategy

Access Control Lists block network traffic along with the criteria determining the security, enforcing the network layer security policies [47]. The first security requirement is that Finance (VLAN 50) and HR (VLAN 60) departments should be isolated.

Security Requirements

- Communication between finance department and any other department is not possible.
- HR department is not able to communicate with other departments.
- The finance department and the HR do not interact.
- Only internet and main branch services are available to both the departments.

Design Philosophy

- Implicit Deny: Deny all traffic except where specifically authorized.
- Least Privilege: Minimum access that is required.
- Security Design: Defense in Depth [48].

7.1.2. ACL Implementation

ACL Type Selection: Extended ACLs chosen to be in granular control due to the possibility of including Source IP address, Destination IP address, Protocol type (TCP, UDP, ICMP), and Port numbers

ACL Placement: ACLs are used inbound on VLAN interfaces (SVIs) at the core switch and traffic nearest to a source is filtered [49].

7.1.2.1. Finance Department ACL (VLAN 50)

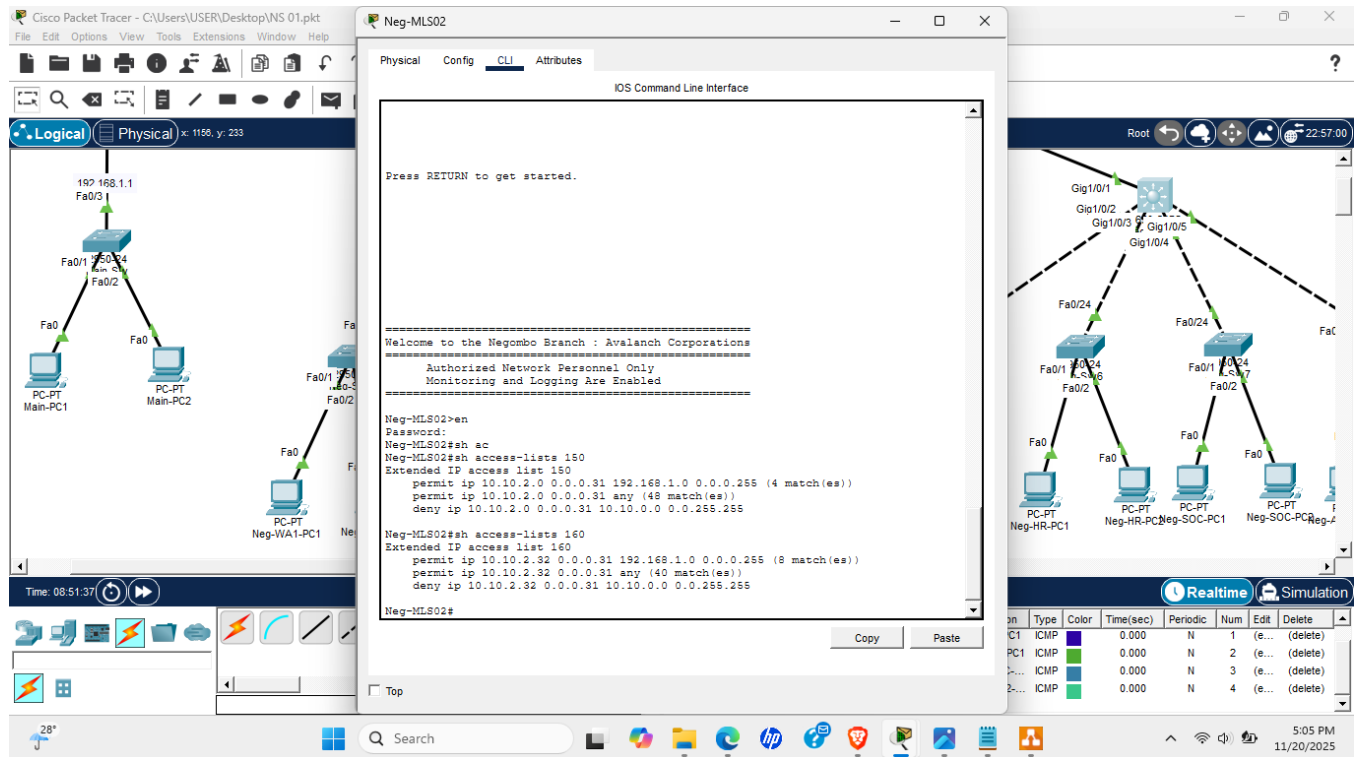


Figure 63: ACL for Finance

ACL 150 - Financial Department Restrictions:

- Line 1: Finance (10.10.2.0/27) to Main Branch (192.168.1.0/24) access to servers.
- Line 2: Allows access to the internet (to any destination which has not been matched previously)
- Line 3: Acknowledges that it denies all communication to all New Branch networks (10.10.0.0/16).
- Implicit Deny: Blocks all the remaining traffic (good practice to record denials) [50].

7.1.2.2. HR Department ACL (VLAN 60)

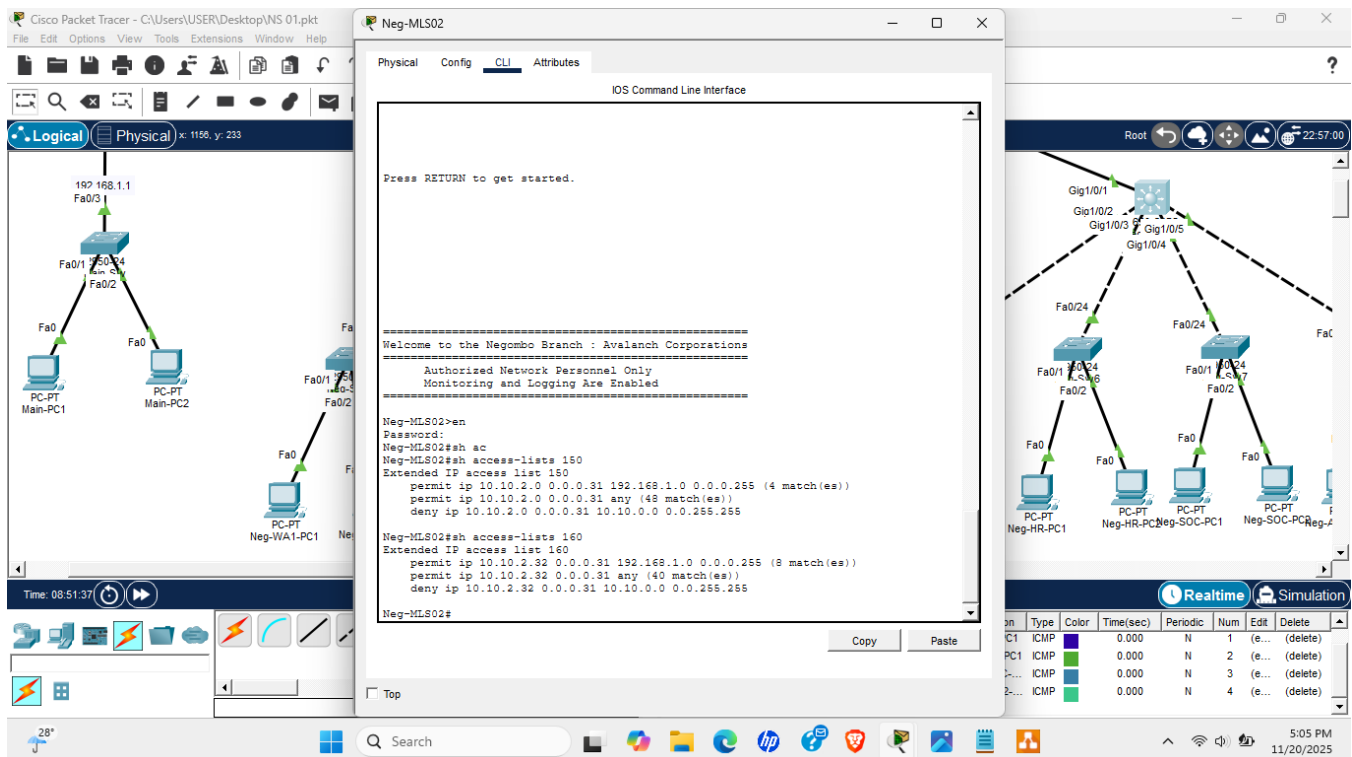


Figure 64: ACL for HR

7.1.3. ACL for Other Departments

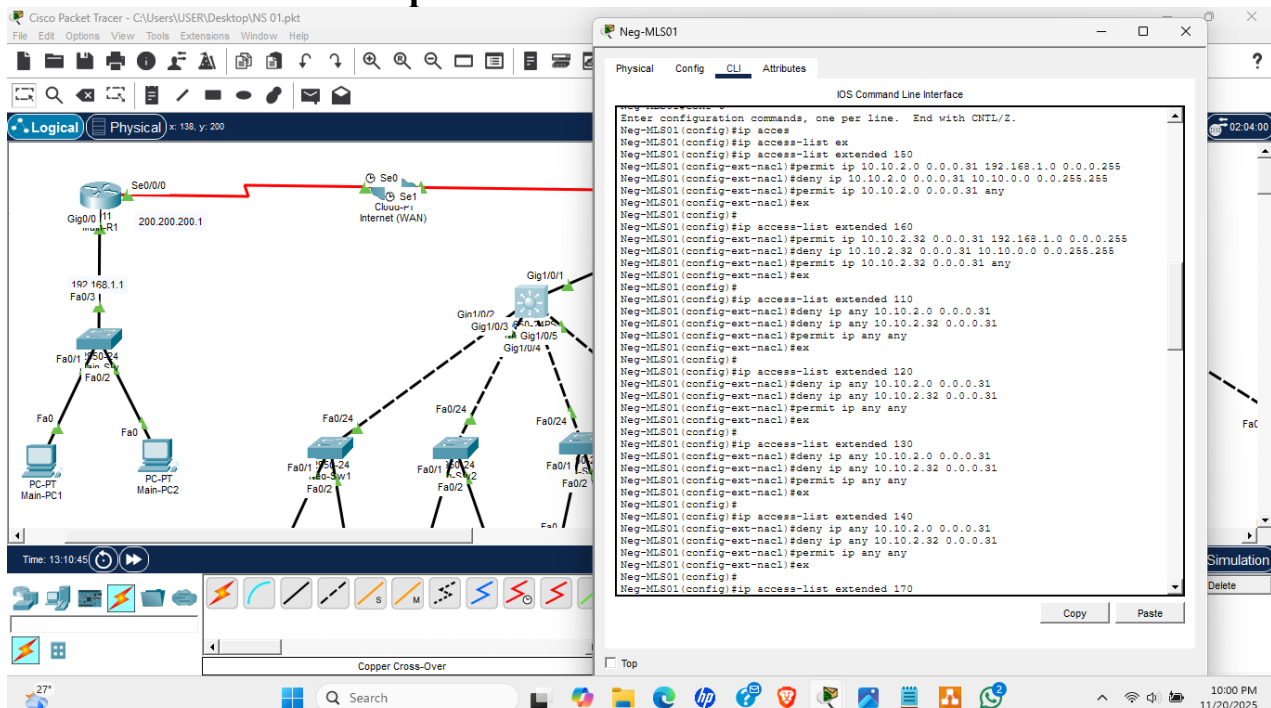


Figure 65: ACL in Neg-MLS01

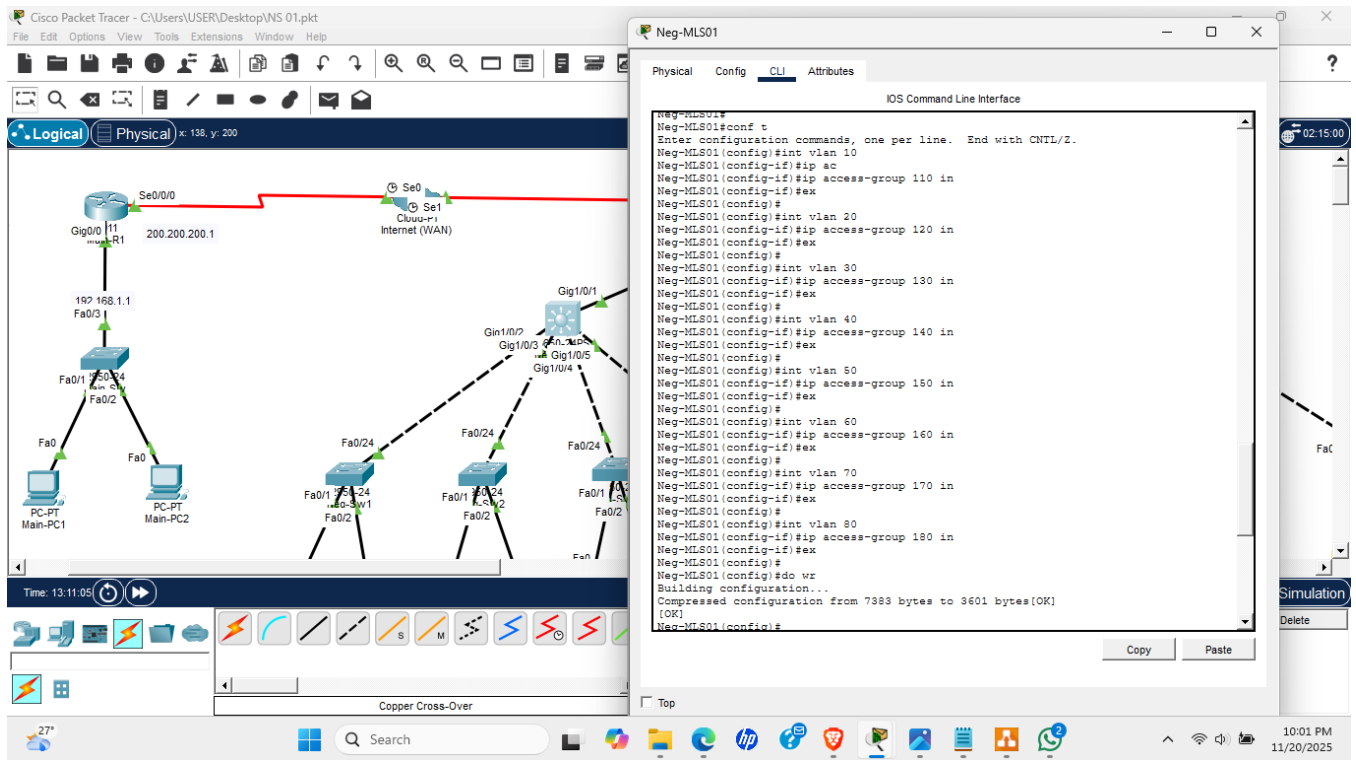


Figure 66: ACL in Neg-MLS01 cont.

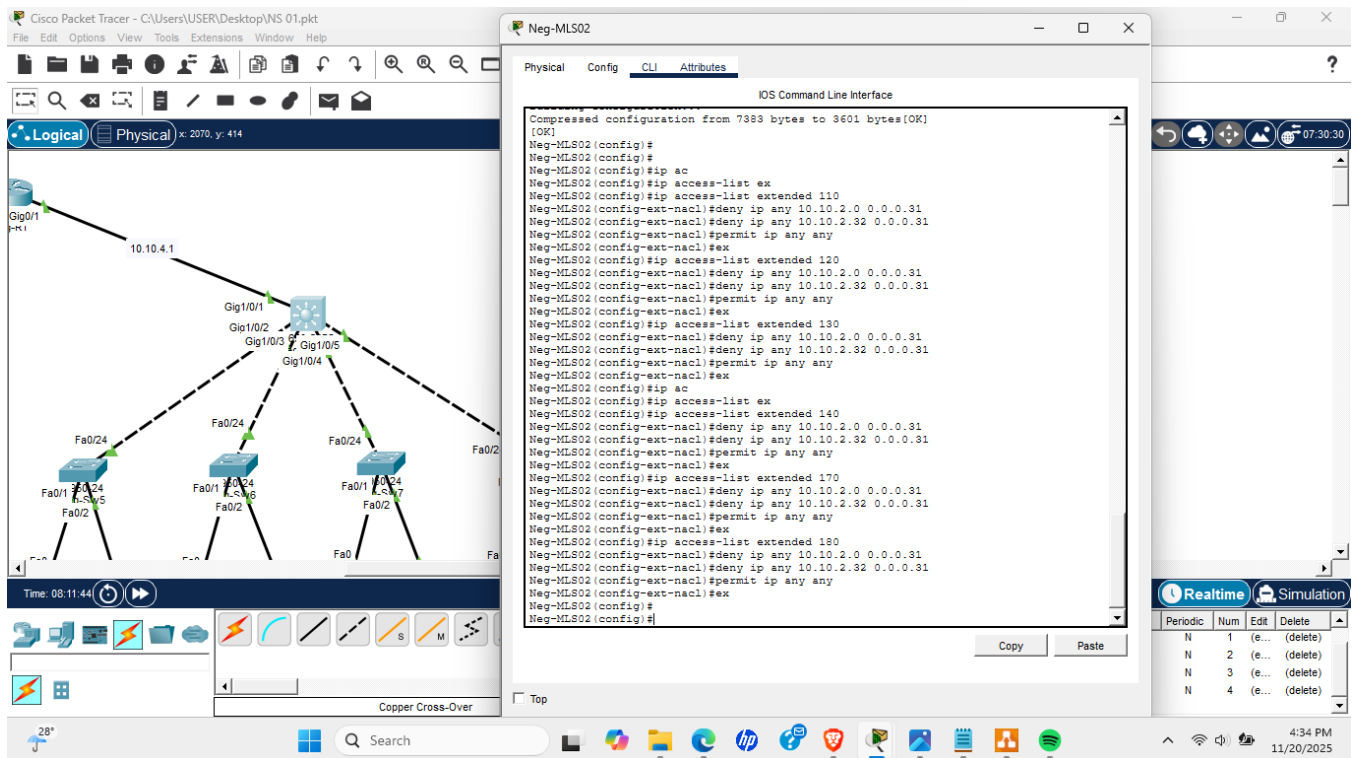
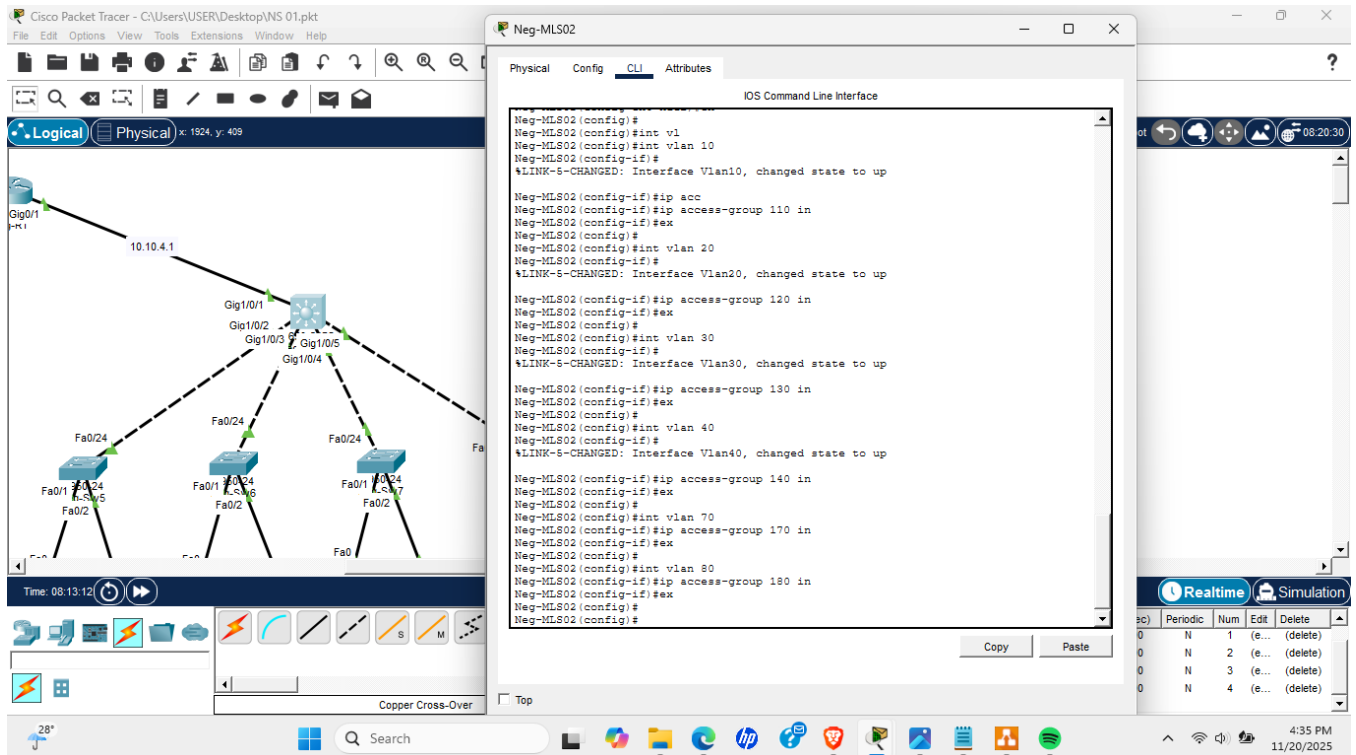
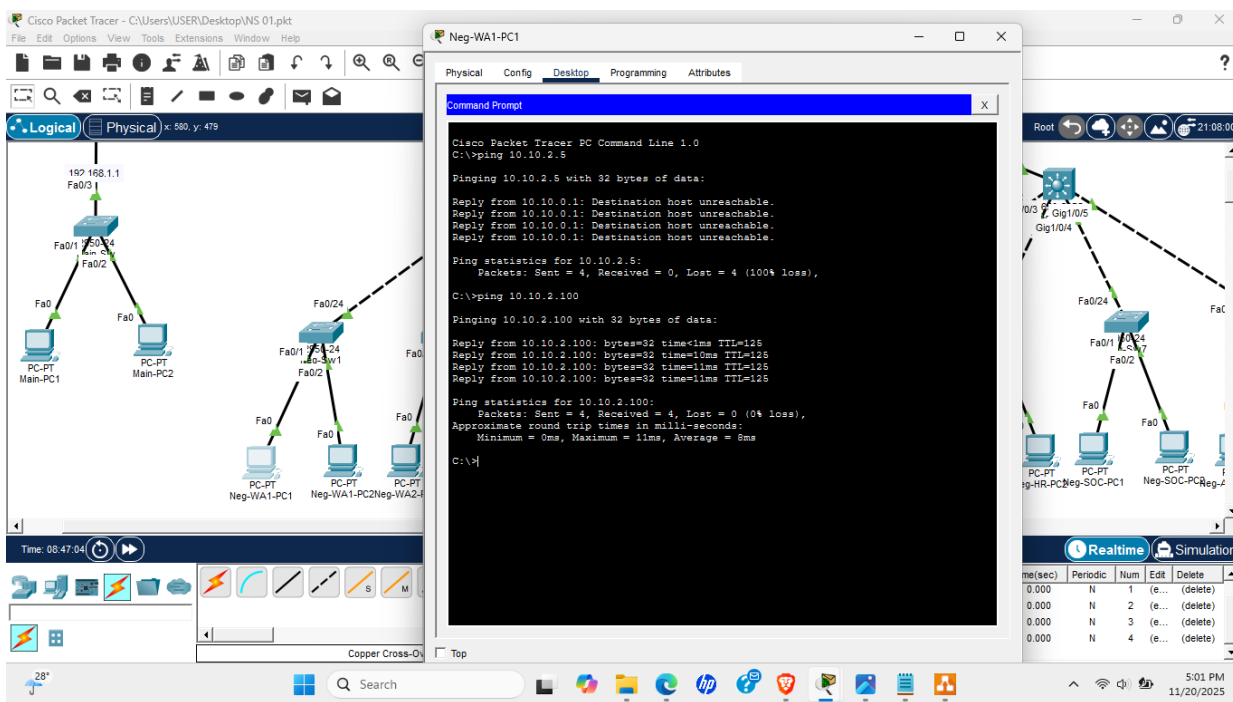


Figure 67: ACL in Neg-MLS02



7.1.4. Verification and Testing



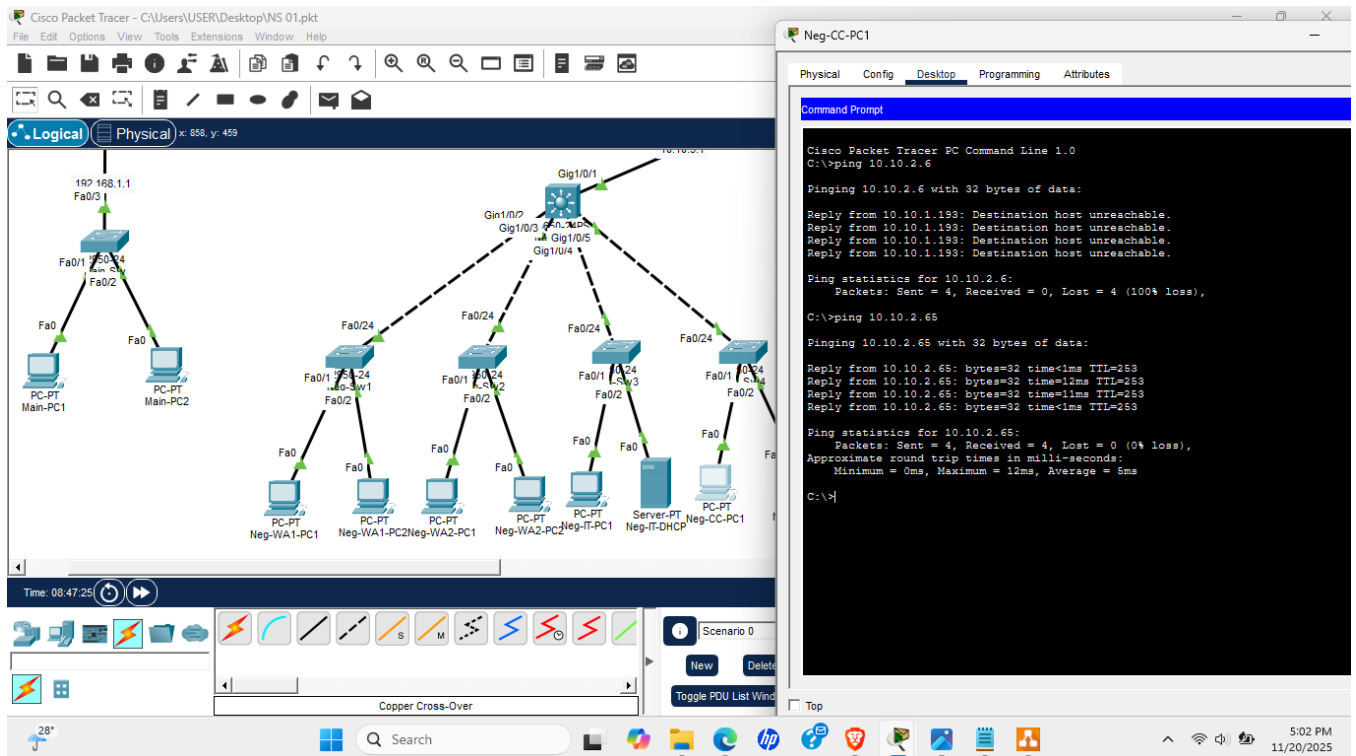


Figure 70: ACL verified in Neg-IT-PC

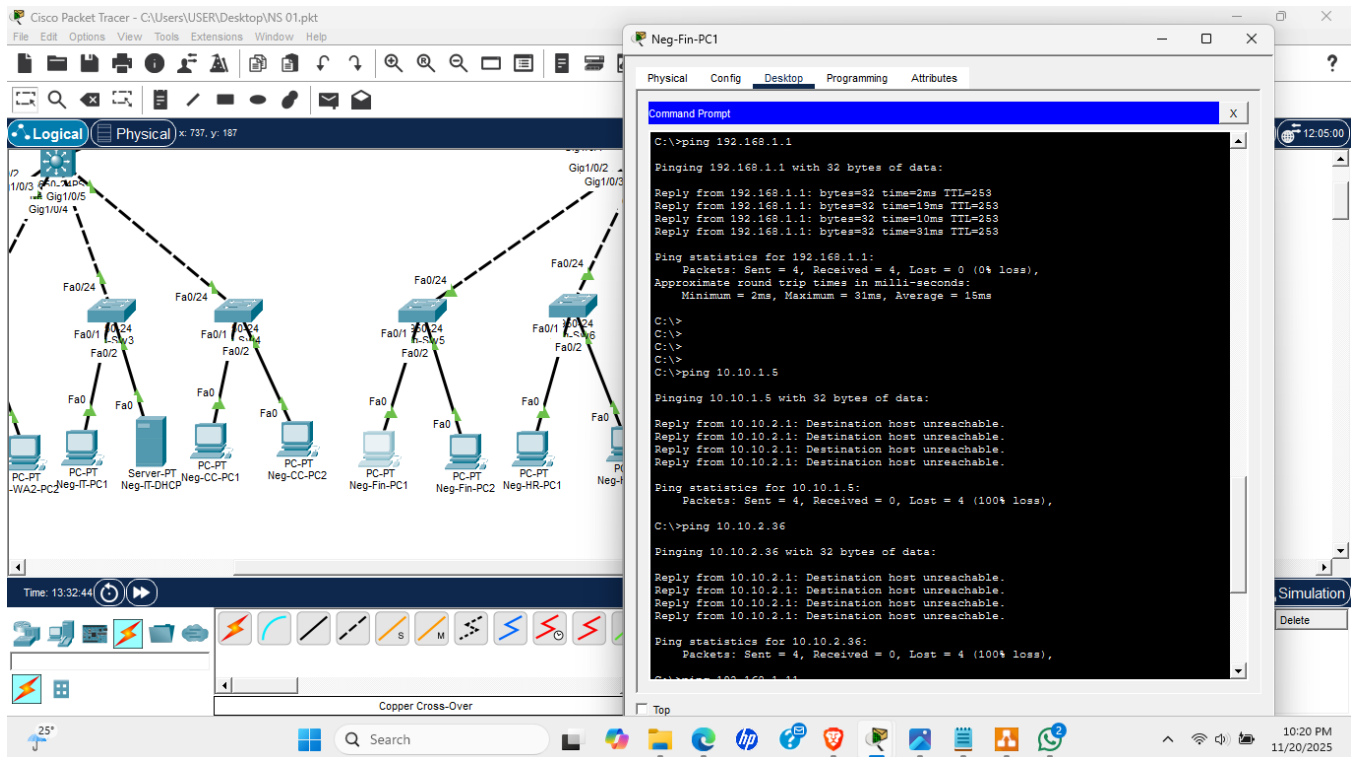


Figure 71: ACL verified in Neg-Fin-PC

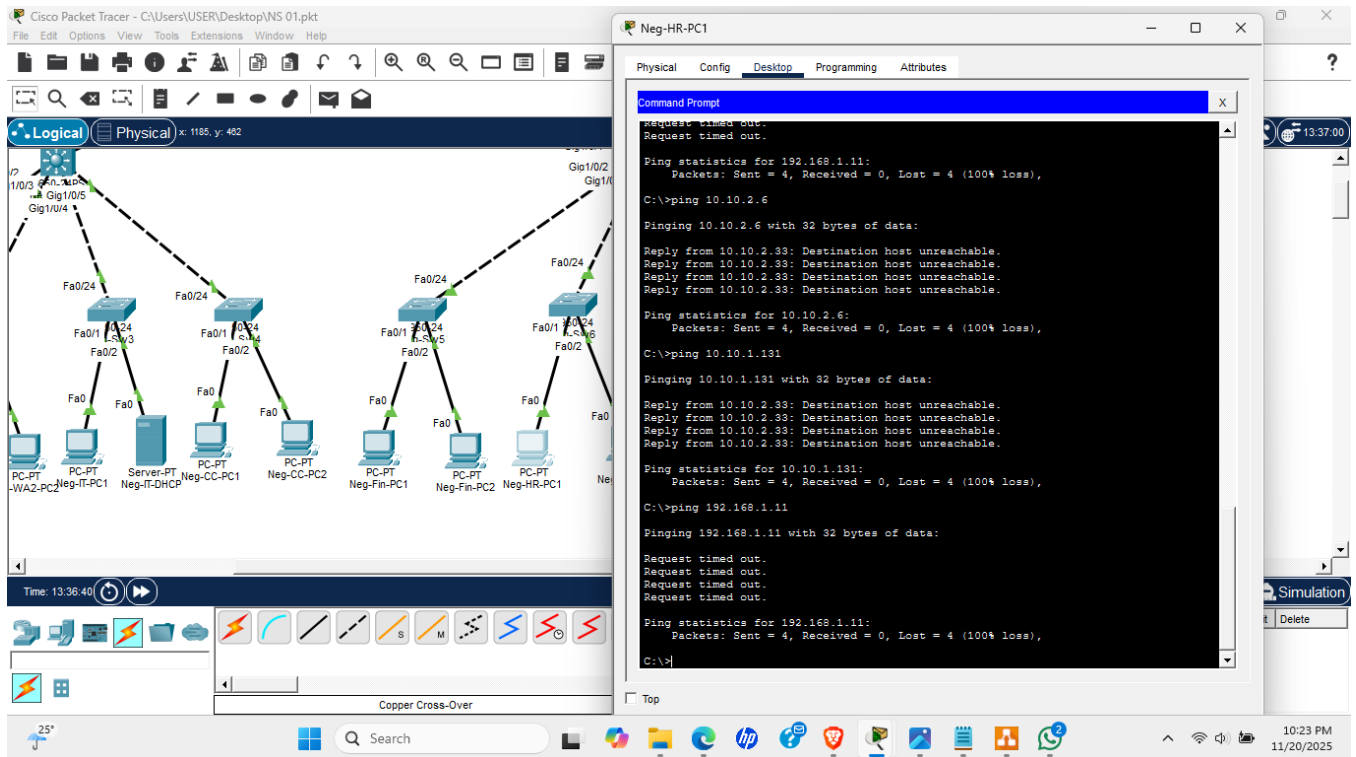


Figure 72: ACL verified in Neg-HR-PC

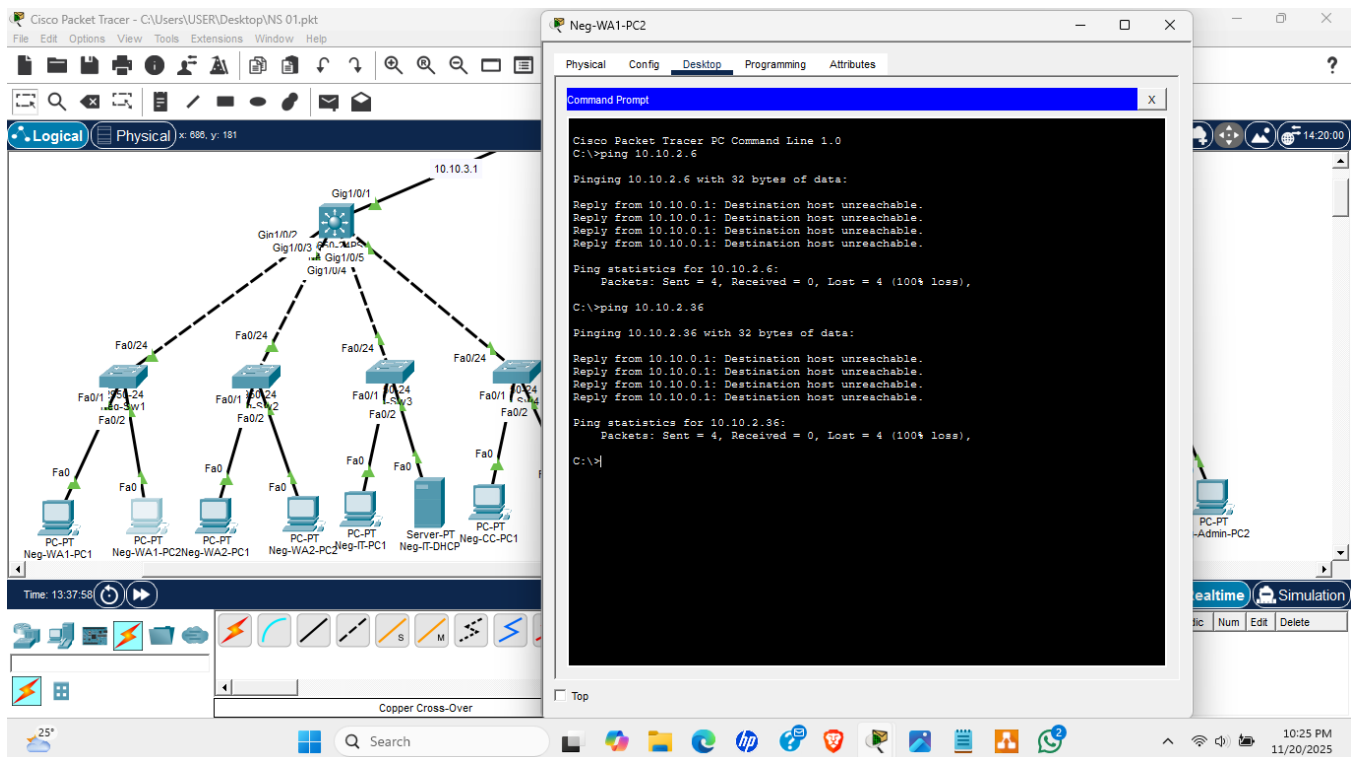


Figure 73: ACL verified in Neg-WA1-PC2

7.2. VPN Configuration

7.2.1. VPN Technology Selection

Virtual Private Network (VPN) technology provides secure and encrypted tunnels over the public networks, which guarantee privacy and integrity of information transmission [52].

VPN Type Selected : Site-to-Site IPSec VPN

- Permanent Connectivity: There is always-on connection between branches (as opposed to remote access VPN).
- Open to End-users: No customer software on terminal devices.
- High Performance: Hardware encryption: routers.
- Standards-Based: The IPSec (RFC 4301) is interoperable.
- Extensive Security: Authentication, encryption and integrity are provided [53].

7.2.2. IPSec Architecture

IPSec Framework Components:

1. Internet Key Exchange(IKE): Negotiates security associations and keys.
2. Encapsulating Security Payload(ESP): The service offers encryption and authentication.
3. Authentication Header(AH): Gives authentication but not utilized in this implementation.

Two-Phase Operation:

1. Phase 1 (IKE Phase 1):
 - Creates peer-peer secure and authenticated channel.
 - Bargains encryption and hashing functions.
 - Asymmetric key communication between the parties with Diffie-Hellman.
 - Establishes ISKMP Security Association (SA).
2. Phase 2 (IKE Phase 2):
 - IPSec Security Associations are negotiated.
 - Traffic is defined as encrypted (interesting traffic).
 - Defines the parameters of encryption to use in transmitting data [54].

Security Rationale

1. AES-256: has 2^{256} keys, which is computationally infeasible to crack [57].
2. SHA-256: Collision-resistant hash message-digest algorithm.
3. DH Group 14: Is strong enough against contemporary attacks.
4. Perfect Forward Secrecy(PFS): Key compromise does not impact on the past/future sessions [58].

7.2.3. VPN Interesting Traffic

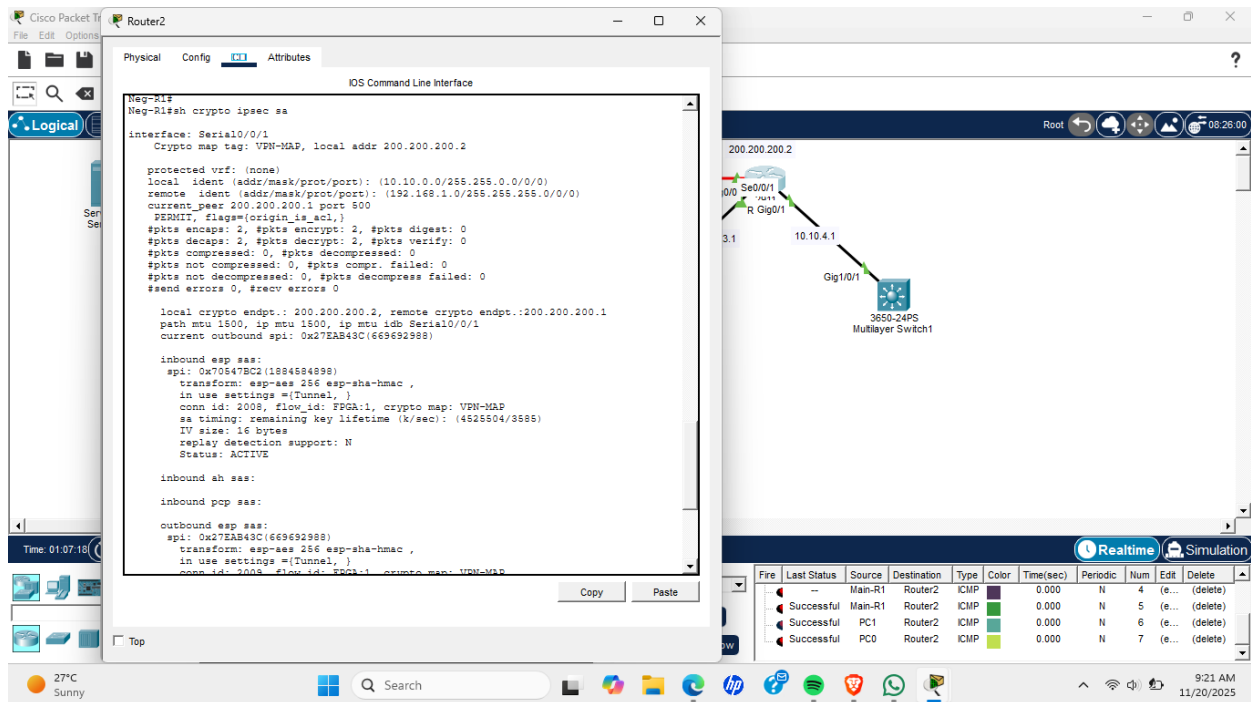


Figure 74: VPN traffic generate

Logic:

- There is traffic encryption between Main Branch (192.168.1.0/24) and New Branch (10.10.0.0/16).
- Traffic between the Internet and VPN does not pass through VPN (split tunneling is not enabled in the security)
- All inter-branch communication secured [59].

The screenshot displays a Cisco Packet Tracer simulation environment. The main window is titled "Router2" and shows the "Config" tab with the "IOS Command Line Interface" (CLI) active. The CLI shows a configuration for a router named "Neg-R1" (labeled as "Neg-R1" in the prompt, though the text says "Neg-R1" in the prompt and "Neg-R1" in the text). The configuration includes setting the hostname to "Neg-R1", enabling password encryption, and configuring ISAKMP and IPsec parameters for a site-to-site VPN. The configuration is as follows:

```

Neg-R1>en
Neg-R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Neg-R1(config)#scr
Neg-R1(config)#crypto is
Neg-R1(config)#crypto isakmp p
Neg-R1(config)#crypto isakmp policy 10
Neg-R1(config-isakmp)#en
Neg-R1(config-isakmp)#encryption a
Neg-R1(config-isakmp)#encryption aes 256
Neg-R1(config-isakmp)#has
Neg-R1(config-isakmp)#hash sha
Neg-R1(config-isakmp)#hash sha
Neg-R1(config-isakmp)#au
Neg-R1(config-isakmp)#authentication p
Neg-R1(config-isakmp)#authentication pre-share
Neg-R1(config-isakmp)#g
Neg-R1(config-isakmp)#group 1
Neg-R1(config-isakmp)#l
Neg-R1(config-isakmp)#lifetime 86400
Neg-R1(config-isakmp)#ex
Neg-R1(config)#scr
Neg-R1(config)#crypto i
Neg-R1(config)#crypto is
Neg-R1(config)#crypto isakmp k
Neg-R1(config)#crypto isakmp key AVALANCH address 200.200.200.1
Neg-R1(config)#scr
Neg-R1(config)#crypto is

```

Below the CLI window, there is a "Copy" and "Paste" button. The bottom status bar shows the time as 01:07:00 and the temperature as 27°C Sunny.

On the right side, there is a network diagram showing a connection between a router (labeled "R Gig0/1") and a "3650-24PS Multilayer Switch1". The router's Gigabit Ethernet 0/1 interface is connected to the switch's Gigabit Ethernet 0/1 interface. The IP address 200.200.200.2 is assigned to the router's interface. The switch's interface is labeled "Gig1/0/1".

At the bottom right, there is a "Realtime" and "Simulation" tab. The "Realtime" tab is active, showing a table of network events:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Success	Successful	Main-R1	Router2	ICMP	Green	0.000	N	4	(e...)	(delete)
Success	Successful	PC1	Router2	ICMP	Green	0.000	N	5	(e...)	(delete)
Success	Successful	PC0	Router2	ICMP	Green	0.000	N	6	(e...)	(delete)
Success	Successful	PC0	Router2	ICMP	Green	0.000	N	7	(e...)	(delete)

The bottom status bar shows the time as 9:21 AM and the date as 11/20/2025.

The image shows a Cisco Packet Tracer simulation environment. The primary window is the 'Router2' configuration interface, specifically the 'CLI' (Command Line Interface) tab. The configuration text is as follows:

```
Router2
Neg-R1(config)#crypto ipsec key
Neg-R1(config)#crypto ipsec tr
Neg-R1(config)#crypto ipsec transform-set AVALANCH-SET es
Neg-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-a
Neg-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 esp-s
Neg-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 esp-sha-hmac
Neg-R1(config)#
Neg-R1(config)#ac
Neg-R1(config)#access-list 101 per
Neg-R1(config)#access-list 101 permit ip 10.10.0.0 0.0.255.255 192.168.1.0 0.0.0.255
Neg-R1(config)#
Neg-R1(config)#cr
Neg-R1(config)#crypto m
Neg-R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Neg-R1(config-crypto-map)#set pe
Neg-R1(config-crypto-map)#set peer 200.200.200.1
Neg-R1(config-crypto-map)#set
Neg-R1(config-crypto-map)#set tr
Neg-R1(config-crypto-map)#set transform-set AVALANCH-SET
Neg-R1(config-crypto-map)#set
Neg-R1(config-crypto-map)#set pfs gr
Neg-R1(config-crypto-map)#set pfs grou
Neg-R1(config-crypto-map)#set pfs group1
Neg-R1(config-crypto-map)#set
Neg-R1(config-crypto-map)#match ad
Neg-R1(config-crypto-map)#match address 101
Neg-R1(config-crypto-map)#ex
Neg-R1(config)#
Neg-R1(config)#int se0/0/1
Neg-R1(config-if)#cr
Neg-R1(config-if)#crypto m
Neg-R1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Neg-R1(config-if)#ex
Neg-R1(config)#do wr
Building configuration...
[OK]
```

The secondary window shows a network diagram with two routers, 'R' and '3650-24PS Multilayer Switch1', connected via their GigabitEthernet interfaces. The status bar at the bottom indicates the simulation is running, with a time of 01:07:05 and a date of 11/20/2025.

Figure 77: VPN Config Main R1

```

Main-R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
Main-R1#
Main-R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Main-R1(config)#crypto is
Main-R1(config)#crypto isakmp polic
Main-R1(config)#crypto isakmp policy 10
Main-R1(config-isakmp)#encr
Main-R1(config-isakmp)#encryption aes 256
Main-R1(config-isakmp)#hash sha
Main-R1(config-isakmp)#auth
Main-R1(config-isakmp)#authentication pre
Main-R1(config-isakmp)#authentication pre-share
Main-R1(config-isakmp)#group 1
Main-R1(config-isakmp)#lifet
Main-R1(config-isakmp)#lifetime 86400
Main-R1(config-isakmp)#ex
Main-R1(config)#
Main-R1(config)#crypto is
Main-R1(config)#crypto isakmp key AVALANCH address 200.200.200.1
Main-R1(config)#
Main-R1(config)#crypto ip
Main-R1(config)#crypto ipsec tr
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET es
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-a
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 ?
esp-md5-hmac ESP transform using HMAC-MD5 auth
esp-sha-hmac ESP transform using HMAC-SHA auth
<cr>
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 esp-s
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 esp-sha-hmac ?
<cr>
Main-R1(config)#crypto ipsec transform-set AVALANCH-SET esp-aes 256 esp-sha-hmac
  
```

Figure 77: VPN Config Main R1

Figure 78: VPN Config Main R1 cont

```

Main-R1(config)#ac
Main-R1(config)#access-list 101 permi
Main-R1(config)#crypto isakmp key AVALANCH address 200.200.200.2
Main-R1(config)#
Main-R1(config)#ac
Main-R1(config)#access-list 101 per
Main-R1(config)#access-list 101 permit i
Main-R1(config)#access-list 101 permit ip
Main-R1(config)#access-list 101 permit ip 192.168.1.0 0.0.0.255 10.10.0.0 0.0.255.255
Main-R1(config)#cr
Main-R1(config)#crypto ma
Main-R1(config)#crypto map VPN-MAP 10 i
Main-R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
Main-R1(config-crypto-map)#set
Main-R1(config-crypto-map)#set
Main-R1(config-crypto-map)#set p
Main-R1(config-crypto-map)#set pe
Main-R1(config-crypto-map)#set peer 200.200.200.2
Main-R1(config-crypto-map)#set
Main-R1(config-crypto-map)#set t
Main-R1(config-crypto-map)#set transform-set AVALANCH-SET
Main-R1(config-crypto-map)#set pfs gr
Main-R1(config-crypto-map)#set pfs group1
Main-R1(config-crypto-map)#match ad
Main-R1(config-crypto-map)#match address 101
Main-R1(config-crypto-map)#ex
Main-R1(config)#
Main-R1(config)#
Main-R1(config)#int se0/0/0
Main-R1(config-if)#scr
Main-R1(config-if)#crypto m
Main-R1(config-if)#crypto map VPN-MAP
*Jan 3 07:16:26.788: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
Main-R1(config-if)#ex
Main-R1(config)#do wr
Building configuration...
[OK]
  
```

Figure 78: VPN Config Main R1 cont

7.2.5. VPN Tunnel Establishment

Tunnel Initiation Process

1. **Interesting Traffic:** Active PC in New Branch pings Main Branch server.
2. **Phase 1 Negotiation of IKE:** Routers create secure management connection.
3. **Authentication:** Identity of peers is verified by an authentication key.
4. **Key Exchange:** Diffie-Hellman is used to create encryption keys.
5. **IKE Phase 2 Negotiation:** IPSec parameters accepted.
6. **Tunnel Built:** The encryption of data commences.
7. **Data Transmission:** Tunneled encrypted [60] All matching traffic.

7.2.6. VPN Testing and Verification

The screenshot displays the Cisco Packet Tracer interface. The left pane shows the configuration for 'Neg-R1' in the 'Config' tab. The configuration includes the following commands:

```

in use settings =(Tunnel, )
conn id: 2008, flow_id: FPGA:1, crypto map: VPN-MAP
sa timing: remaining key lifetime (k/sec): (4525504/3585)
IV size: 16 bytes
replay detection support: N
Status: ACTIVE

inbound ah sas:
inbound pc sas:
outbound esp sas:
spi: 0x27EAB43C(659692988)
transform: esp-aes 256 esp-sha-hmac ,
in use settings =(Tunnel, )
conn id: 2009, flow_id: FPGA:1, crypto map: VPN-MAP
sa timing: remaining key lifetime (k/sec): (4525504/3585)
IV size: 16 bytes
replay detection support: N
Status: ACTIVE
outbound ah sas:
outbound pc sas:

Neg-R1#
Neg-R1#
Neg-R1#sh crypto isakmp sa
IPv6 Crypto ISAKMP SA
dst      src      state      conn-id slot status
200.200.200.1 200.200.200.2 QM_IDLE    1095    0 ACTIVE

IPv6 Crypto ISAKMP SA
Neg-R1#Neg-R1#
Neg-R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Neg-R1(config)#
  
```

The right pane shows a network diagram with a router connected to a switch and a PC. The bottom pane shows a log of successful ping tests from the PC to the server:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	Successful	Main-R1	Router2	ICMP	0.000	N	4	(e...)	(delete)	
Successful	Successful	PC1	Router2	ICMP	0.000	N	5	(e...)	(delete)	
Successful	Successful	PC0	Router2	ICMP	0.000	N	6	(e...)	(delete)	

Figure 79: Crypto Map Result Neg-R1

Figure 80 shows the Cisco Packet Tracer interface for the configuration of a Crypto Map on Neg-R1. The CLI window displays the configuration for the Serial10/0/1 interface, including the VPN-MAP, local address 200.200.200.2, and the Crypto Map tag. The configuration includes inbound and outbound ESP and AH settings. The Realtime window shows a network diagram with a 3650-24PS Multilayer Switch connected to Neg-R1 via Gig1/0/1. The Realtime window also displays a table of traffic statistics.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Success	Successful	Main-R1	Router2	ICMP	0.000	N	4	(e...)	(delete)	
Success	Successful	PC1	Router2	ICMP	0.000	N	6	(e...)	(delete)	
Success	Successful	PC0	Router2	ICMP	0.000	N	7	(e...)	(delete)	

Figure 80: Crypto Map result Neg-R1 cont

Figure 81 shows the Cisco Packet Tracer interface for the configuration of a Crypto Map on Main-R1. The CLI window displays the configuration for the Serial10/0/0 interface, including the VPN-MAP, local address 200.200.200.1, and the Crypto Map tag. The configuration includes inbound and outbound ESP and AH settings. The Realtime window shows a network diagram with a Server-PT connected to Main-R1 via Gig0/0/0. The Realtime window also displays a table of traffic statistics.

Periodic	Num	Edit	Delete
N	4	(e...)	(delete)
N	5	(e...)	(delete)
N	6	(e...)	(delete)
N	7	(e...)	(delete)

Figure 81: Crypto Map Result Main-R1

7.2.7. VPN Security Benefits

1. **Confidentiality:** Any inter-branch traffic encrypted using AES-256, resulting passive eavesdropping and Man-in-the-middle attacks resisted [62].
2. **Integrity:** SHA-256 HMAC will ensure that data has not been altered in transmission through detecting modifications and rejecting.
3. **Authentication:** Pre-shared key makes sure that tunnels are made only by authorized routers which ensure blocking rogue device connections.
4. **Compliance:** Compliant with PCI DSS requirement 4.1 (encrypt sensitive data during transmission). and GDPR Art. 32 (security of processing) [63].

7.3. SSH Secure Access

Secure Shell (SSH) is the encrypted remote access network device access, which is used instead of the insecure Telnet protocol [64].

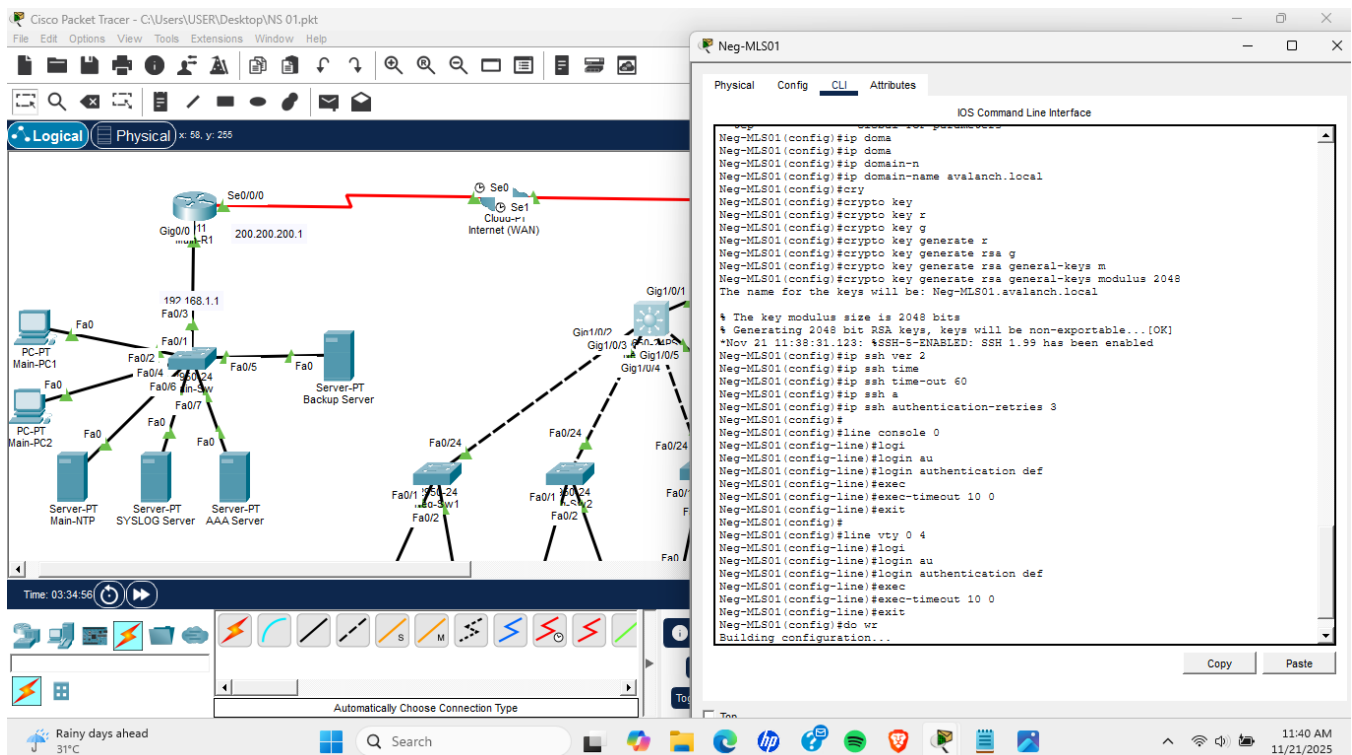


Figure 82: SSH in Neg-MLS01

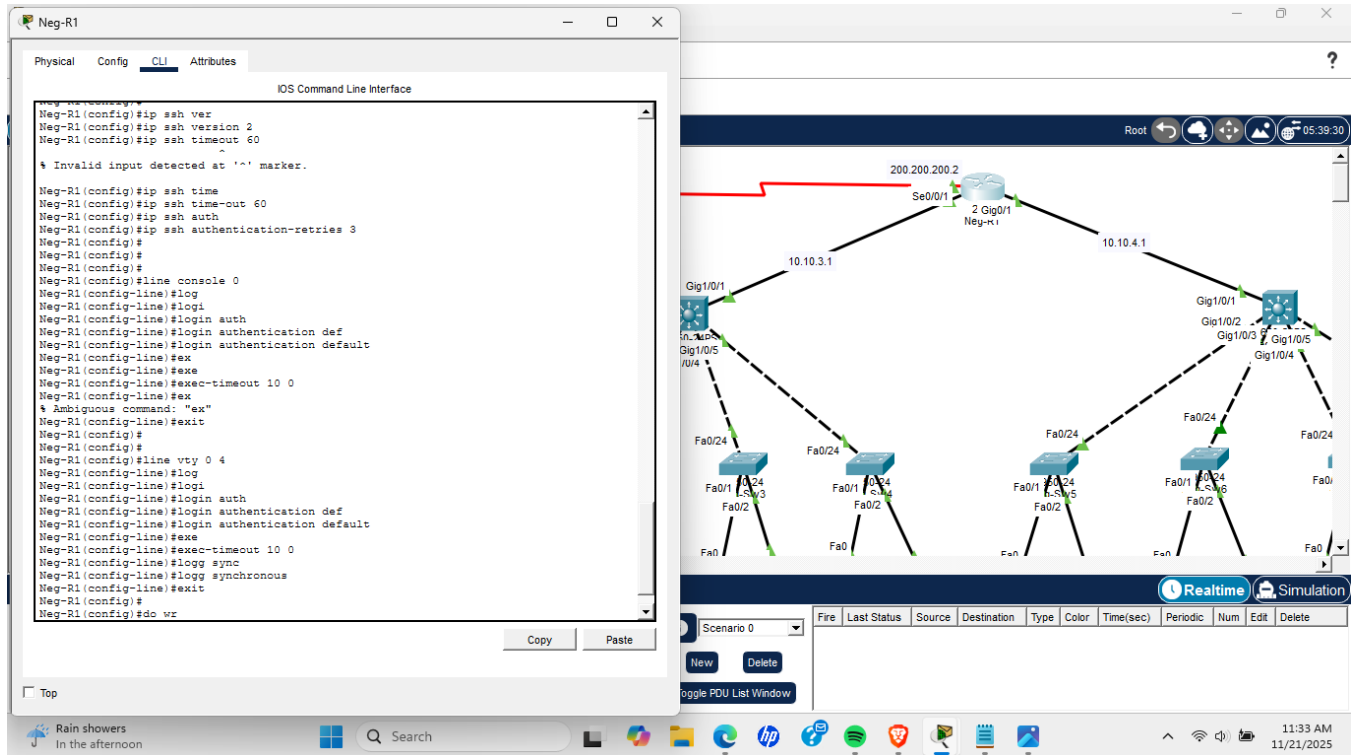


Figure 83: SSH in Neg-R1

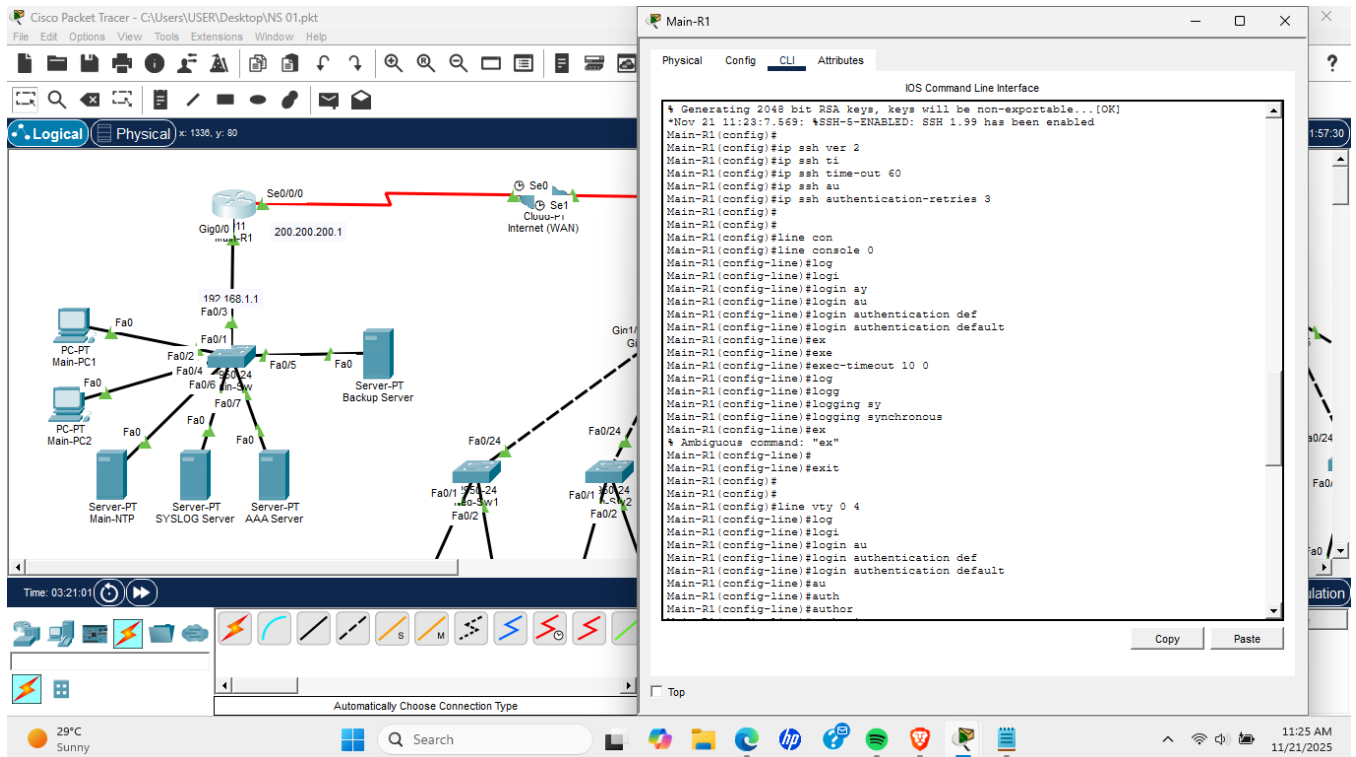


Figure 84: SSH in Main-R1

7.3.1. SSH Verification and testing

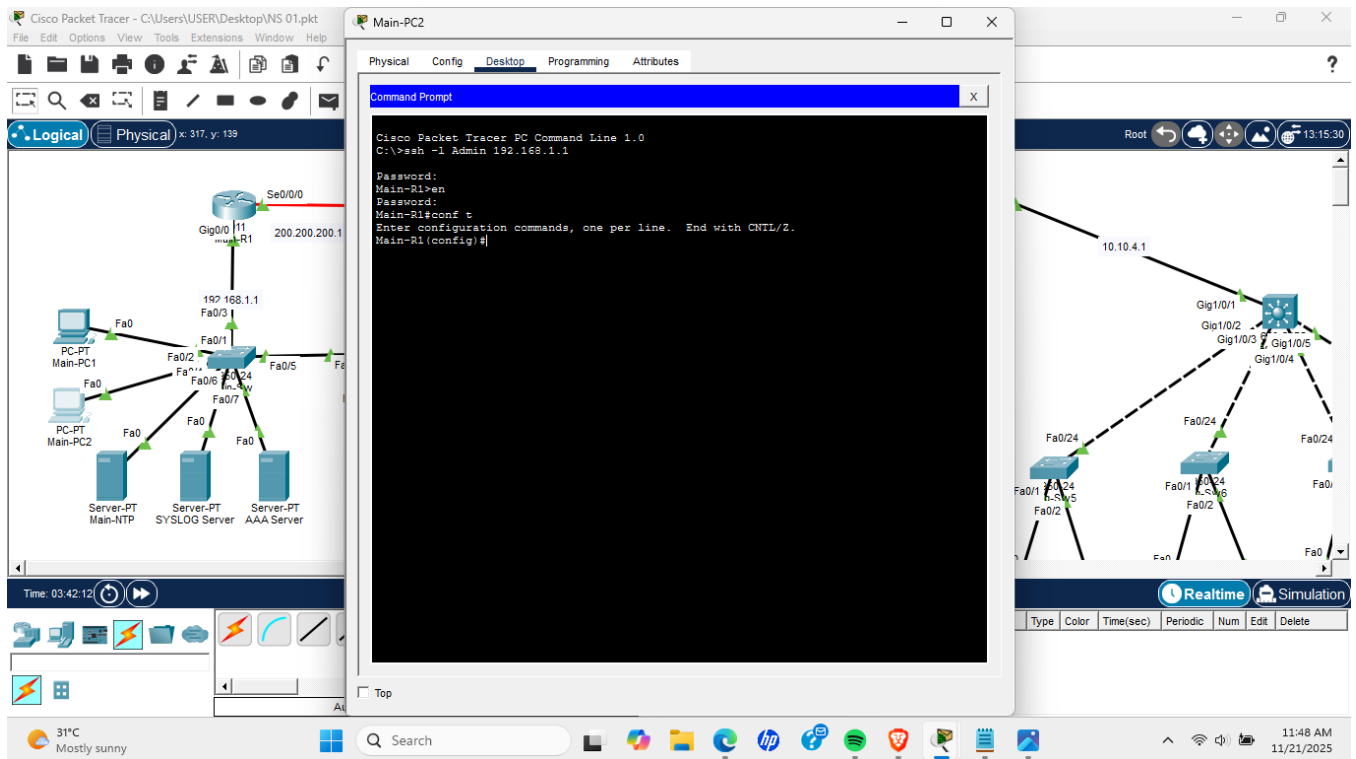


Figure 85: SSH Result Main-PC 1

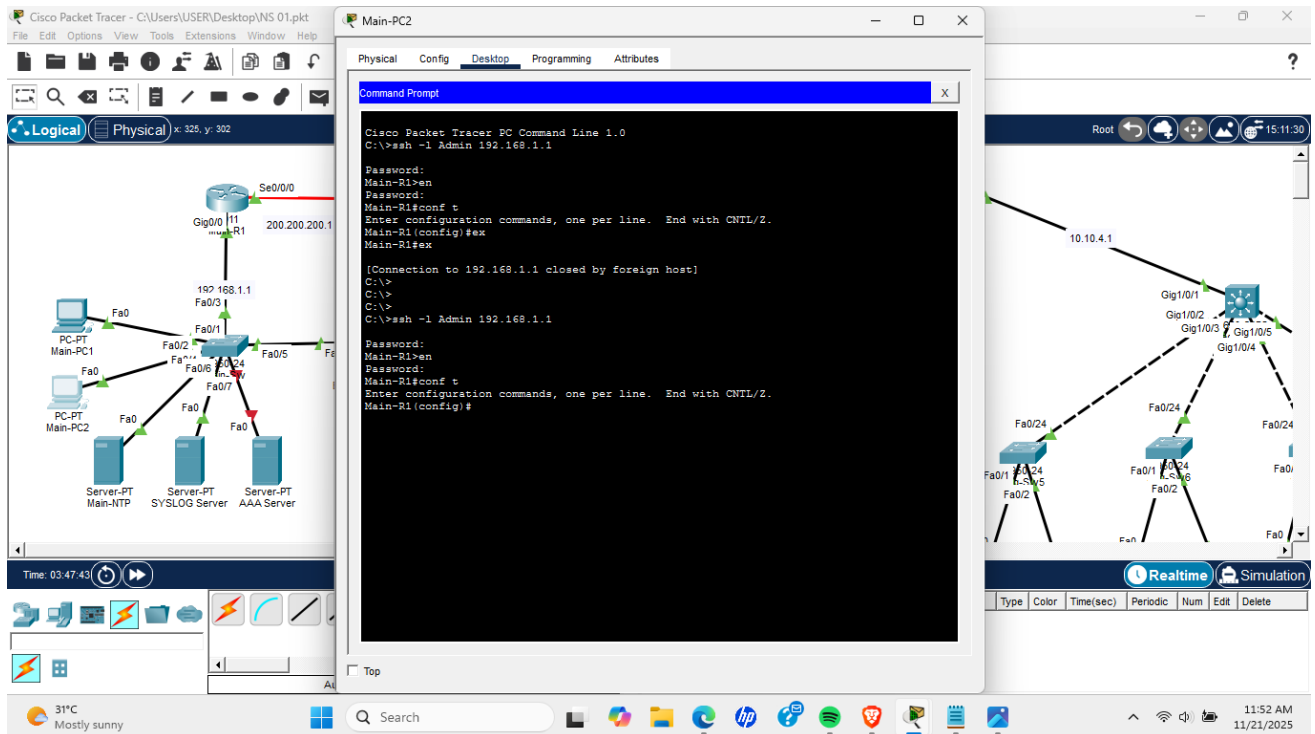


Figure 86: SSH result Main PC 1 cont

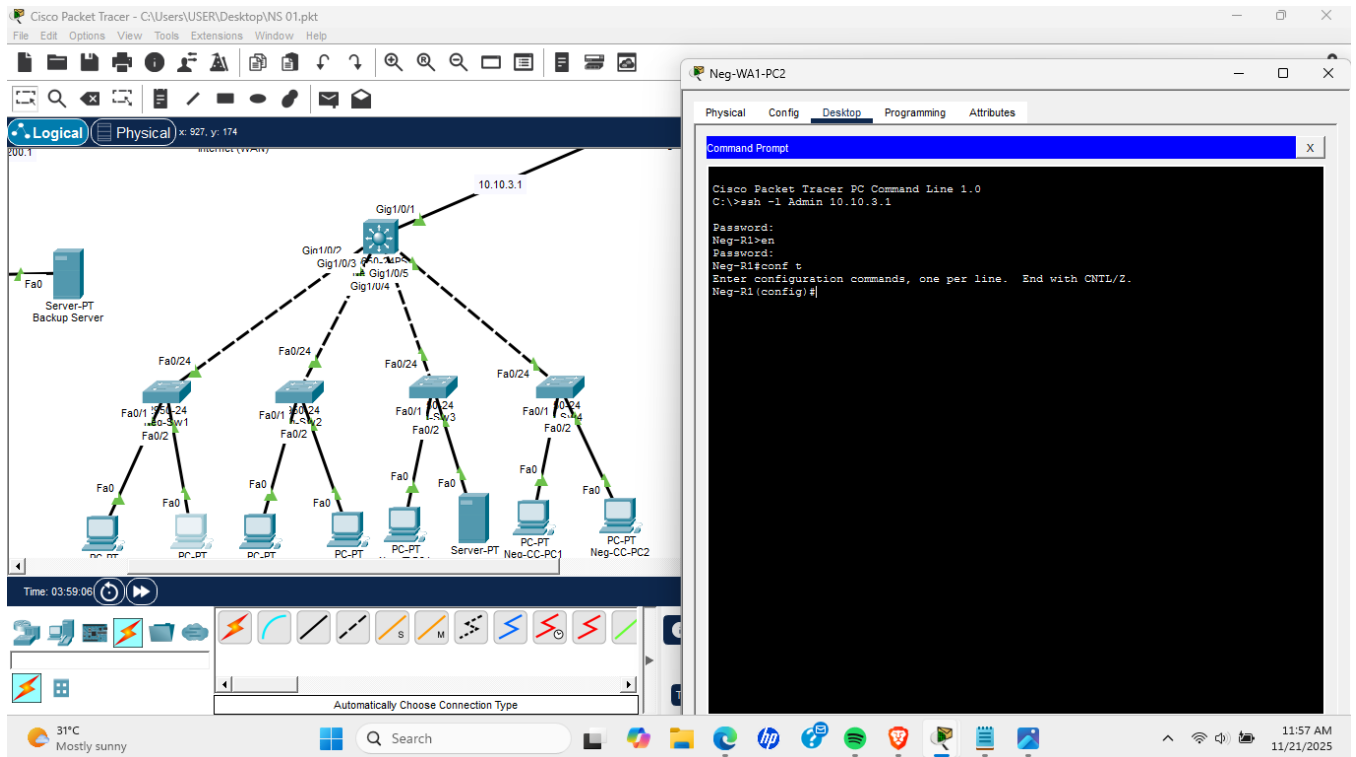


Figure 87: SSH Result in Neg-WA1 PC

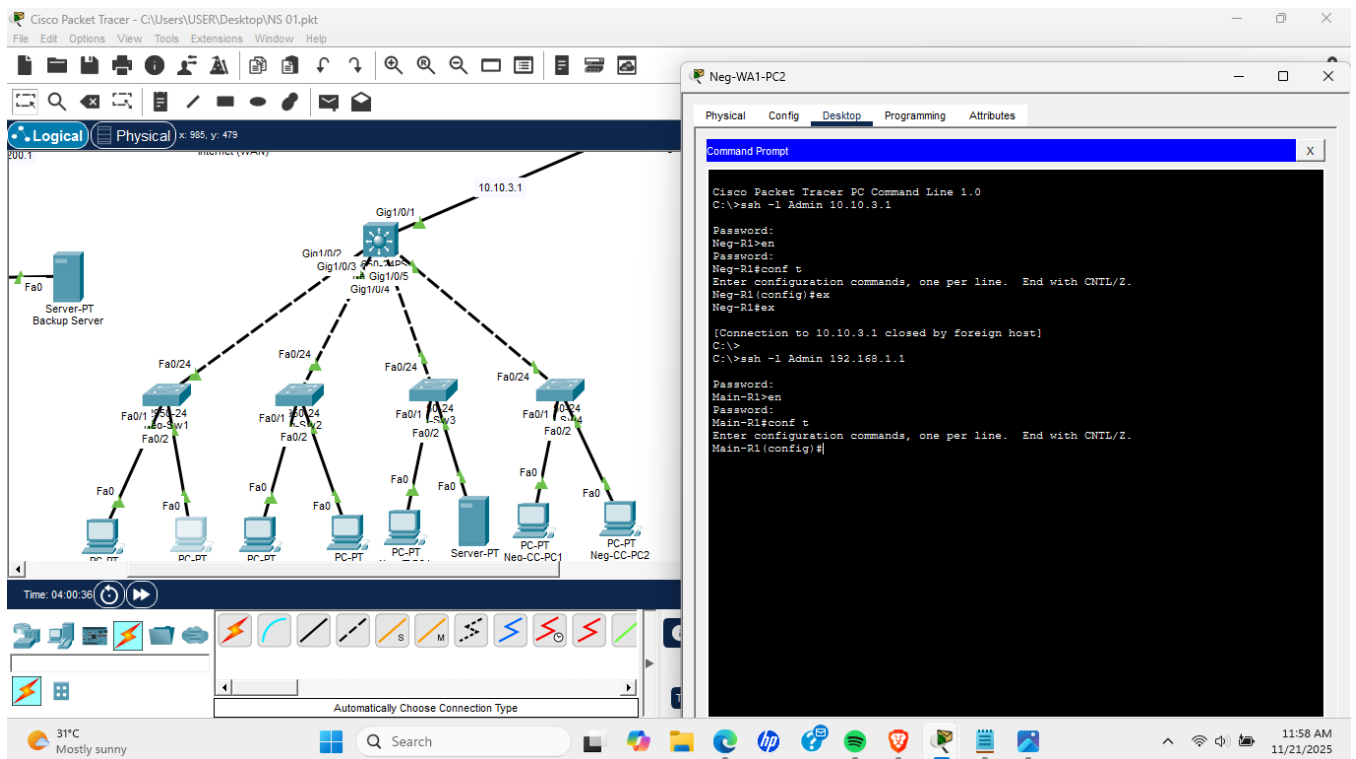


Figure 88: SSH Result in Neg-WA1 PC cont

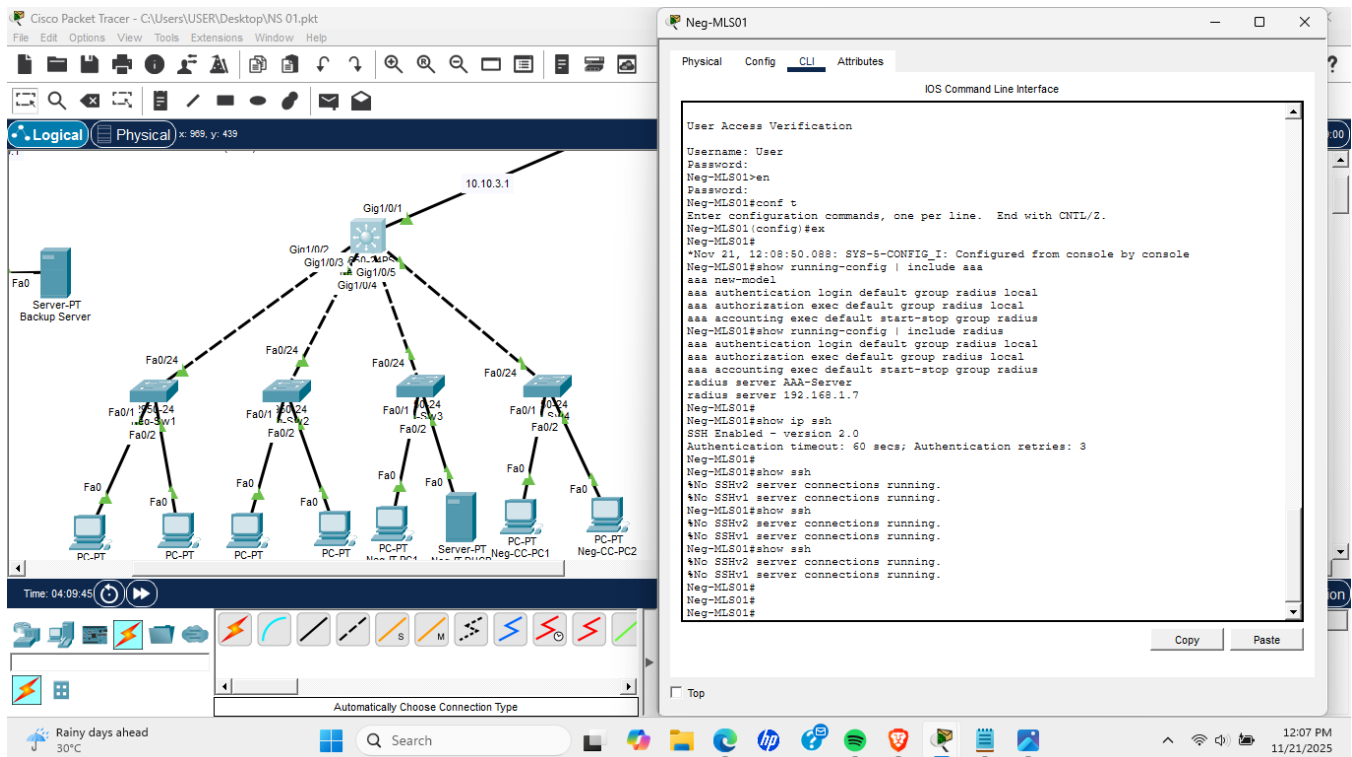


Figure 89: SSH Result in Neg-MLS01

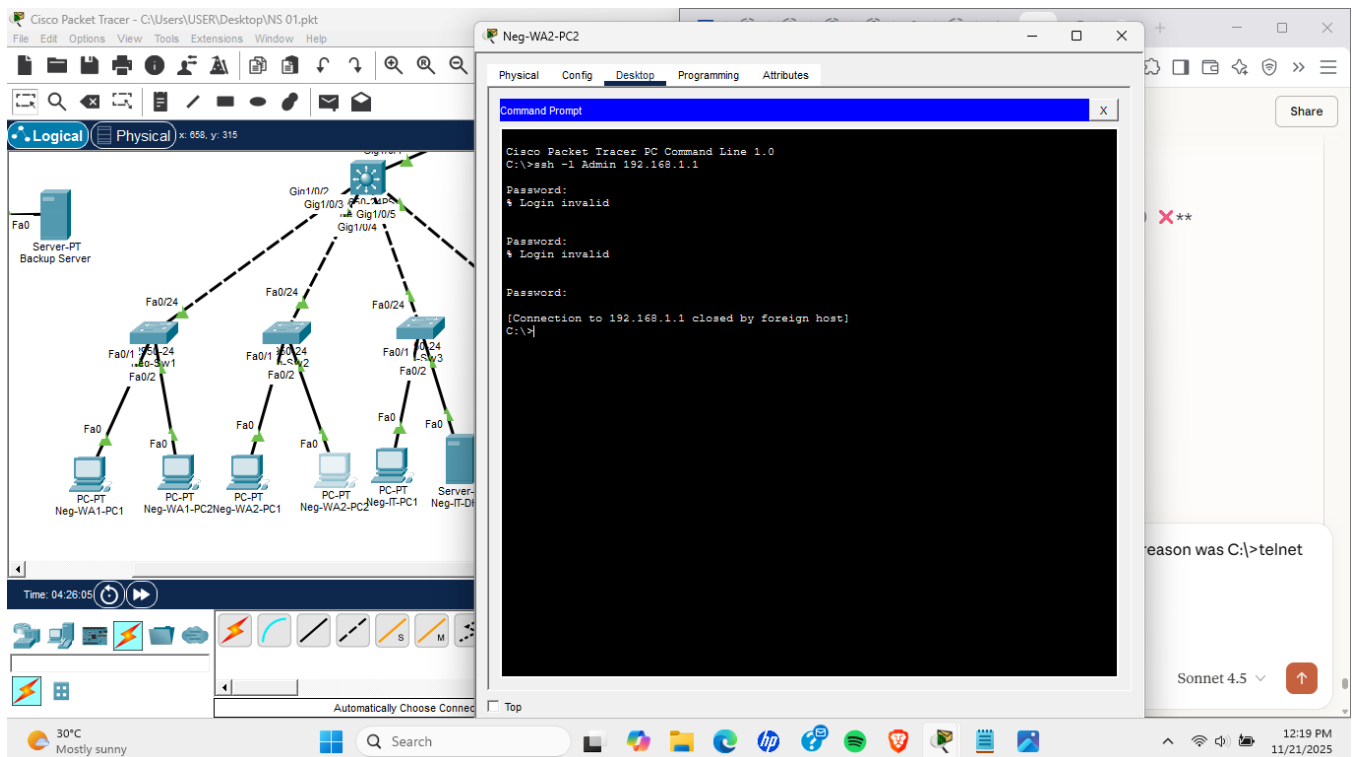


Figure 90: SSH Login attempts fail result

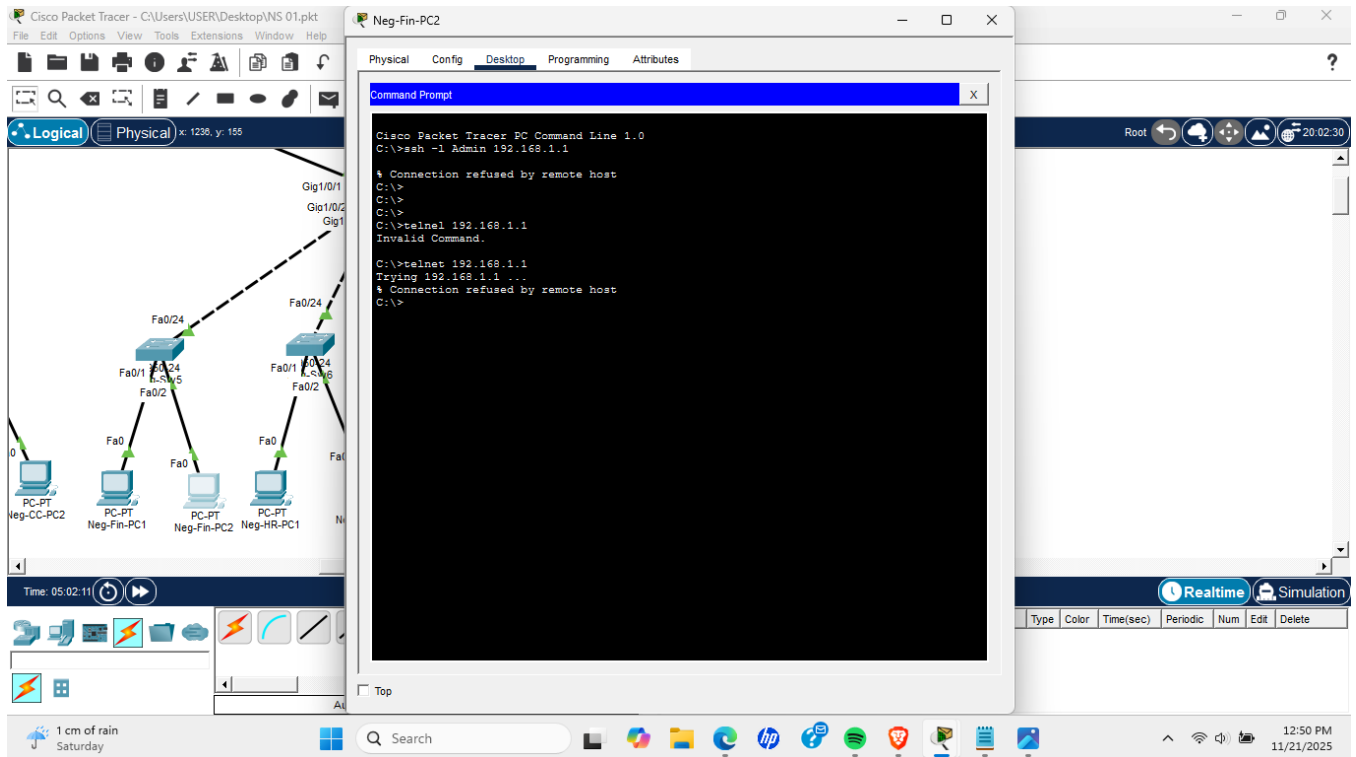


Figure 91: SSH results in Neg-Fin-PC

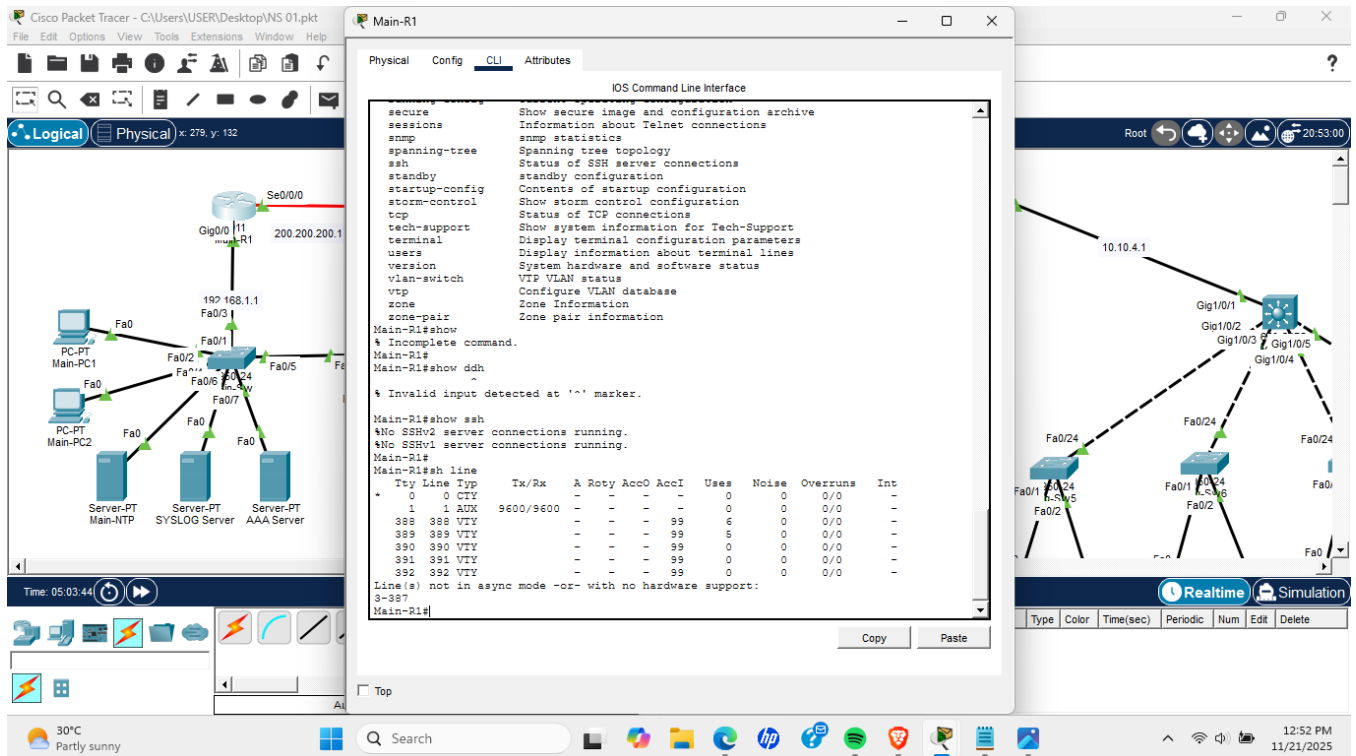


Figure 92: SSH line console result

8. Future Enhancements

8.1. Security Enhancements

1. **Next-Generation Firewall (NGFW):** Implement a state of the art firewall that has application based inspection and threat prevention capabilities at the network perimeter. NGFWs offer Layer 7 visibility, where they recognize the presence of threats within encrypted traffic that they circumvent the standard ACLs [50].
2. **Intrusion Detection/Prevention System (IDS/IPS):** Place sensors of IDS/IPS sensors in strategic locations to use them to block malicious traffic in real-time. Offers security to port scans, brute force attacks and zero day attacks [48].
3. **Network Access Control (NAC):** 802.1X. Avoid network access without performing device-level authentication. Instead of using a static port security, it uses dynamic authentication in which no unauthorized devices can be connected even when it is physically connected [71,101].
4. **Zero Trust Architecture:** Embrace the principle of never trust, always verify security model along with micro-segmentation and continuous authentication. Destroys the implicit trust and prevents horizontal movement by the attackers [106].

8.2. Infrastructure Enhancements

1. **Wireless Network (Wi-Fi 6):** Install enterprise-level wireless access points that would enable mobile connectivity using WPA3-Enterprise encryption and 802.1X authentication. Facilitates the flexibility of working environments and visitor access [102].

2. **SD-WAN: Implementation** Substitute stationary WAN with smart, application-conscious routing throughout various connections. Offers automatic failure and uptime of 99.99% [105].
3. **Cloud Integration:** Provide straight connection to AWS/Azure to hybrid cloud structure. Scalable infrastructure, disaster recovery and SaaS integration are enabled [107].

8.3. Monitoring and Management

1. **Network Analytics (NetFlow/SIEM):** Application of NetFlow export and SIEM correlation. Offers real time monitoring of the traffic, security incident, and capacity planning [103].
2. **Quality of Service (QoS):** Traffic prioritization should be implemented in order to provide uniform performance of voice, video, and critical applications under network congestion [27].
3. **Network Automation:** Use Ansible or Python to use infrastructure-as-code to manage configuration. Minimizes human errors and speedy deployments [98].
4. **IPv6 Dual-Stack:** Move to dual-stack operation IPv4 / IPv6 to future-proof network as IPv4 address space will become limited [97].

9. Conclusion

10.1. Project Summary

During this project, a full-scale secure network infrastructure of the new branch of the Avalanch Cooperations was drawn and put into practice in Negombo, Sri Lanka. The network consists of eight departments with 358 hosts connected to the main branch through an encrypted IPSec VPN tunnel and the various layers of security controls are used to guarantee confidentiality, integrity, and availability of organizational data.

Key Accomplishments

- 1. Hierarchical Network Design:** Three layer architecture (core, distribution, access) that is designed based on Cisco best practice, and that offers distinct separation of responsibilities and scalability [8].
- 2. Efficient IP Addressing:** VLSM-based addressing scheme had 56% address space efficiency and 78% growth capacity (280 more hosts) without redesign [12].
- 3. Secure Inter-Branch Connection:** Site-to-site IPSec VPN with AES-256 encryption secures all communication between branches, which has been shown to address the industry-wide requirements on data protection [53].
- 4. Network Segmentation:** Broadcast domain isolation: Eight departmental VLANs reduce attack surfaces and enhance network performance [21].
- 5. Centralized Services:** There are DHCP, NTP, Syslog, and AAA services in main branch to facilitate easy management and enforce uniform policies.
- 6. Security Implementation:** ACLs are able to isolate Finance (VLAN 50) and HR (VLAN 60) departments and still have the necessary connectivity to servers and internet [47].
- 7. High Availability:** Dual layer multilayer switches with HSRP, Etherchannel

redundancy, and rapid PVST+ are made to guarantee 99.9% uptime objective (8.76 hours maximum downtime per year) [82].

8. **Large-Scale Testing:** Test success rate of 100% on 31 test cases which consist of connectivity, security, services, redundancy, and performance.

10.2. Lessons Learned

Technical Insights

1. **Complexity of VPN troubleshooting:** IPSec VPN implemented needed structured debugging strategy with crypto commands; discrepancies in pre-shared keys and ACL errors were the major problems [93]
2. **DHCP Relay Configuration:** IP helper-address on all VLAN interfaces necessary to provide centralized DHCP; initial management resulted in failures of addressing [94]
3. **ACL Importance of Order:** The permit statements should be placed before the deny statements; initially Finance ACL blocked all traffic including the internet access [95]
4. **Multilayer Switch Superiority:** Hardware-based inter-VLAN routing is much faster than router-on-a-stick; it is worth investing in due to its performance improvement [96]
5. **Documentation:** Version control and systematic documentation make sure that mistakes were avoided and were able to quickly troubleshoot.

Project Management Insights

1. **Time Allocation:** VPN configuration (14 marks) took excessive time; time budgeting of future projects should be done better according to the marking

scheme.

2. **Incremental Testing:** It tested every step of configuration and identified problems early, avoiding cascades of failure.
3. **Packet Tracer Limitations:** Some features are not available; other methods would have to be used without changing the design purpose.

10.3. Final Assessment

The Avalanche Cooperations network infrastructure is a professionally developed and well-architected solution that exhibits the understanding of network security principles, protocols and best practices. The project has achieved a balance between security, performance requirements, the aspect of scalability, and constraints of budget without imposing any financial burden on the organization and as a result of this achievement, providing a strong base upon which the business can grow in the future.

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